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DOCUMENT TITLE:	
Installation, Operation and Maintenance Manual	

Content

Item	Description
1	DSCU Installation & Operators Manual
2	DSL-800 Installation Manual
3	DSL-800 Operators Manual
4	Purge Air Requirements



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Installation, Operation and Maintenance Manual

1.DSCU Installation & Operators Manual

Installation & Operators Manual V1.6



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Precautionary Statements

Throughout this manual, precautionary statements are included to draw attention to potential hazards.

These statements are always boxed and include the signal words DANGER, WARNING, CAUTION, or NOTE, to indicate the degree of hazard that may be encountered. These words are defined below:

DANGER: *Imminent hazard and failure to avoid hazard will result in death.*

WARNING: *Hazard that may result in injury or death.*

CAUTION: *Hazard that may result in minor or moderate injury.*

NOTE: *Hazard that may result in either, damage to equipment that will require repair or replacement; or a reduction in performance that renders the equipment unable to meet its specification.*

General Precautions

DANGER: *Never enter an active traffic tunnel bore. Fast moving traffic represents an imminent threat that will result in death, whilst hazardous atmospheres may result in injury or death.*

WARNING: *No electrical or mechanical work must be permitted on this device unless it is completely isolated from any source of voltage. Failure to comply with this requirement may result in serious injury or death.*

CAUTION: *All installations and connections must be made by suitably qualified and experienced engineers with proper regard to good practice and compliance with applicable legislation.*

CAUTION: *Appropriate Protective Personal Equipment (PPE) must be worn when performing electrical and mechanical installation.*

CAUTION: *Appropriate access equipment, operated by qualified personnel, must always be used when working at height.*

NOTE: *Prior to use, this equipment must be stored in its original packaging in a dry and sheltered area that is not exposed to extremes of temperature or humidity.*

NOTE: *This device is particularly vulnerable during transportation and manual handling. Always transport the instrument in its original packaging and handle with utmost care.*

NOTE: *This device contains sensitive electronics and must be protected from physical shock, excessive vibration, and ingress, at all times. Always handle with care; replace lids and covers when not in use; and protect from ingress.*

NOTE: *This device contains sensitive electronics which are vulnerable to ESD and therefore care should be taken to avoid making contact with electronic circuit boards & components. Use ESD handling precautions.*

NOTE: *Cables connected to this device may carry sensitive electronic signals which can be affected by RFI and EMI. Care must be taken to run cables away from high voltage power cables, high frequency signal cables, or switch gear cabling. Ideally any cables connecting to the device, or between instrument components, will follow an independent route and will not be installed in bundles or cable trays with other cabling.*

NOTE: *Do not make or break electrical connections whilst power is connected to this device, or any device connected to it. “Hot plugging” as this is known may damage the electronics. Such damage is NOT covered under warranty.*

NOTE: *Applications vary and all installations will involve a degree of application engineering which is the responsibility of the customer, not Dynoptic Systems.*

NOTE: *Read this manual in entirety before installing, commissioning, or otherwise using this device. If in doubt, contact Dynoptic Systems directly for advice.*

Product Overview

The DSCU (Dynoptic Systems Control Unit) is an OI (Operator Interface) that consists of a numeric / directional keypad, a two line LCD display and a terminal compartment.

The keypad and display combine to form an interface which can be used to setup and control connected instruments with similar functionality to the PC based Utility software supplied with each instrument.

The terminal compartment is used for electrical connections. The DSCU can be connected directly to a single instrument or to a network of instruments via a single umbilical cable which carries power and communications (RS485). The SCADA interface connections can be made directly to the DSCU (located at ground level, or in a control room) and / or to the instrument. This versatility offers the flexibility to accommodate a wide range of wiring schemes.

The DSCU offers a variety of industry standard interface options. Scalable analogue outputs are available in the form of isolated 0 / 2 / 4-20mA loops. Digital outputs are available in the form of relay contacts for service alarm (data valid) and / or programmable level alarms.

The DSCU is available in three variants which provide a different number of analogue outputs and relays. These variants are summarised in the following table. The wiring diagrams, performance descriptions and screenshots in this document all show the DSCU-8 variant.

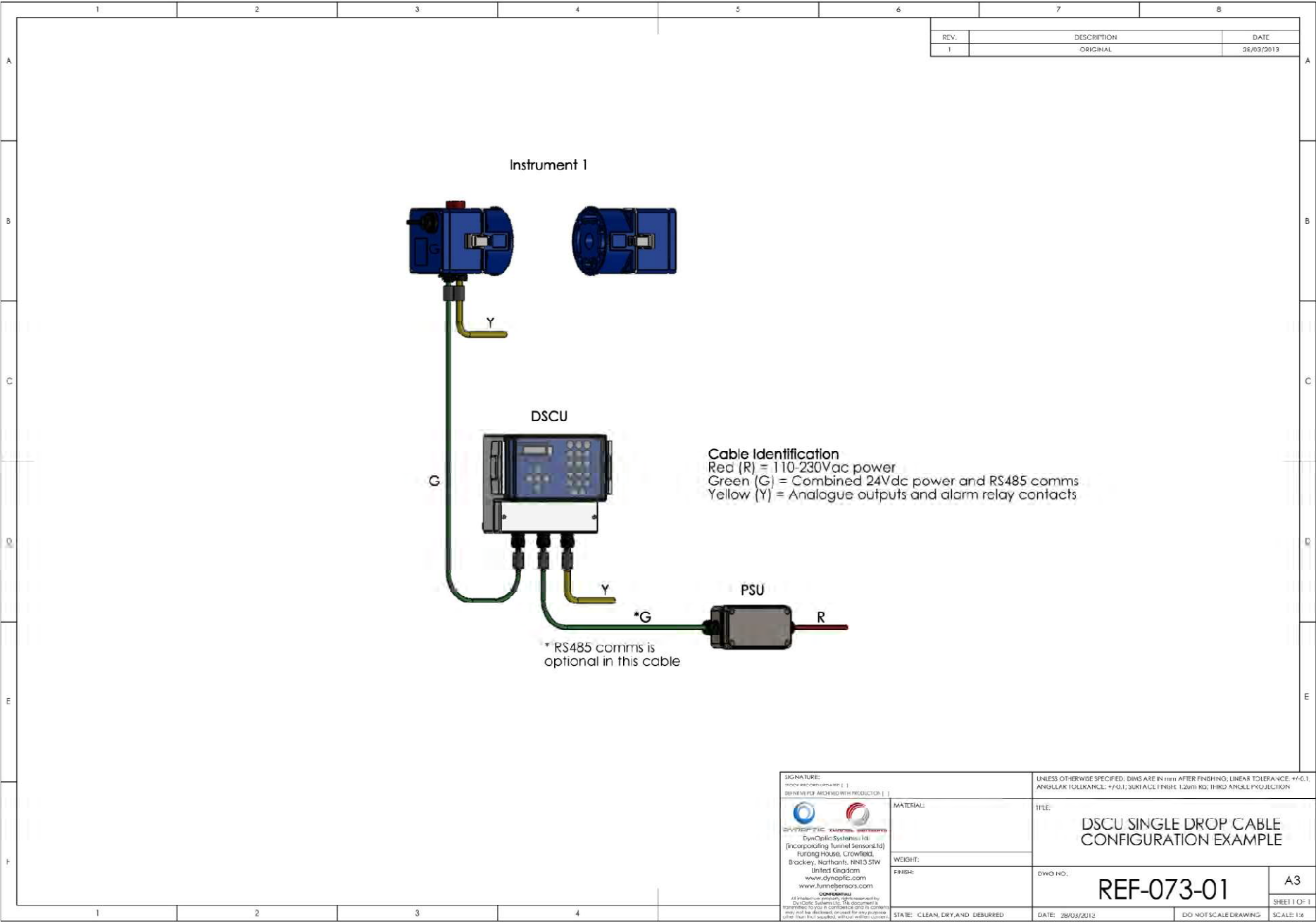
Model	No. Analogues	No. Relays
DSCU-1	1	1
DSCU-4	4	4
DSCU-8	8	8

A serial communications channel is available in the DSCU. RS485 is available as standard. Please enquire with us directly if you have any specific requirements for serial communications.

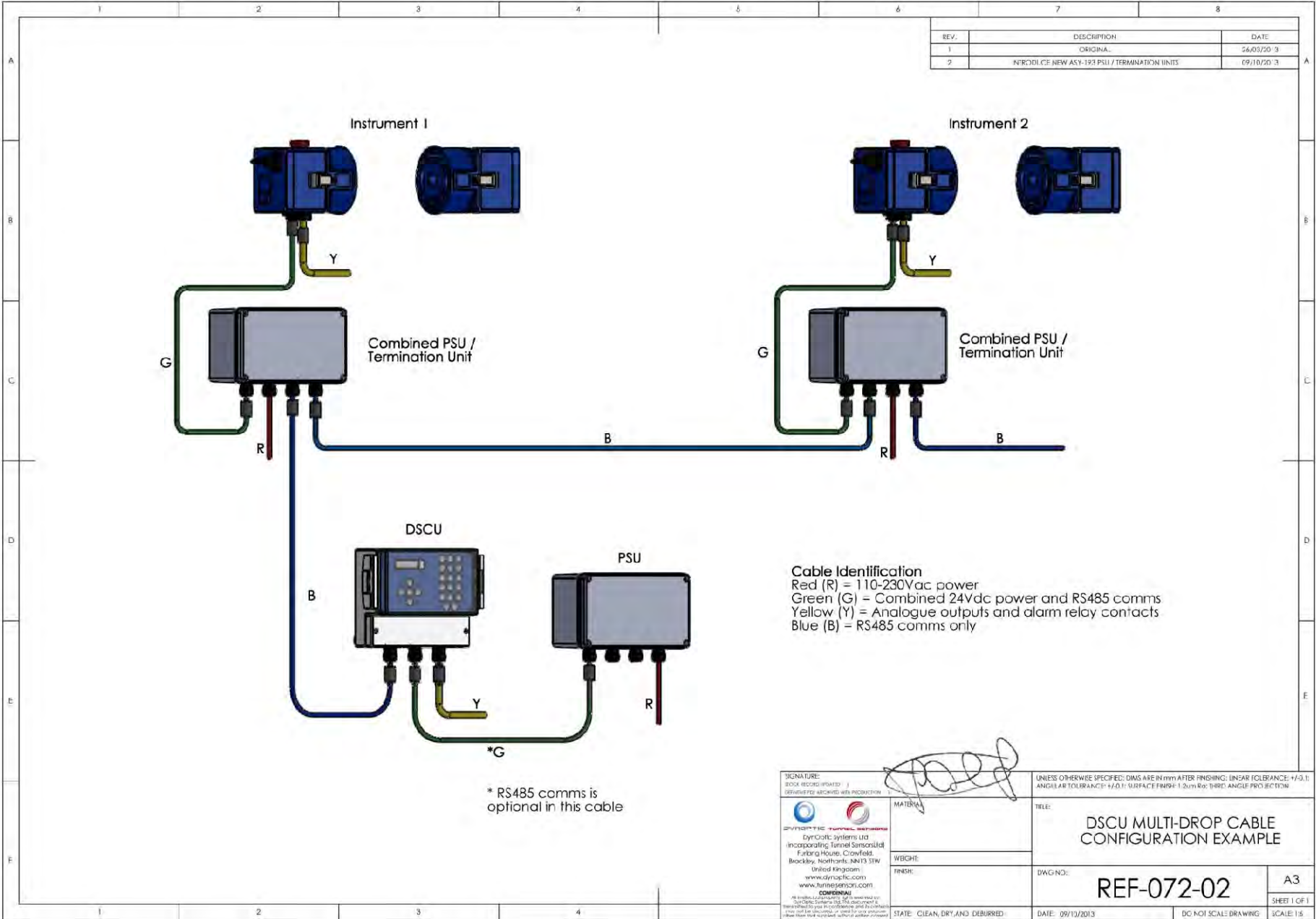
The following pages show example cable configurations for a DSCU connected to a single instrument, and connected to multiple instruments.

The first sections of this manual describe the physical and electrical installation of the DSCU. The remaining sections describe the detailed operation of the DSCU menu structure and the Utility software. A basic description of how to use the Utility to set up the DSCU and how to work with multiple instruments is provided in the Appendix.

Dynoptic Systems Control Unit – DSCU – Installation & Operators Manual



Dynoptic Systems Control Unit – DSCU – Installation & Operators Manual

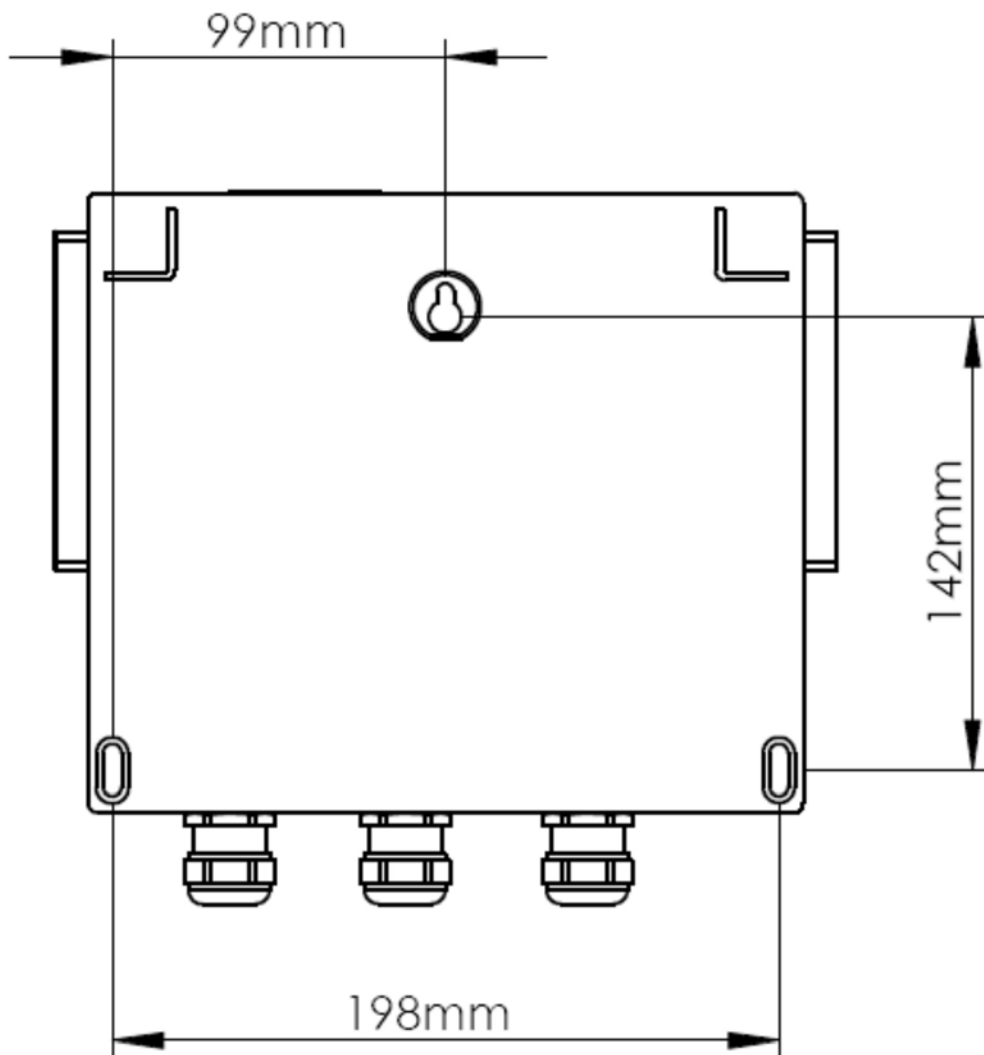


Physical Installation of the DSCU

Wall Mounting the DSCU

The DSCU can be fixed to a wall or panel using 3 off M5 bolts or no.10 sized self-tapping screws.

There is a single central fixing point by which the DSCU enclosure is “hung” on the wall. Two further fixing points are located at the base of the terminal compartment. These two fixings are accessed from the front of the enclosure (through the terminal compartment) and serve to secure the enclosure tight to the wall.



Mark, drill and if necessary tap the mounting holes. Insert a screw into the top hole in the wall, screwing almost all the way down but leaving just a few millimetres of thread protruding from the wall.

Hang the DSCU on this screw using the top “keyhole” mounting point and ensure that the DSCU has dropped down properly over the screw head. Adjust the hanging screw as necessary to ensure the enclosure is pulled gently against the wall.

Open the terminal compartment of the DSCU and insert screws into the two (2) slotted fixing holes to secure the DSCU against the wall and to prevent it from being “lifted” off the central hanging screw.

Electrical Installation

NOTE: *Electrical power must be disconnected from the instrument when making or breaking electrical connections of any kind.*

Installing the Umbilical Cable (DSCU to Instrument)

Due to the variable lengths possible, this cable is not normally supplied with the instruments. Suitable cable can be organised locally or purchased separately from Dynoptic Systems. This cable must have a minimum of 4 cores and meet the specification shown below.

When the DSCU is connected to a single instrument this cable can be used to power the instrument as well as provide the RS485 communication between the DSCU and the instrument head. However, when the DSCU is connected to multiple instruments these instruments must be powered locally and the umbilical cable to the DSCU only carries the RS485 communications.

When the DSCU is used to power a single instrument the maximum length of this cable is limited by the voltage drop likely to occur on the 24Vdc supply carried in this cable. Always calculate the potential voltage drop when specifying the type/length of this cable, taking care to allow for the output tolerance of the power supply. The 24Vdc supply to the instrument must not fall below 22Vdc.

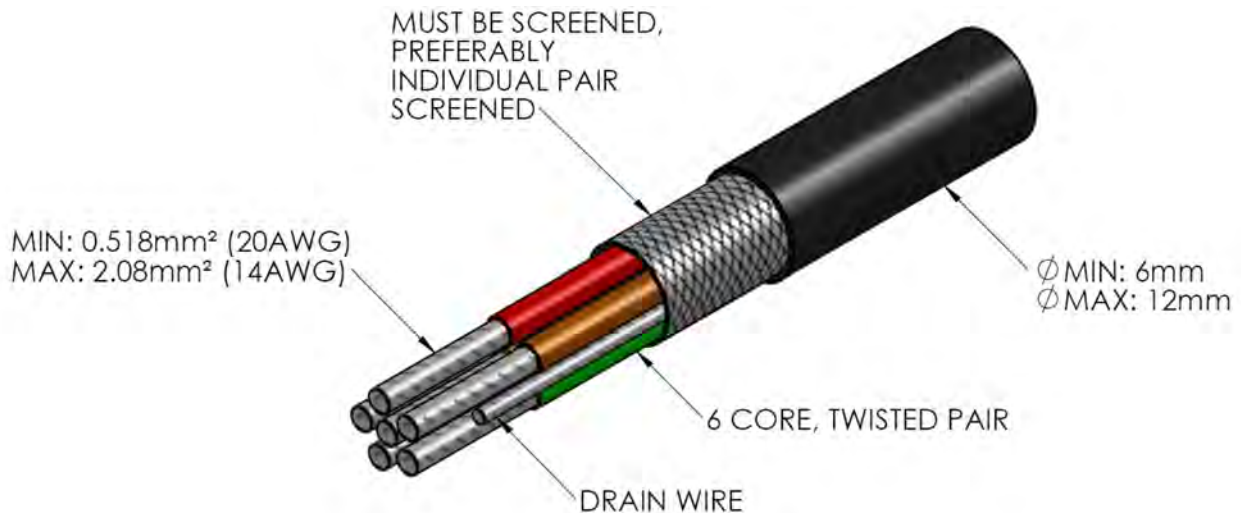
If separate power supplies are used the voltage drop is no longer the limiting factor, the maximum cable length is then 1km which is the recommended limit for the RS485 communications. When the DSCU is connected to multiple instruments the total cable length to the furthest instrument must be less than 1km.

When using long cable runs it is essential to check for differences in earth potential between the different locations and to ensure that the difference is no more than +/- 5V AC or DC.

All cables should have an individual core size of between 0.518mm² (20AWG) and 2.08mm² (14AWG).

All cables must be screened, ideally individually screened twisted pairs with a common drain wire, such as Belden 9873 (6 core, screened pair, 20AWG).

The overall diameter of the cable should be between 6 and 12mm.



The umbilical cable carries sensitive electronic signals which can be affected by RFI and EMI. Care should be taken to run the cables away from high voltage power cables, high frequency signal cables, or switch gear cabling.

Ideally any cables connecting to the instrument, or between instrument components, will follow an independent route and will not be installed in bundles or cable trays with other cabling.

The supplied ferrite tubes should be fitted on the cables where the cables enter any instrument or DSCU enclosure. The ferrites should be secured in place so that they remain fixed adjacent to the cable gland entry.

To fit the ferrite, pass the cable through the centre of the ferrite and then pass the cable through the cable gland and into the enclosure. Fix the ferrite in place by fitting a cable tie either side of the ferrite, thus preventing it from sliding along the cable.

Installation Wiring Diagram

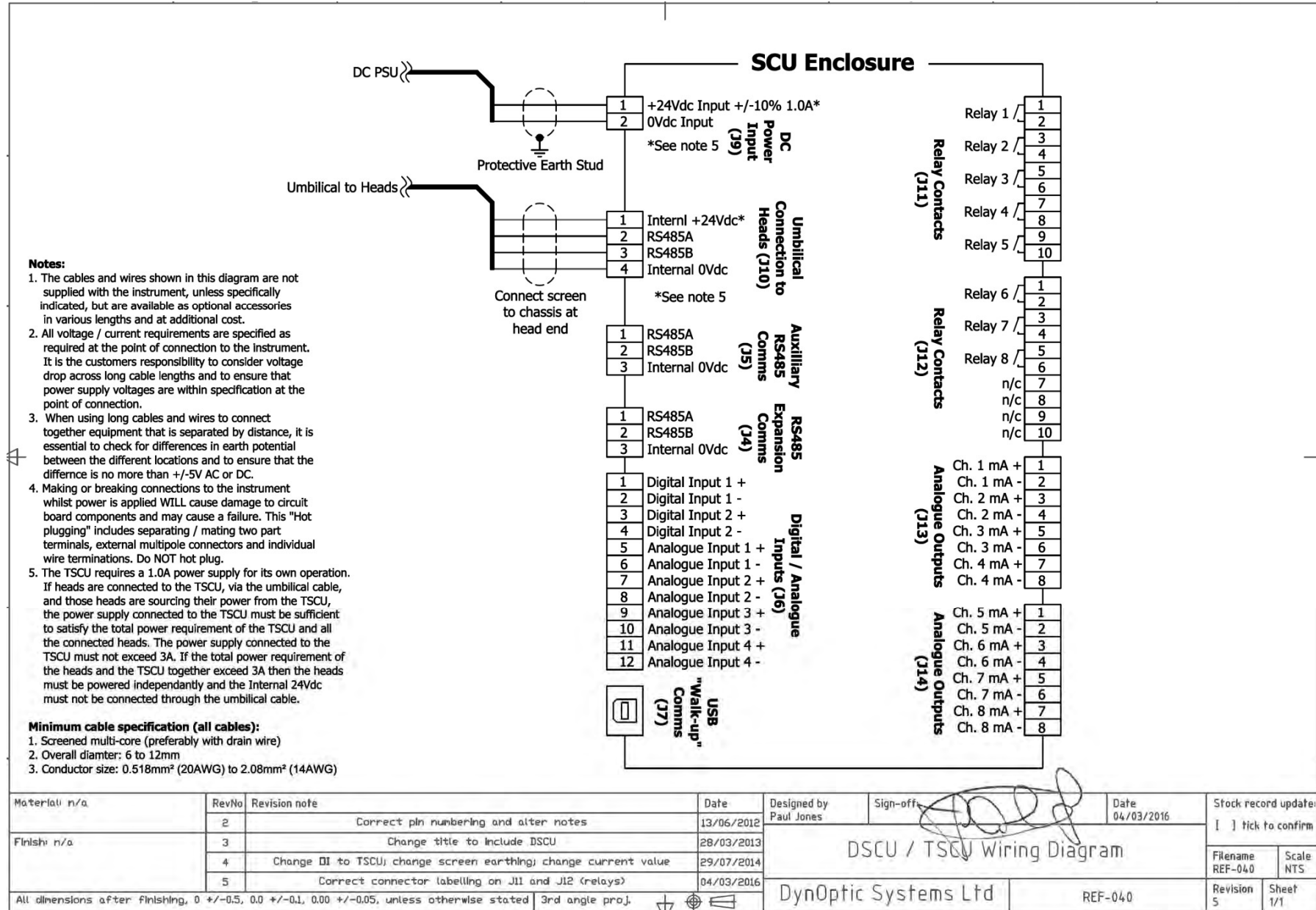
The following drawing shows the installation wiring diagram for the DSCU.

The connection terminals are located in a separate terminal compartment at the bottom of the enclosure.

This diagram includes details of common interface wiring, but not all these connections are essential for functional operation. The drawing shows 8 relays and 8 analogue outputs. The actual number of outputs supplied is dependent on the option selected (1, 4 or 8).

Some interface connections available in the DSCU may also be available within the connected instruments as well. In this case, interface connections can be made at either the DSCU or the instrument, or even both.

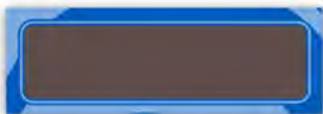
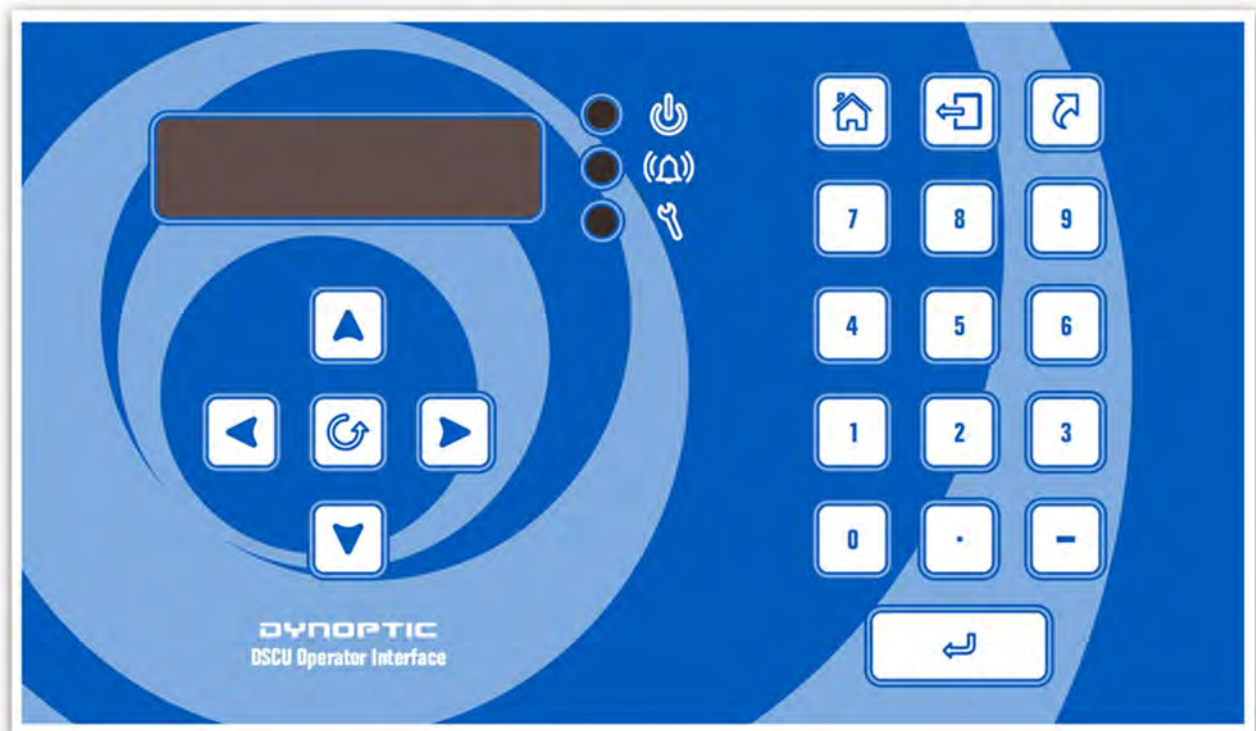
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DSCU Interface & Operation

The DSCU can be set up and operated via the keypad / display or via the Utility software.

The DSCU keypad / display interface is shown below: -



Two-line LCD Display.



The **Power** LED will illuminate (green) when the DSCU is powered on.



The **System Alarm** LED will illuminate (red) when the threshold of any level alarm is exceeded.



The **Service Fault** LED will illuminate (red) when any connected instrument detects a fault condition that requires attention.



Direction buttons allow navigation of the DSCU menus. The right direction button and the cycle button also act as an Enter button where applicable. The left and right arrows are used to navigate between connected instruments (for multi-heading).



The **Home** button returns you to the Readings screen from anywhere in the menu structure. If pressed twice it exits the password entry.



The **Exit** button cancels data entry.



The **Menu** button. You will need to enter a password before being allowed access to the menus.



The numeric keypad for data input.



The **Enter** button is used to end data entry and store the altered value.

DSCU Operation

Readings Screen

On power up the DSCU will search for all of the connected instruments that it has previously been set to find. Note that the default setting is one instrument with ID=1. Appendix A describes how to set the DSCU to find the actual attached instruments. If an instrument is found then it is displayed on the Readings screen.

The Readings screen menu structure is shown below. The DSCU tab shows the name and ID of the DSCU, and the status of the DSCU and any instrument connected to it. If there are no faults or warnings 'System OK' is displayed.

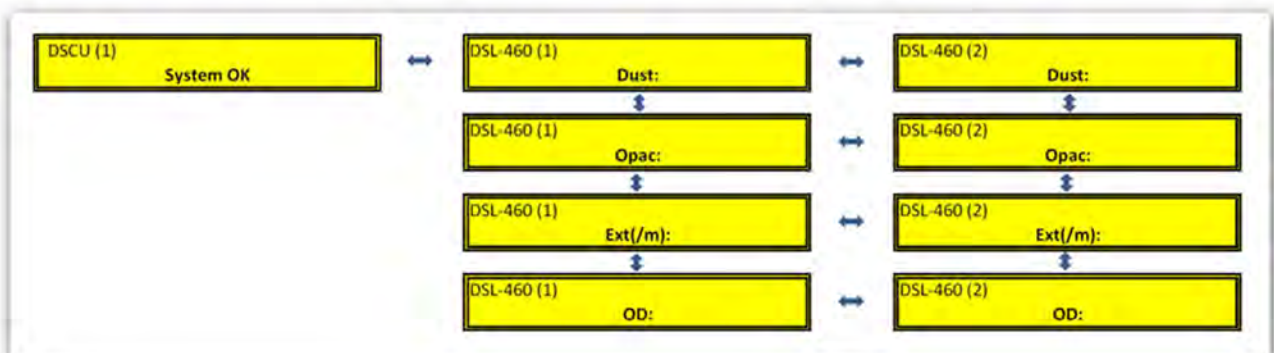
The left / right arrow keys are used to scroll between the DSCU tab and the connected instruments.

When an instrument tab is displayed, the top line of the LCD will display the name of the instrument followed by the ID of the instrument in brackets.

The bottom line of the display shows the measurement name, reading and units e.g. Opacity: 10.0%. If the instrument detects any faults or warnings they will be displayed here. If there is a fault or warning the display will flick between showing the measurement value and the fault(s).

Pressing the down button scrolls through the different measurement readings for the instrument (if more than one is available).

The following example illustrates the display option for a DSCU connected to two DSL-460s, one with ID=1 the other with ID=2.



Menu Structure



Press the Menu button to access the menus.

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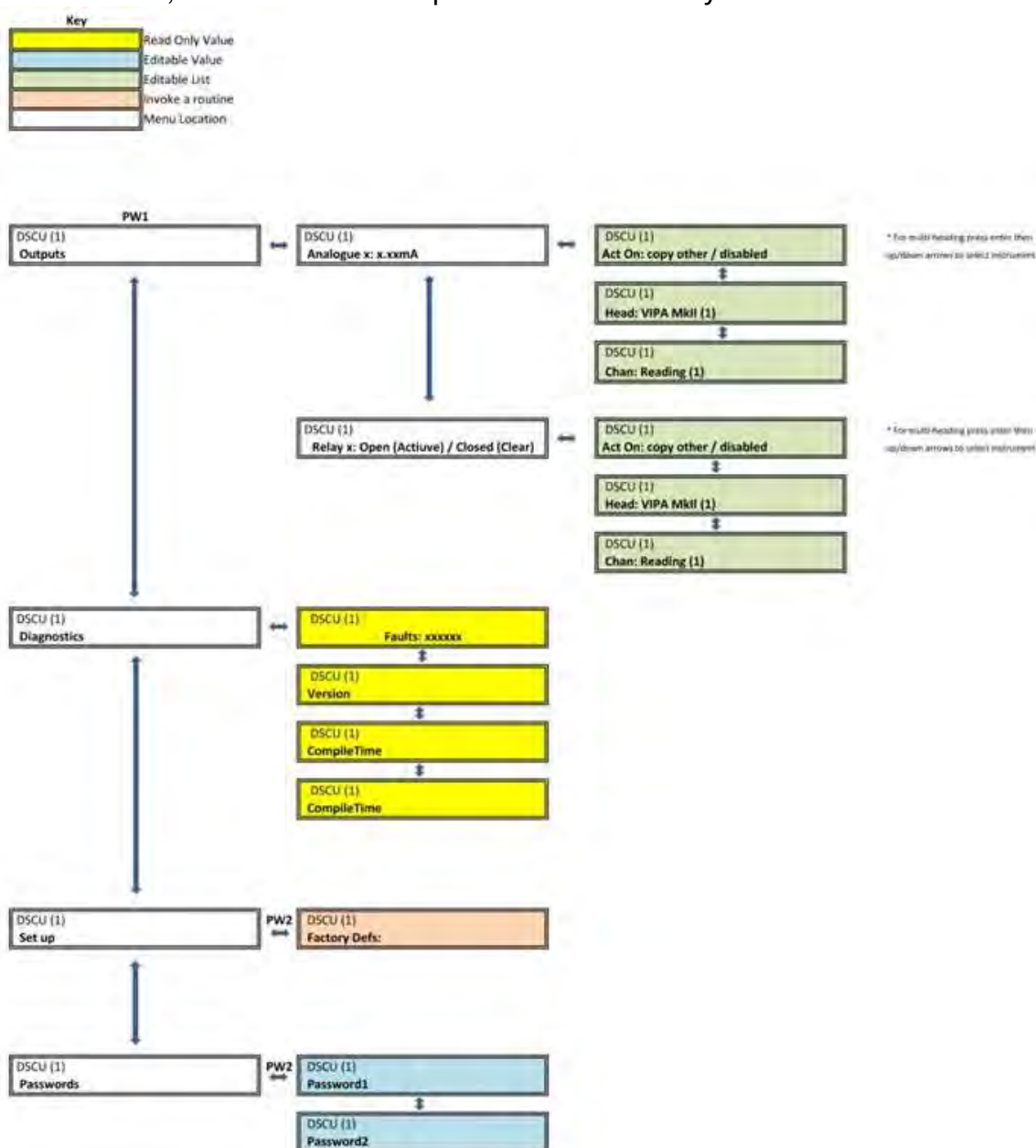
You will be asked to enter a password at this point. The default passwords are as follows: –

Password 1 (PW1) = 0000

Password 2 (PW2) = 1111

For more information on passwords – see the Password section below.

The diagrams below show the menu structure for the DSCU and, as an example, the menu structure for a connected DSL-460 instrument. The basic layout is the same for all DSL instruments, but the individual parameters will vary.



Menus

The menus for the DSCU and any connected instrument comprise of the following sections: -

Outputs

All parameters relating to the analogue and digital outputs from the instrument are shown here. For details on setting up the analogue and digital outputs please refer to the instrument Operators Manual.

Diagnostics

The instrument diagnostics are provided here, such as voltages, gains, temperatures etc. All the values in the section are read only, i.e. they cannot be edited.

Setup

Requires a level 2 password (PW2) to access. This section allows changes to be made to parameters, such as selecting units and setting gains, that are normally performed as part of the setup, commissioning and calibration of the instrument.

Passwords

There are two levels of password – PW1 (password level 1) for users to be able to view the data from the instrument, including checking diagnostics, parameters, analogue and digital output settings etc. PW2 (password level 2) is for engineers or experienced users. This allows the same access as PW1 users plus the ability to edit settings and perform setup and calibration functions.

The default passwords are as follows: -

Password 1 (PW1) = 0000

Password 2 (PW2) = 1111

DSCU Analogue and Digital Outputs

The number of analogue outputs and relays fitted to the DSCU depends on the instrument variant. It can be either one (1), four (4) or eight (8).

The analogue outputs can be configured to track any of the analogue outputs of the connected instruments or any measured parameter from any connected instrument. Note analogue outputs can be taken from the DSCU, or from the instrument or from both.

All relays are fully configurable i.e. it is possible to select the unit and select the parameter that each relay acts on.

The first relay by default is set as a system Data Valid. If any instrument connected to the DSCU goes data invalid (i.e. when multi-heading) this will trigger the relay.

For more information on setting up the relays please see 'Using Utility Software' section.

Installing the Utility Software

NOTE: *The Utility Software and the USB Driver must be installed on the PC before connecting the DSCU. Failure to comply with this requirement is likely to result in a failed installation.*

Minimum hardware/software specification for host PC

The Utility Software supplied with the DSCU is not a large or particularly complex program, and therefore has no requirement for a high performance host PC. Any reasonably modern PC/laptop will be suitable. The only “must have” requirements are:

- Windows 7, or newer, operating system.
- All recent operating system Service Packs and upgrades (visit the Microsoft website) must have been installed.
- At least one free USB slot (assigned to a COM Port between COM1 and COM8 – see later section on Assigning a Valid COM Port) for direct connection to the DSCU.

Installing the USB Driver

The USB Driver software is supplied in electronic format (CD or flash drive) with each instrument. A link to a download file can be provided by Dynoptic Systems on request.

1. Ensure that the PC is logged on to a user account with Administrative rights, so that the necessary permissions are granted to install software.
2. Copy the USB Driver “CDM xxxxx.exe” file to any suitable location on the host PC (“My Documents” is a good location).
3. Double click on the .exe file and follow the onscreen installation prompts.

Installing the Utility Software

The DSCU uses the “SCU Utility Software” which is supplied in electronic format (CD or flash drive) with each device. A link to a download file can be provided by Dynoptic Systems on request.

1. Ensure that the PC is logged on to a user account with Administrative rights, so that the necessary permissions are granted to install software.
2. Copy the file “SCU Utility Install xxx.exe” to any suitable location on the host PC (“My Documents” is a good location).
3. Double click on the .exe file and follow the onscreen installation prompts.

The software should now be installed and should appear in the “All Programs” list in the folder “SCU UTILITY”. The software is called “SCU Utility.exe”.

Updating the Utility Software

From time-to-time Dynoptic Systems may release updated versions of the DSCU Utility Software. If you receive an updated version, it is essential that you remove any existing version from your PC before installing the new one.

Assigning a valid COM Port

When a PC is first connected to the DSCU, it is possible that the PC will assign the USB connection to a COM Port number outside the acceptable range (COM1 to COM8), in which case the Utility Software will be unable to communicate with the instrument.

It is possible to check which COM Port your PC has assigned to the DSCU by opening Windows “Device Manager” before connecting your DSCU. When the “Device Manager” window is open scroll to “Ports (COM & LPT)”, then click the “>” symbol to look at the list of COM ports.

With the list of current COM ports visible on the PC screen, connect the USB to the DSCU. The PC should indicate that it has “Found New Hardware” and a new COM Port number should pop up in the “Ports (COM & LPT)” list.

If the new COM Port is between COM1 and COM8 then your PC has allocated a valid COM Port and no further action is required.

However, if the COM Port number that was assigned to the connection is outside the acceptable range, then you must change the assigned COM Port to a number between COM1 and COM8. Follow the procedure below to assign a valid COM Port:

1. Right click on the COM Port (in the “Ports (COM & LPT)” list) that appeared when DSCU was connected.
2. Click “Properties”
3. Select the “Port Settings” tab
4. Click on the “Advanced” button
5. Click the “COM Port” and select a new “COM Port Number” between COM1 and COM8 (avoiding any COM Ports which are marked as in use) from the drop-down list

Once a valid COM Port number has been assigned to the USB connection, the PC will save this COM Port assignment and there will be no need to repeat the process again, when connecting to the DSCU in the future, provided that the connection is made using the same PC and the same USB slot.

NOTE: *If an alternative computer is used it will be necessary to repeat the process above to assign a valid COM Port.*

NOTE: *Always connect to the DSCU using the same USB slot on the PC. If an alternative USB slot is used, it will be necessary to repeat the process above and re-assign a valid COM Port, because the PC identifies each USB slot as a separate entity.*

Using the Utility Software

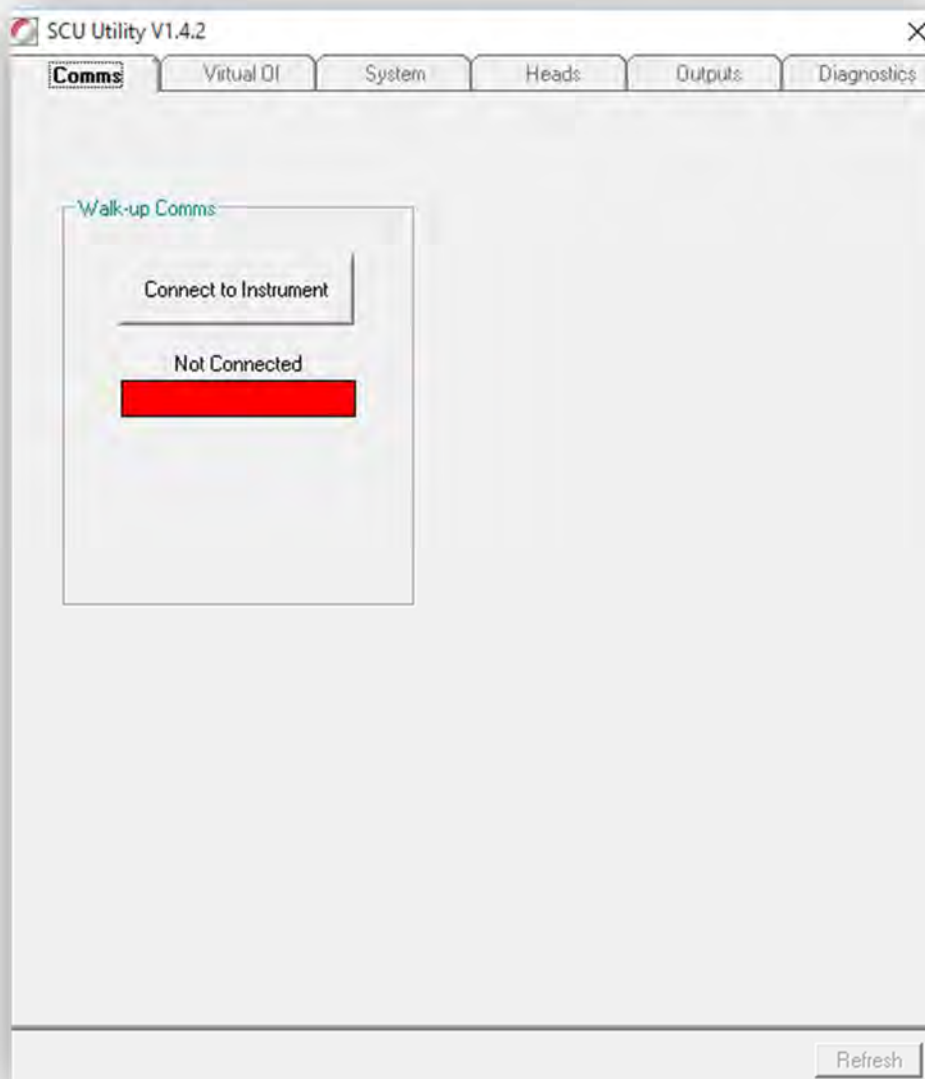
NOTE: *The Utility Software and the USB Driver must be installed on the PC before connecting the DSCU. Failure to comply with this requirement is likely to result in a failed installation.*

Connecting a PC to the DSCU

The USB connection can be found inside the wiring enclosure of the DSCU (on the circuit board inside).

The Comms tab

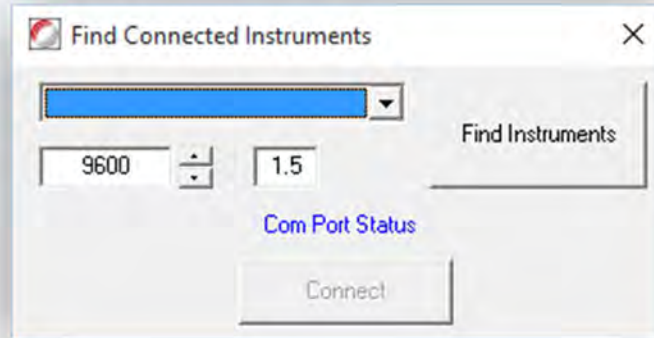
On running the Utility software it will open on the Comms tab (see below).



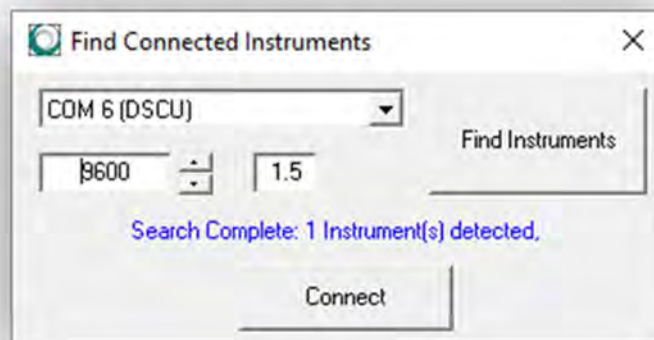
The Comms tab allows the user to establish a connection with the DSCU. When the utility is first opened there will be no connection so the status bar will be red (indicating no connection). When a comms connection is established, additional settings become available on the Comms tab and all other Utility tabs become accessible.

Connect to DSCU

To establish a connection with the DSCU click the “Connect to Instrument” button. The following pop-up window will appear.



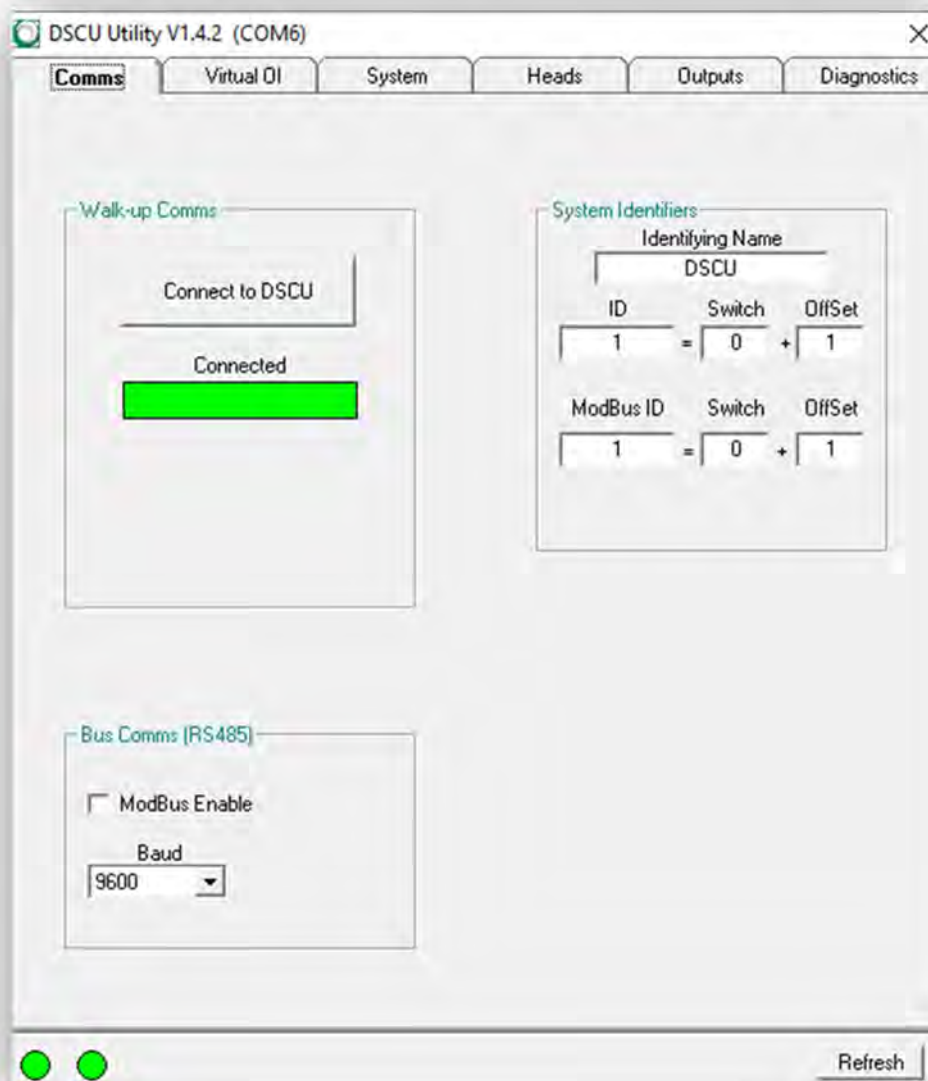
Click the “Find Instruments” button. The software will scan all available computer ports (up to COM 8) for any connected DSCUs (see below).



Select the required DSCU from the list (there will generally only be one) and click ‘Connect’, you will then be returned to the Comms tab where the status indicator bar will now be green (indicating a positive connection). Note that it may take a few seconds to complete the connection, depending on the number of instruments connected to the DSCU.

To illustrate the multi-heading functionality of the DSCU and the Utility software the following descriptions and screenshots show a DSCU with eight (8) analogues and relays, connected to two DSL-460 MkIIs.

Once a comms connection has been established the connected status bar of the Comms tab will turn green. When first connected the software will scan take a few seconds to identify all of the connected instruments. A Comms tab correctly connected to two instruments is shown below.



Once a comms connection is established additional settings become available on the Comms tab and you will be able to select any of the other tabs.

Bus Comms (RS485)

Checking the 'ModBus Enable' tick box allows communication with the DSCU via Modbus. For further information please contact Dynoptic Systems.

System Identifiers

This section is for use when multiple DSCUs are being connected together.

DSCUs can be connected together via RS485 using Modbus or Dynoptic Systems own ASCII protocols. The name and address can be set here. For more information please contact Dynoptic Systems.

Traffic Light Identifiers

The coloured circles at the bottom left of the tab indicates the number of instruments connected to the DSCU and their current status. The status is shown by the following colour codes: –

- Green indicates that the instrument is communicating OK
- Amber indicates that communications with the instrument has failed
- Red indicates that no instrument has been detected

These identifiers are visible on all tabs. The example above shows two instruments communicating successfully.

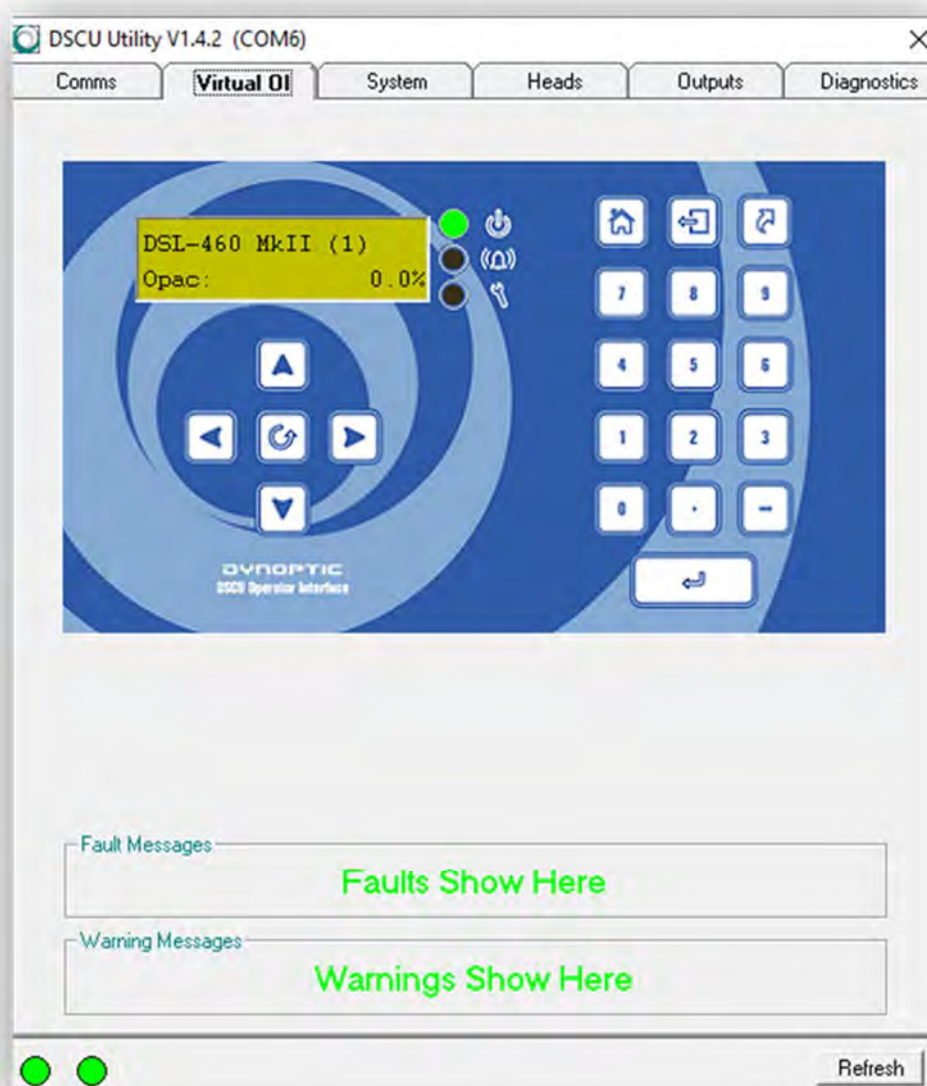
Refresh

The refresh button is shown on each tab, clicking this will refresh the data from the DSCU to the Utility and will perform a scan for connected instruments, this can take several seconds, depending on how many instruments are connected to the DSCU.

The Virtual OI Tab

The Virtual OI tab shows a representation of the DSCU panel and display that allows full virtual control of the DSCU.

This tab also displays the Fault and Warning Messages from the DSCU or from any instrument connected to the DSCU.



Please refer to section “DSCU Interface & Operation” for details on using the keypad and display

NOTE: *Response time is slower than using the keypad directly as the data has to be called back from the DSCU and all instruments connected to it.*

Fault Messages

If the DSCU, or an instrument connected to the DSCU, self-detects a fault, it will display a brief message, describing the fault on this tab. Most fault messages will continue to be displayed on this tab until an operator clicks on the message to acknowledge it. If the fault is no longer active the message will disappear, but if the fault status is still active the message will re-appear immediately.

Refer to the “Fault Messages and Troubleshooting” section of the manual for DSCU faults and the instrument’s Operators Manual for instrument faults.

Warning Messages

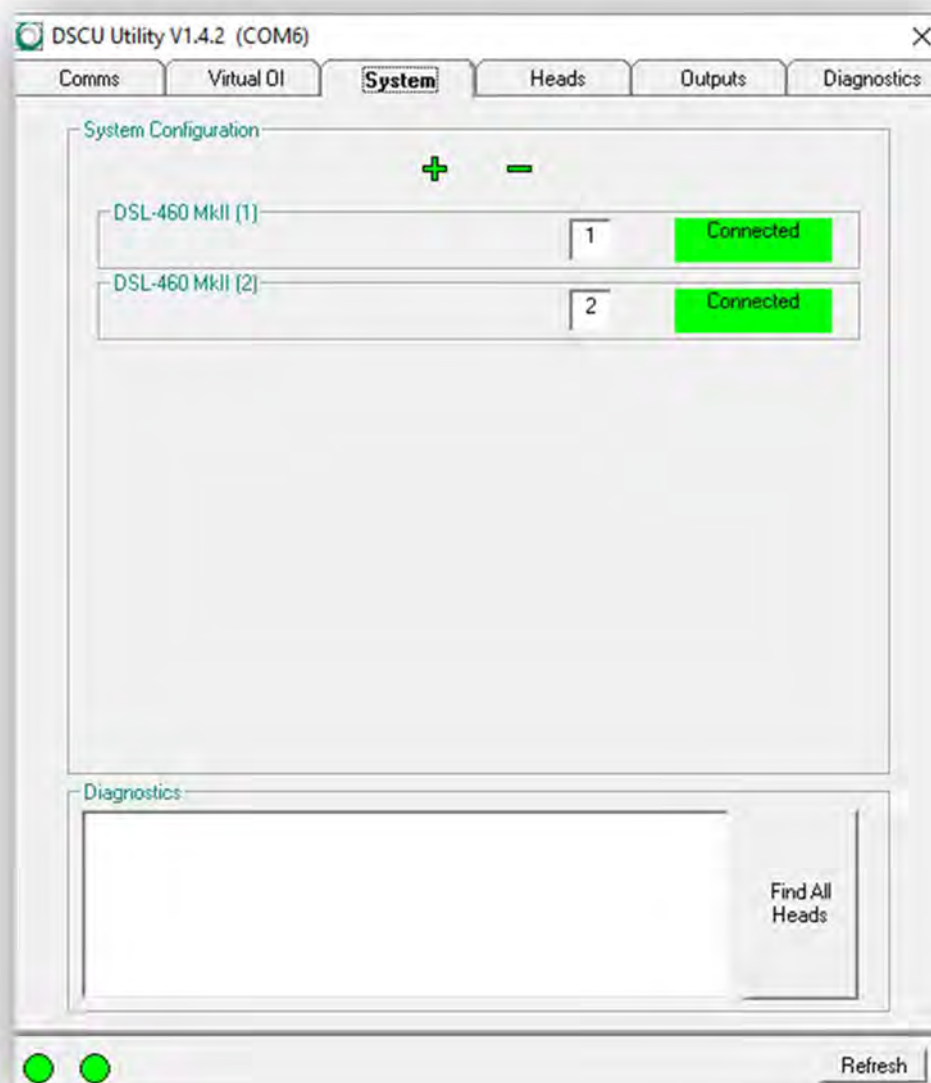
If the DSCU, or an instrument connected to the DSCU, self-detects a parameter that is close to or out of range that needs investigating, a brief warning message will be displayed describing the problem. The warning message will continue to be displayed on this tab until an operator clicks on the message to acknowledge it. If the warning is no longer active the message will disappear, but if the problem causing the warning is still active the message will re-appear immediately.

Refer to the “Fault & Warnings” section of this manual for DSCU warnings and refer to the instrument’s Operators Manual for instrument warnings.

The System Tab

The System Tab allows instruments connected to the DSCU to be added and removed, and shows communication diagnostic data.

The example below shows two DSL-460 MkII instruments connected to the DSCU. Before being connected to the DSCU these two instruments had to be setup to have different RS485 ID's (ID=1 and ID=2). If any two instruments connected to the DSCU have the same ID the connection will fail.

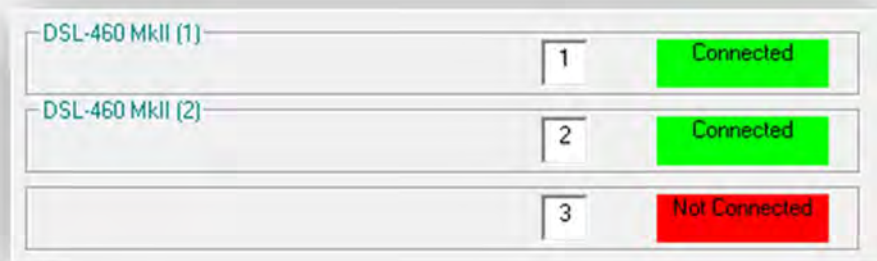


System Configuration

By default the DSCU will be setup to look for only one connected instrument with ID=1. For single instrument operation this is generally all that is required.

The ‘+’ sign is used to tell the DSCU to try and connected to an additional instrument, it is possible to connect up to eight (8) separate instruments to one DSCU. Clicking on the ‘+’ sign adds another instrument connection line with the next ID number. If there is no connected instrument with this ID number the status will be shown as “Not Connected” as shown below.

To connect to an actual instrument with a known ID number you need to manually enter the instrument ID number into the ID box.

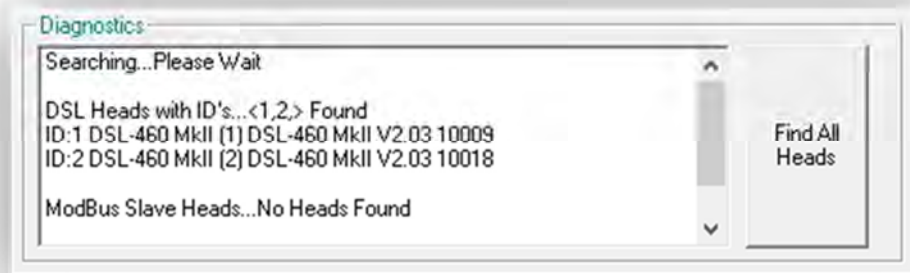


DSL-460 MkII (1)	1	Connected
DSL-460 MkII (2)	2	Connected
	3	Not Connected

The ‘-’ sign is used to remove a line from the instrument connection list. Clicking on the ‘-’ sign will remove the last instrument connection line from the list.

Diagnostics

To the right of the Diagnostics box is the ‘Find All Heads’ button, clicking this button will make the DSCU scans all the addresses available to determine what instruments are actually connected. The results will be shown in the Diagnostics box giving the instrument ID and Type. This information can be used to populate the instrument connection list.

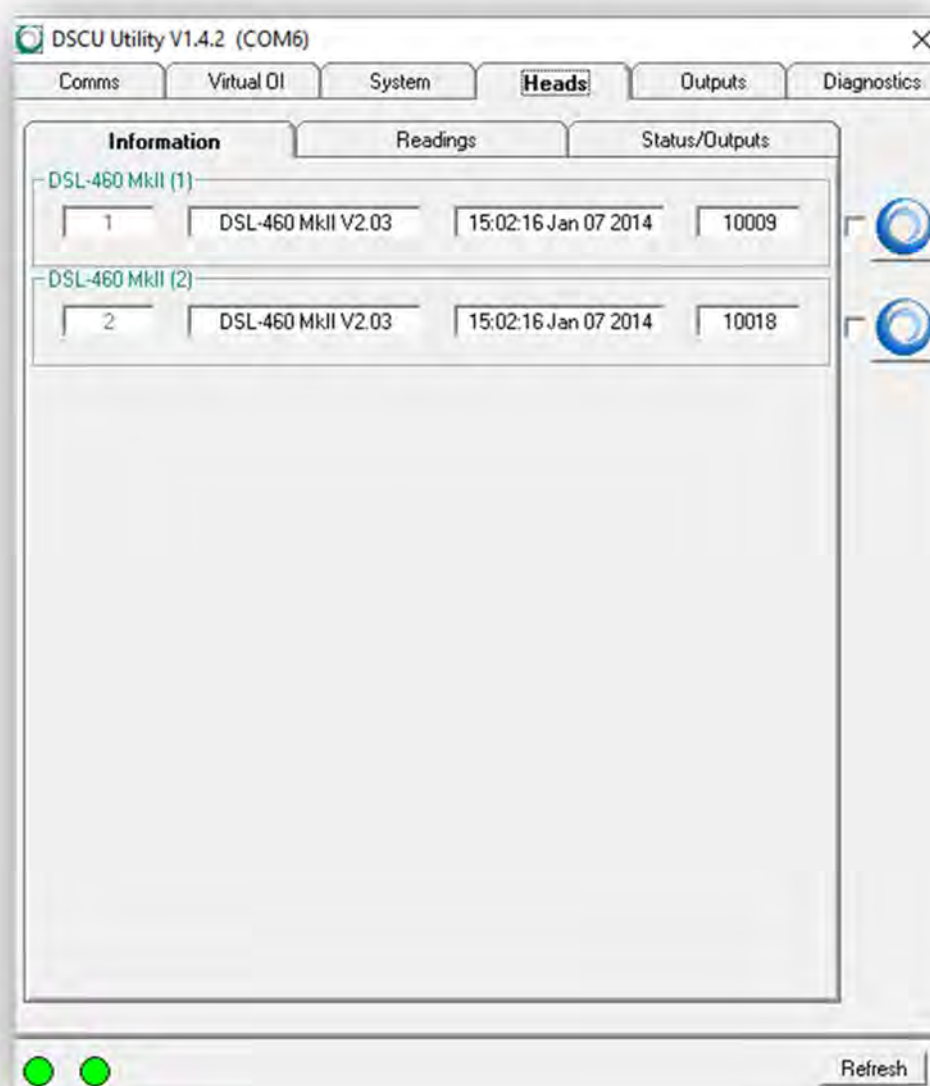


Double click the text in the Diagnostics box to clear it.

The Heads Tab

The Heads tab shows information about the attached instruments including the readings and the status of their outputs. There is also a quick launch button for the instrument's Utility program and a disable 'service fault' checkbox next to this button.

This tab contains of three sub-tabs, Information, Readings and Status/Outputs.



Information

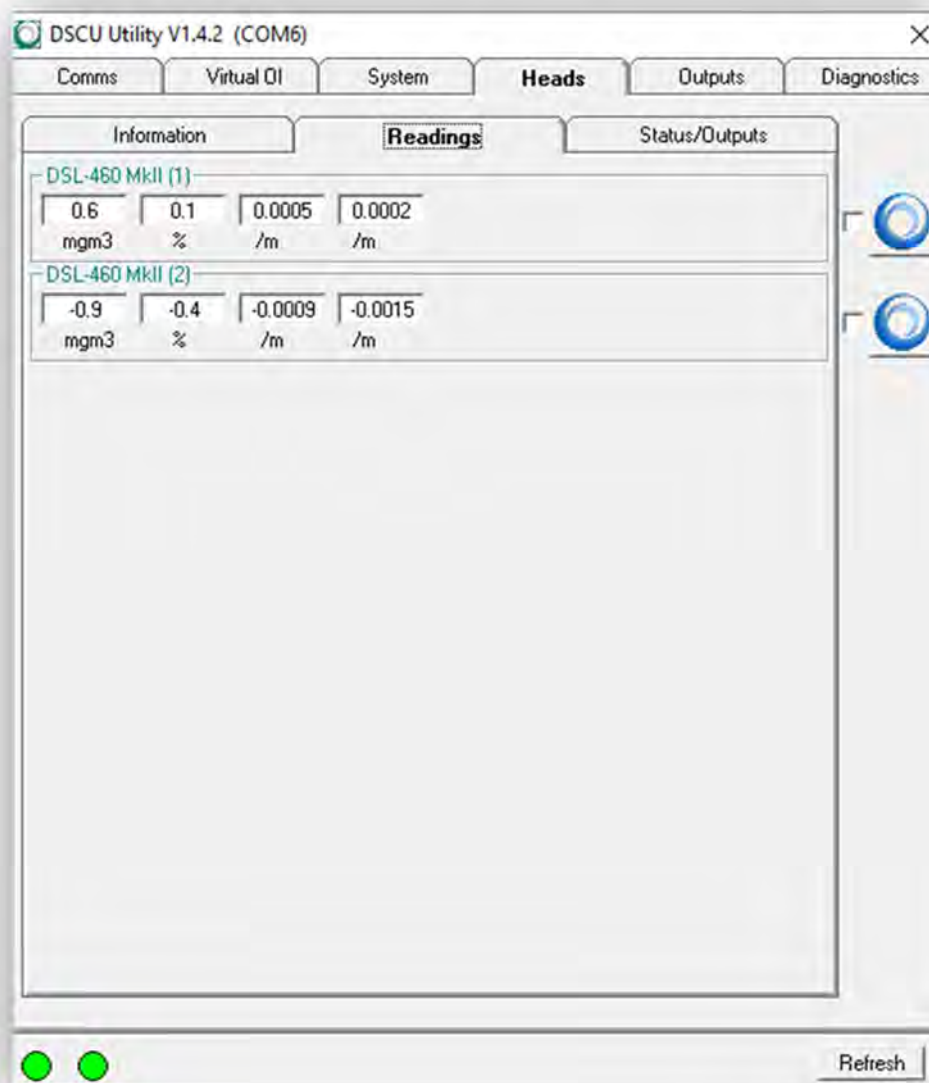
This sub-tab shows information relating to each instrument connected to the DSCU. Each instrument has a descriptive name and the following information is displayed; ID number, instrument description and firmware version, compile time/date of software and the serial number of the instrument's CPU board.



Readings

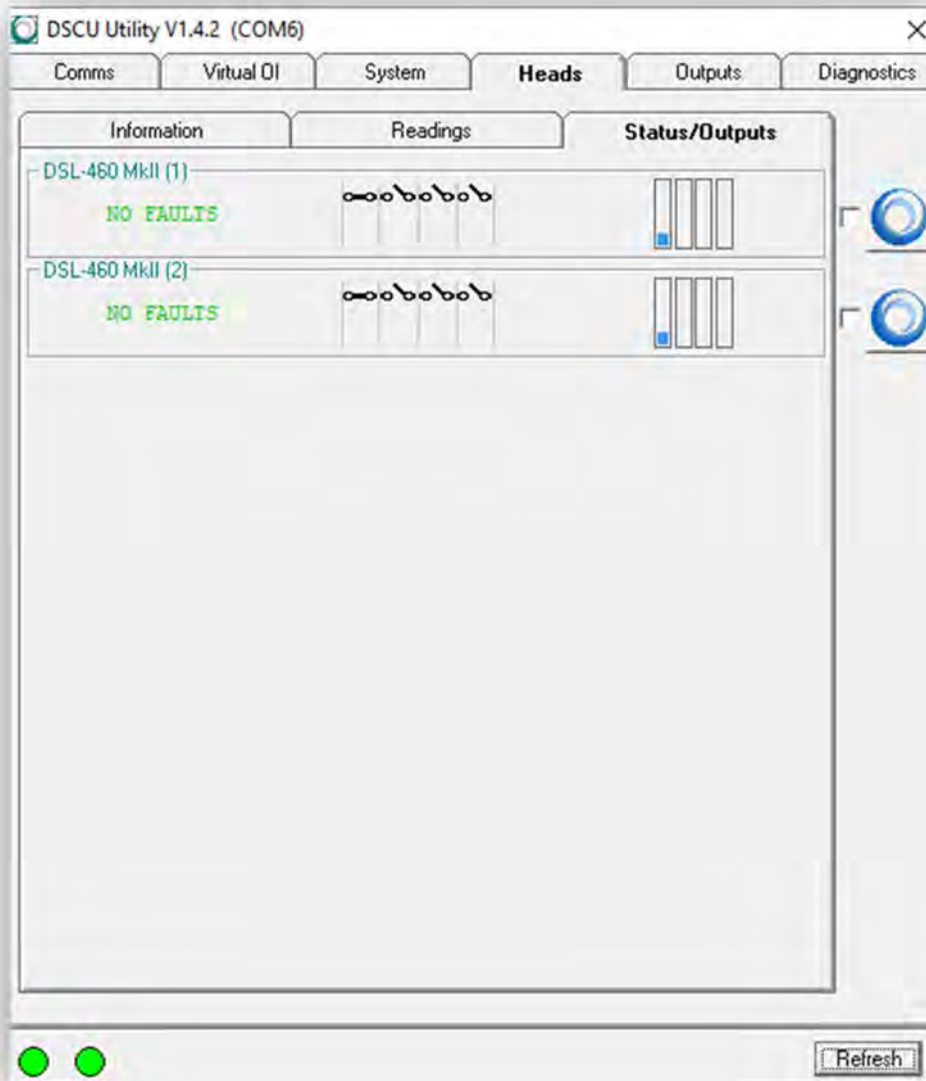
The readings sub-tab shows all of the readings available from each connected instrument. The number of readings shown will depend on which instrument is connected to the DSCU and how many measurement outputs it can provide.

Hover the cursor over the reading for a description of the measurement units.



Status/Outputs

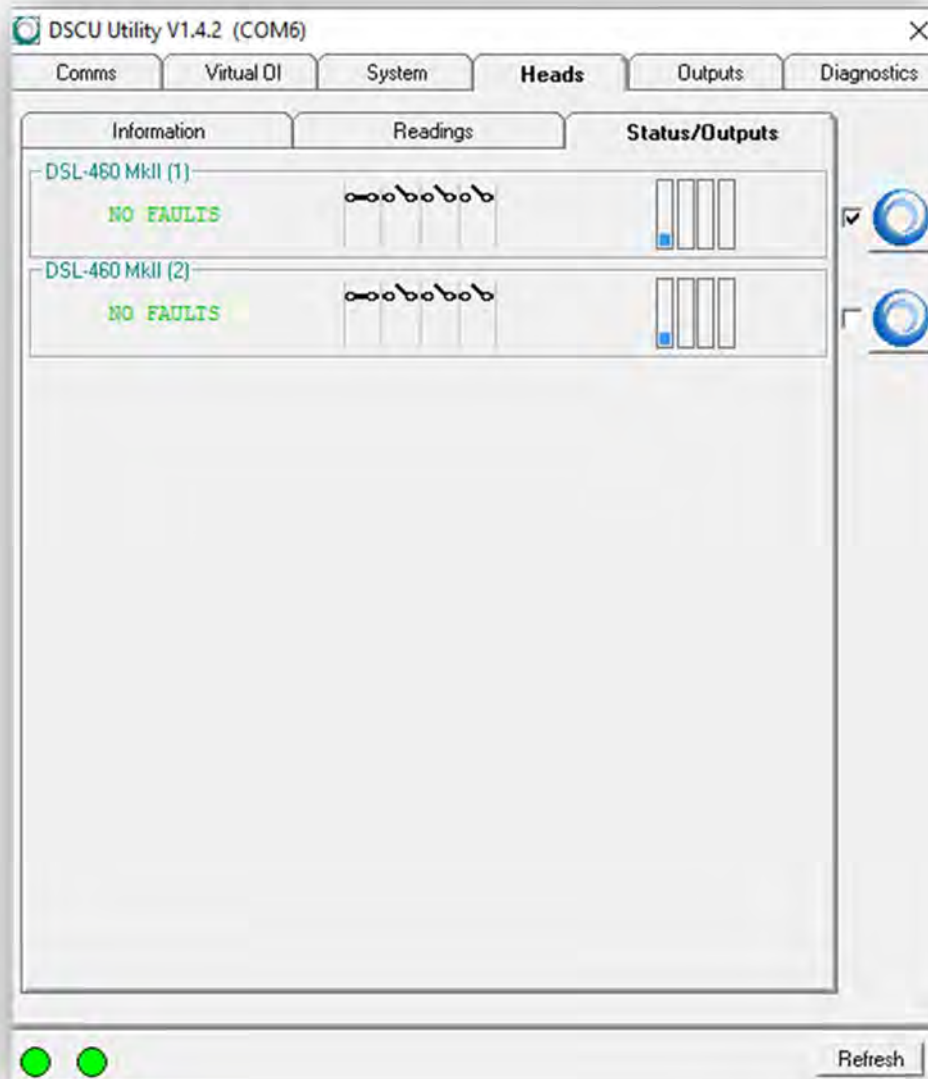
For each connected instrument the Status/Outputs tab shows any fault or warning messages, the status of the relays and the calculated analogue outputs. For more detailed information about the instrument status and setup it will be necessary to open the instrument Utility by clicking on the Dynoptic System logo button to the right of the tab (see later).



In the above example both DSL-460 MkII instruments are showing no faults. They both have four (4) relays the first relay is closed and the rest are open. The DSL-460 MkII head can also configure up to four analogue outputs (one real, three virtual), only the first is configured to an instrument reading.

Service Fault Checkbox

On the right hand side of the Heads tab, next to each instrument line, there is an unlabelled Service Fault checkbox. Ticking this checkbox stops the DSCU Service fault form triggering when there is a fault with this instrument. This can be useful in a multiheading system when one instrument needs to be taken offline, for example powered down to be checked or repaired



When a Service Fault checkbox is ticked a 'Service Override' warning message will be displayed on the DSCU and the DSCU Utility (on the Virtual OI Tab) to make operators aware that an instrument has been taken offline.

This feature is used to temporarily isolate an attached instrument. The DSCU will still search for the missing head. To remove the head completely from the DSCU use the

‘System Configuration’ section to move the unused instrument to the bottom of the list then press the ‘-’ button to remove it.

Utility Quick Launch Buttons

On the right hand side of the Heads tab, for each instrument connected, there is a button with the Dynoptic Systems logo on it. Click this button to launch the Utility program for that instrument.

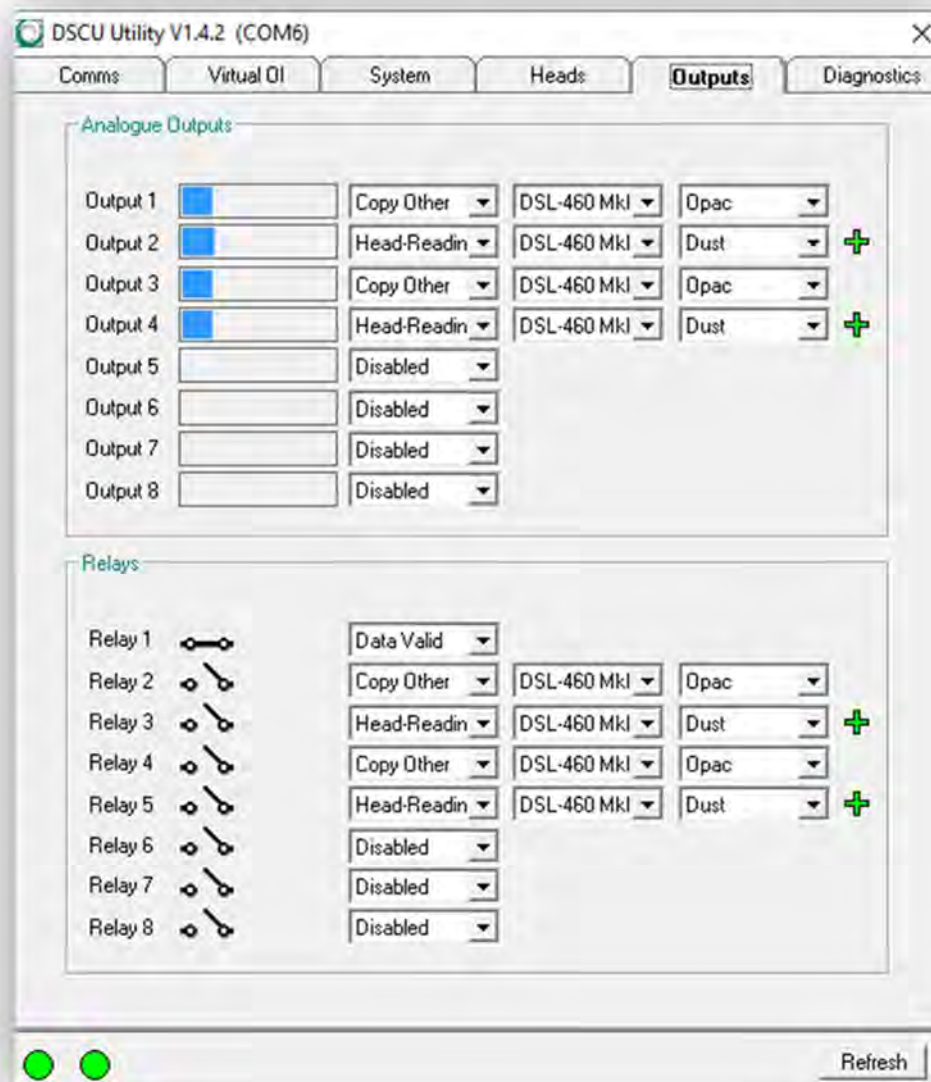
Please refer to the instrument’s Operator Manual for full details and instructions on the use of the instrument Utility.

NOTE: *Only one Utility program can be launched at a time.*

NOTE: *If the instrument settings are changed through the Utility quick launch, on closing the instrument Utility perform a DSCU refresh to ensure that all of the changed settings are updated into the DSCU.*

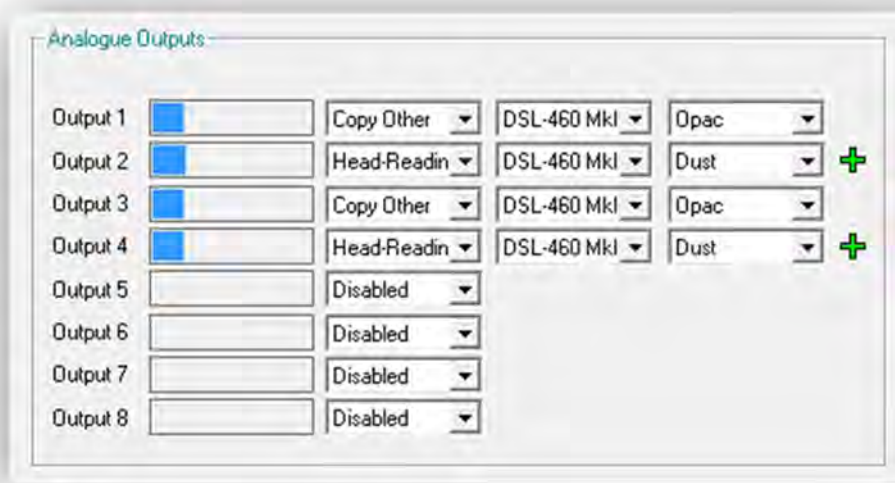
The Outputs Tab

The Outputs Tab shows the status of all of the analogue and relay outputs from the DSCU. The following example is for a DSCU-8 and so there are eight (8) analogues and eight (8) relays available. Each output can be configured to operate in several different ways.



Analogue Outputs

The DSCU analogue outputs can be configured to output the readings from any instrument that is connected to it. The following example illustrates the different configuration available.



The first part displays graphically the value of the currently selected analogue output.

The first drop down menu allows the selection of 'Copy Other', 'Disabled' or 'Head-Reading'

NOTE: Any analogue output not being used should be set to 'Disabled'.

Copy Other allows the DSCU analogue settings to just copy any analogue settings that have already configured in a connected instrument.

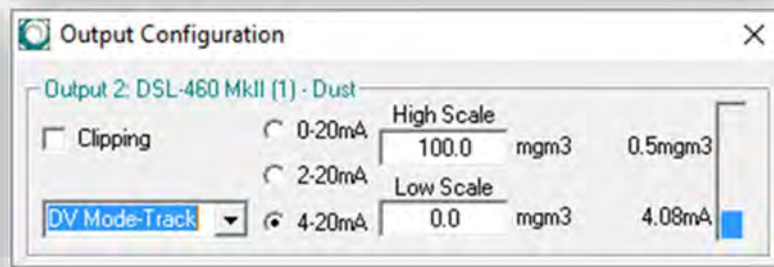
Disabled will deactivate the output (i.e., set it to 0.00mA).

Head-Reading allows the DSCU analogue settings to be manually configured to any measured reading from any connected instrument. This is the most flexible option and does not need the analogues in the instrument to have been configured first.

The second drop down menu allows the selection of one of the connected instruments (not available if 'Disabled' has been selected). There can be up to eight (8) separate instruments. In this example there are only two DSL 460 MkII [1] and DSL-460 MkII [2].

The third drop down menu allows you to select which measurement reading is used for the analogue channel. The options available depends on what has been selected in the first drop down menu. If 'Copy Other' has been selected the list will be limited to only the measurements configured for the analogues in the instrument. If 'Head-Reading' has been selected the list will include all available measured readings from the selected instrument.

If 'Head-Reading' has been selected then it is necessary to set the range for the analogue output (for 'Copy Other' this information is just copied from the instrument settings). To do this click on the '+' symbol that appears at the right-hand side, this opens a new popup window with the following settings.



Clipping

If this box is CHECKED the analogue output will not respond to readings below the value set as "Low Scale", but will send an analogue output equivalent to the lowest current for the scale selected.

If the box is UNCHECKED then the analogue output will track the measured signal even if it is below the value set as "Low Scale", but only as low as 0mA.

If the instrument reading already has negative clipping applied to it (see the Instrument operators manual) then since the analogue output tracks this reading it will also be clipped to zero.

Data Valid (DV) Mode

This drop-down menu allows you to select what happens to the analogue output in the event of the data being invalid. Both the actual and calculated analogue outputs are acted on. The available options are: -

DV Mode-Track – In the event of data invalid, the analogue output will continue to track the reading. This is the default setting.

DV Mode-Low – In the event of data invalid, the analogue output will be forced to the low scale value.

The analogue output will continue to be held at this value for the duration of the data invalid period.

When the data valid status is returned, the output will return to tracking the current measurement reading.

DV Mode-High – In the event of data invalid, the analogue output will be forced to the high scale value.

The analogue output will continue to be held at this value for the duration of the data invalid period.

When the data valid status is returned, the output will return to tracking the current measurement reading.

DV Mode-Hold – In the event of data invalid, the analogue output will be held at the last known output current.

The analogue output will continue to be held at this value for the duration of the data invalid period.

When the data valid status is returned, the output will return to tracking the current measurement reading.

0-20mA / 2-20mA / 4-20mA

The current loop analogue outputs used on the DSCU can be configured to operate across any one of three scales: 0-20mA, 2-20mA or 4-20mA. Select the required scale using these radio buttons.

High Scale and Low Scale

These two parameters define the upper and lower scaling points of the 0/2/4-20mA output.

Set the low scale value to the reading at which the analogue output should generate 0mA, 2mA or 4mA (depending on selected scale point).

Set the high scale value to the reading at which the analogue output should generate 20mA.

Note: If you change the measurement units, (on the Settings tab), you will need to change the low scale and high scale settings here, as there is no automatic adjustment of scaling values when you change between units.

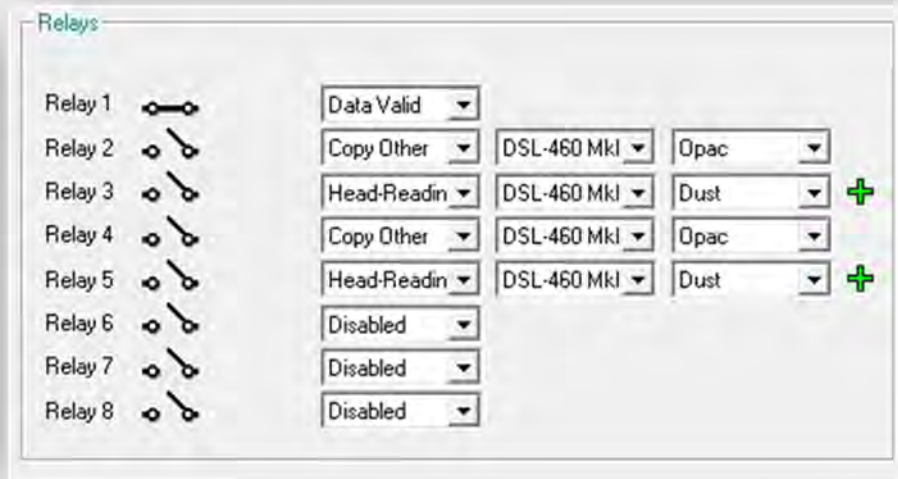
Output

The output bar is a calculated indication of the expected current output in mA, taking into account the low scale and high scale settings and also the 0/2/4mA scale selection.

The value shown here is a calculated value only. It is not an electronically measured value from the actual analogue output circuit. There is no direct connection between the calculated value shown and the actual output. However, unless there is a fault with the instrument, the correlation between the calculated value and the electrical output should be very good.

Relays

The DSCU relays can be configured to operate on the status or value of any readings from any instrument that is connected to it. The following example illustrates the different configuration available.



The first part displays the current status of the relay. This graphically shows whether the relay is open or closed and a bell is displayed if the relay is in alarm status.

The first drop down menu allows the selection of 'Data Valid', 'Copy Other', 'Disabled', 'Head-Reading' or 'Head-Fault/Warning'.

NOTE: Any Relay not being used should be set to 'Disabled'.

Data Valid is the default setting for relay 1. Any fault that appears on the DSCU will trigger this relay. This includes DSCU faults and faults from any connected instrument.

Copy Other allows the DSCU relay settings to just copy any relay settings that have already been configured in a connected instrument.

Disabled will deactivate the relay.

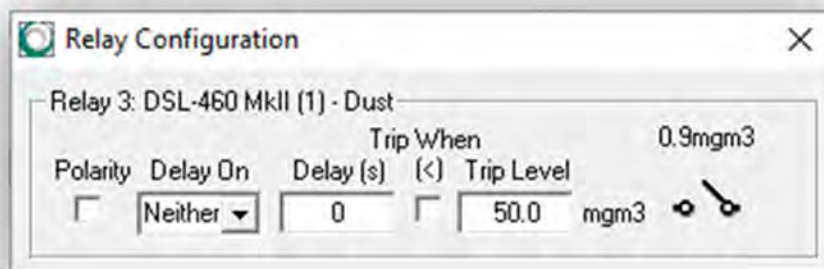
Head-Reading allows the DSCU relay to be manually configured to trigger on any measured reading from any connected instrument. This does not require the relays to have been configured in the instrument.

Head-Fault/Warning allows the DSCU relay to be manually configured to trigger on any fault or warning from any connected instrument. This does not require the relays to have been configured in the instrument.

The second drop down menu allows the selection of one of the connected instruments (not available if 'Disabled' has been selected). There can be up to eight (8) separate instruments. In this example there are only two DSL 460 MkII [1] and DSL-460 MkII [2].

The third drop down menu allows you to select which measurement reading, fault or warning is used for the relay. The options available depends on what has been selected in the first drop down menu. If 'Copy Other' has been selected the list will be limited to the faults and reading alarm levels that have been configured for the relays in the instrument. If 'Head-Reading' has been selected the list will include all available measured readings from the selected instrument. If 'Head-Fault/Warning' has been selected the list will include all of the faults and warnings available from the selected instrument.

If 'Head-Reading' or 'Head-Fault/Warning' has been selected then it is necessary to configure the relay (for 'Copy Other' this information is just copied from the instrument settings). To do this click on the '+' symbol that appears at the right hand side, this opens a new popup window with the following analogue settings. Note for 'Head-Fault/Warning' option not all of these following features are available.



Relay Polarity checkbox

When UNCHECKED, and acting as fault or warning, the relay will be energised when the data is valid, and the relay status indicator will show a closed contact in the data

valid condition. The relay will de-energise when the data becomes invalid (i.e., a fault or warning is present). This condition represents the recommended failsafe operation.

When UNCHECKED, and acting as a threshold relay, the relay will be de-energised in a below threshold condition and the relay status indicator will show an open contact. In this polarity the relay will be energised when the reading exceeds the threshold.

With the checkbox CHECKED, the associated relay polarity will be reversed.

The operation of the relays can be delayed using the Delay setting.

Delay On

The activation of the relay can be delayed before triggering. The following options are available: -

Neither – The triggering of the alarm condition is not delayed at the start, or at the end, of the alarm condition.

Clear – The triggering of the alarm condition is not delayed at the start of the alarm condition, but is delayed at the end i.e., when the alarm condition is clear.

Active (default) – The triggering of the alarm condition is delayed by the time set in the 'Delay' box. At the end of the alarm condition, the relay returns immediately to its previous state.

Both – The triggering of the alarm condition is delayed at the start and end of the alarm condition.

Delay

This value is the continuous length of time (in seconds) that the relay is delayed by. The exact function is determined by the 'Delay On' condition.

Delaying the activation of the relay can be used to prevent borderline level changes from “dithering” the relay state. Only genuine, sustained readings in excess of the threshold will actually trigger the alarm.

Trip When (<)

This is only available for a level alarm. Ticking the ‘trip when less than’ checkbox will cause the alarm condition to be activated when the measurement reading is less than the ‘Trip Level’.

For example, this could be used to activate an alarm when the temperature drops below a certain value.

Trip Level

This is only available for a level alarm. This parameter defines the measurement reading at which the alarm level relay will trip.

The Diagnostics Tab

The Diagnostics tab has a number of information only boxes and some features for backup and restore. See the following screen shot.



DSCU Information

The first three boxes display the DSCU firmware version, compile time and PCB serial number.

Backup to PC

Clicking this button will cause the current DSCU settings to be saved in a backup text file in the location: 'C:\Dynoptic Systems\DSCU Utility\BACKUPS' on the PC. The filename of the backup file will incorporate the date, time and electronic serial number of the DSCU at time of backup.

It is highly recommended that you use this button to take a backup of your DSCU settings after successfully completing your installation and set-up. The backup file can be used to restore the DTCU settings, should the live settings ever be lost or become corrupted.

NOTE: *A backup file should be downloaded and archived immediately after the successful set up and commissioning of this device, or whenever the settings and/or parameters are permanently altered.*

Restore from PC

Clicking this button will allow you to choose a backup file from which to restore your DSCU settings.

NOTE: *Restoring from a backup file will permanently overwrite ALL settings and parameters in the instrument.*

NOTE: *A reserve backup file should be downloaded and archived immediately prior to restoring a previous backup file.*

Restore Factory Defaults

Clicking this button will restore all settings and parameters to default values as determined at our factory.

NOTE: *Restoring factory defaults will permanently overwrite ALL settings and parameters in the instrument.*

NOTE: *A reserve backup file should be downloaded and archived immediately prior to restoring factory defaults.*

Log Reading

This area allows the user to log readings in real time onto the PC running the Utility and, if required, display the readings on a graph

Log Reading – Enabled

When this box is CHECKED the measured values (defined by the 'LogMe.txt' file) will be logged in a CSV file. The files are stored in a folder in called \DSCU Utility\READING LOGS, the location of this folder on the PC is given on the Diagnostics tab. The user must be logged onto the PC as an administrator for the logs to be stored and the logs will be created at intervals defined by the interval setting.

The default value logged in the CSV file is the reading measured by analogue channel 1. Contact Dynoptic Systems to log other parameters.

Log Reading – Append to File

This box is CHECKED by default. When checked the log file created will be a single file using the electronic serial number of the DSCU as a reference in the filename and each new log is appended within that file. A new file is created each day with the date forming part of the filename

When the 'Append to File' box is UN-CHECKED, the logging will generate a new file at each logging interval, using the time, date and electronic serial number of the DSCU as the file name.

The log file(s) can be opened in a text editor or spreadsheet and can be used for data logging, trend graphing or for diagnostics.

Log Reading – Log Q and A

The logging function uses a "question and answer" system to retrieve data for the log. In this system the DSCU Utility Software sends a question (an ASCII text string) to the DSCU, which responds with an answer (usually a number).

When the 'Log Reading' checkbox is un-checked, the log file (in whichever format) only includes the answer, but when this box is checked the log file includes both the question and answer.

By default, the question that is asked relates to the measurement reading of the first analogue output of the DSCU only. However, with advice from Dynoptic Systems, advanced users may modify the question so that the log file records additional or alternative data, for example other instrument readings, relay thresholds or analogue output scaling values.

In these circumstances it is beneficial to log both the question and the answer so that the log file is more easily interpreted.

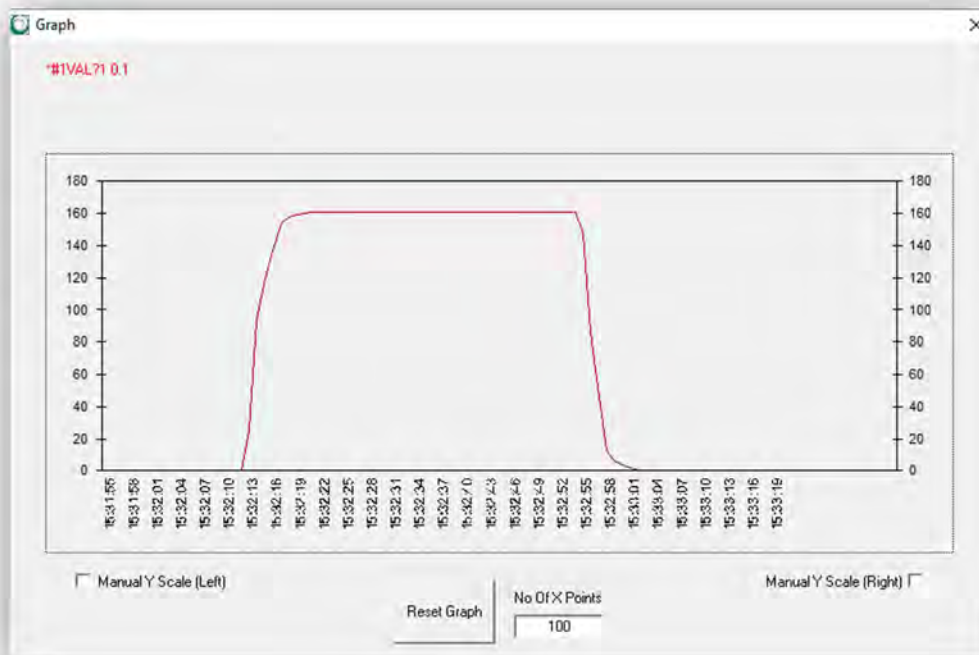
The file that is edited to modify the question is 'DSCU Utility LogMe.txt', Left clicking on the filename opens the location folder and allows users to browse the PC to select a different 'LogMe' file. Right clicking on the filename opens the current file in a text editor. Please seek advice from Dynoptic Systems before making any changes to this file.

Log Reading - Interval

The time interval between the creation / update of log files is determined here.

Graph Button

As well as a log file being created, the data can be viewed using the graphing feature. While logging click on the 'Graph Icon' a new graph window will appear (see screenshot below).

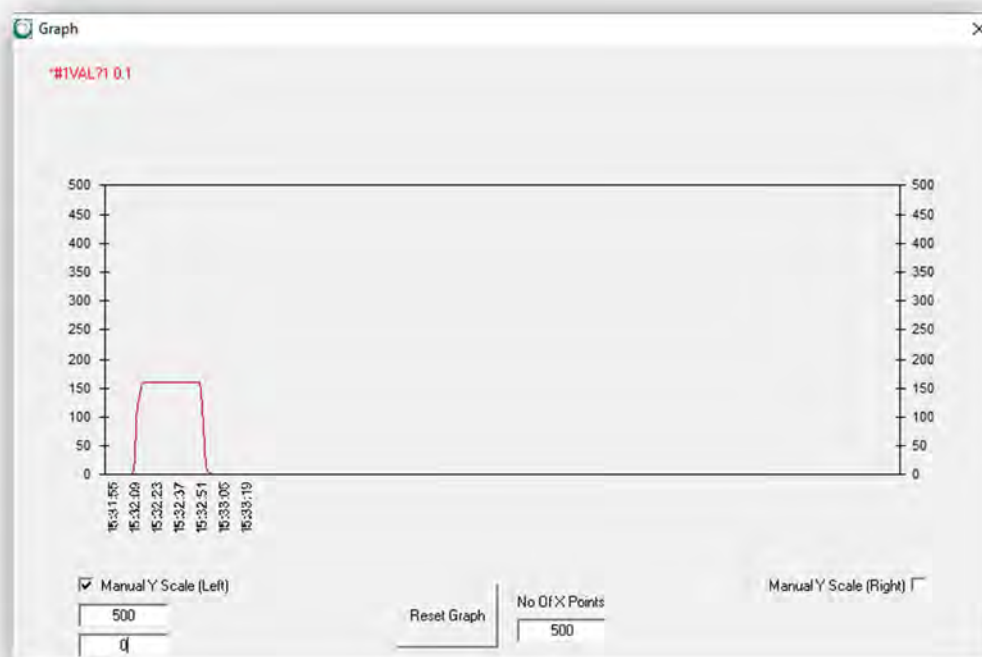


The default file, 'DSCU Utility LogMe.txt', only logs the reading on Channel 1 (*#1VAL?1) and this appears at the top in red '*#1VAL?1'. If the 'DSCU Utility LogMe.txt' file has been changed to log several parameters only the first four lines of results are displayed on the graph.

There are two 'Y' axis, and by clicking on the coloured text at the top '*#1VAL?1', the axis that this data is plotted against can be changed; Left = Left Y-Axis, Centre = Not displayed and Right = Right Y-Axis.

The Y-Axis scales are automatic but can be manually changed by checking the 'Manual Y Scale' tick box and then entering the required minimum and maximum values, as illustrated below.

The number of X coordinates can be changed (between 100 and 3600) by entering the value in the 'No Of X Points' box and pressing enter, as illustrated below.

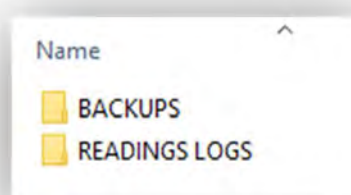


The 'Reset Graph' button clears the currently displayed data.

To exit the graphing feature, close the window using the cross at the top right-hand corner of the window to return to the main Portal.

Folder Location

This box shows the location on the PC of the folder that contains all of the data stored by the Utility. Left clicking on this filename opens the folder to show two subfolders.



BACKUPS: Is where the DSCU backups are stored (see 'Backup to PC')

READINGS LOGS: Is where the reading logs are stored (see 'Log Reading')

Direct Comms

Selecting the “Enable” box opens up an input box, as shown below, which provides direct communication access to the DSCU and through the DSCU to any connected instrument. This can be used to read and edit additional system parameters. This feature should only be used with help from Dynoptic Systems.



Faults & Warnings

Fault and Warning messages can appear either on the Virtual OI tab of the DSCU Utility Software, or in the DSCU display.

Where messages appear in the Utility Software most will remain permanently displayed on the Virtual OI tab until the fault or warning has been cleared and the operator has clicked on the text to acknowledge the fault or warning. This means that even if a fault or warning occurs only temporarily, and has been resolved before the operator views the Utility Software, the message will remain visible until the operator has clicked to acknowledge it. Fault and warning message text can only be cleared if the cause of the fault or warning has been resolved.

Where messages appear on the DSCU display they will flash intermittently over the reading display. This means that the operator can still navigate and use the DSCU, but will be constantly reminded of the fault by the flashing message. Flashing messages will automatically disappear when the fault or warning has been resolved.

Fault messages indicate there is a fault with the instrument and that the readings are not valid. Warning messages indicate that there may be a problem, and the warning should be investigated.

The following section describes the DSCU fault and warning messages, outlines potential causes and highlights actions that can be taken to try and resolve the issue. If the actions described do not solve the problem, then further advice should be sought from your Dynoptic Systems representative. For information on instrument faults and warnings please refer to the instrument operator's manual.

DSCU Fault Messages

SERVICE FAULT

If the DSCU or any instrument connected to the DSCU detects a fault, the Service Fault will be shown.

A fault can be triggered by missing head(s) or a fault with any unit connected.

Check which unit is showing a fault to find out the specific fault message.

MEMORY FAULT

The DSCU employs CRC checking on the EEPROM memory. Each time the instrument is powered up the CRC checking compares the state of EEPROM that it last knew, against the state of the EEPROM memory on power up.

If the DSCU finds a discrepancy between the two EEPROM memory states, a MEM fault is registered.

On registering the MEM fault the instrument performs a self-checking routine which analyses each byte of memory and ensures that the values stored there are within an acceptable range. Values outside the acceptable range are considered to be corrupted values and the memory byte is replaced with a default value from FLASH memory. Essentially the instrument heals itself.

Memory corruption is rare, but can occur in certain circumstances, for example if the instrument suffers a power loss whilst writing to its EEPROM memory.

Discrepancies can also be caused by a firmware upgrade, so they can be an expected result of such actions.

On encountering a MEM fault operators should always power cycle the instrument, i.e., switch it off, then on again. This action should clear the fault because self-healing process will have installed a new value in the corrupted memory byte, and the compared states of the power off memory and the power on memory should now match.

If a power cycle does not clear the fault, it is likely that there is a more serious problem and you should contact Dynoptic Systems or your distributor for further assistance.

Please note that when the instrument self-heals by replacing corrupted data with default data from FLASH memory, operational settings and values may have been overwritten. It is therefore essential that all parameters, settings, and values are checked for validity by the operator immediately after resolving a MEM fault.

ALL HEADS MISSING

If the DSCU cannot detect any instruments connected to it, the All Heads Missing fault will be shown.

Check the wiring between the DSCU and all connected instruments.

Check that each instrument has been set to a unique id, otherwise conflicts will occur.

SOME HEADS MISSING

If the DSCU cannot detect some instruments connected to it, the Some Heads Missing fault will be shown.

Check the wiring between the DSCU and all connected instruments.

Check that each instrument has been set to a unique id, otherwise conflicts will occur.

DSCU Warning Messages

SYSTEM ALARM

If any of the instruments connected to the DSCU flag a threshold alarm, the System Alarm warning will be triggered.

Check the instrument to see the specific threshold alarm that has been flagged.

SERVICE OVERRIDE

If an instrument needs to be removed from the system, the service fault checkbox can be ticked to disable the Service Fault from triggering. If the checkbox is enabled the Service Override warning will be displayed on the DSCU & Utility.

COMMS CRC FAULT

The DSCU employs real time CRC checking on the communications between the DSCU and the connected instruments to maintain data integrity. This checking can determine if any packets of information sent between the DSCU and the connected instruments, or vice versa, have been corrupted.

If the instrument finds an error in a data packet, it will reject that data and request that the data be resent. If the CRC checking identifies a repeatedly high proportion of CRC failures the instrument will flag a CRC FAULT.

CRC failures are caused by corruption of the data passed along the interconnecting cable between the DSCU and the connected instruments. Corruption is rare and is usually caused by RF interference, electromagnetic radiation, or a poor connection in the wiring.

Check that the interconnecting cable is screened and that the screen is connected to Earth as described in the installation manual. Also check all wiring connections for loose wires, poor connections, or loose terminal screws.

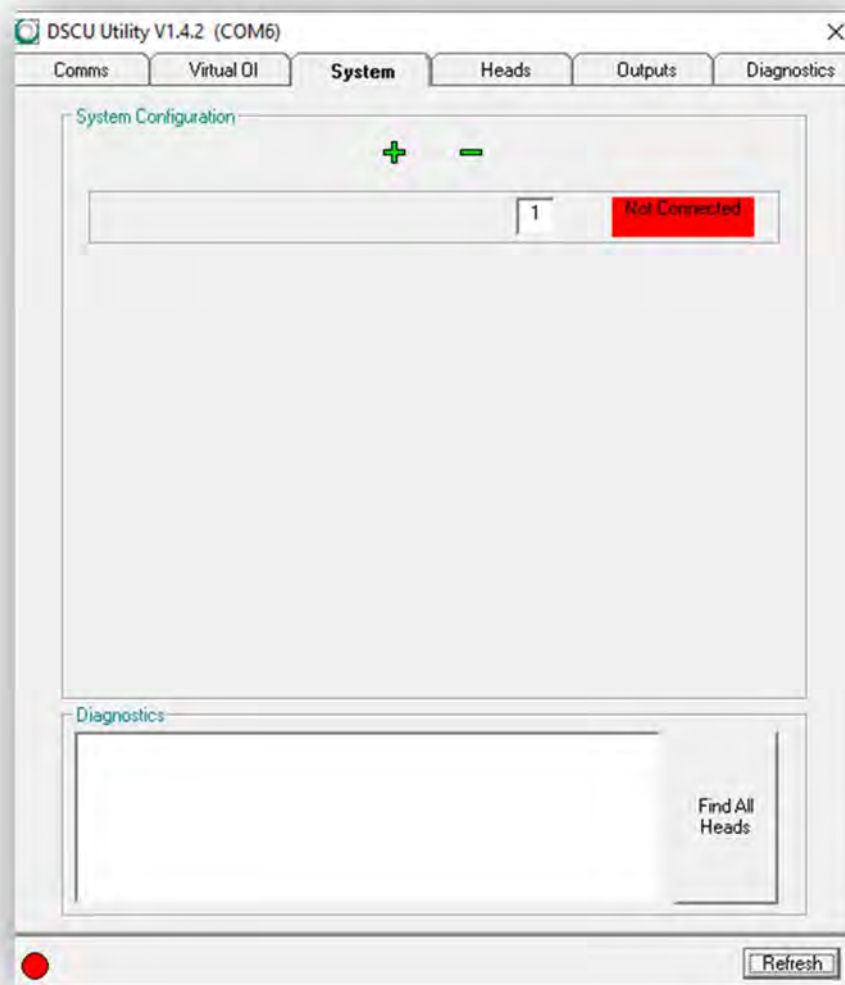
This problem can also be caused by having an incorrect combination of negative and positive bias applied to the RS485 signals in the interconnecting cable, or by missing termination links. Also check that the termination link (LK5) is fitted on the circuit board in the DSCU.

Appendix A – DSCU Initial Set-Up Procedure

The default setting for the DSCU is to have only one head attached and for its ID address to be '1'. When the DSCU is first turned on it will search for one instrument with this ID on the network. If one is found then the reading will be displayed (if no faults are returned). For single instrument use this is generally all that is required.

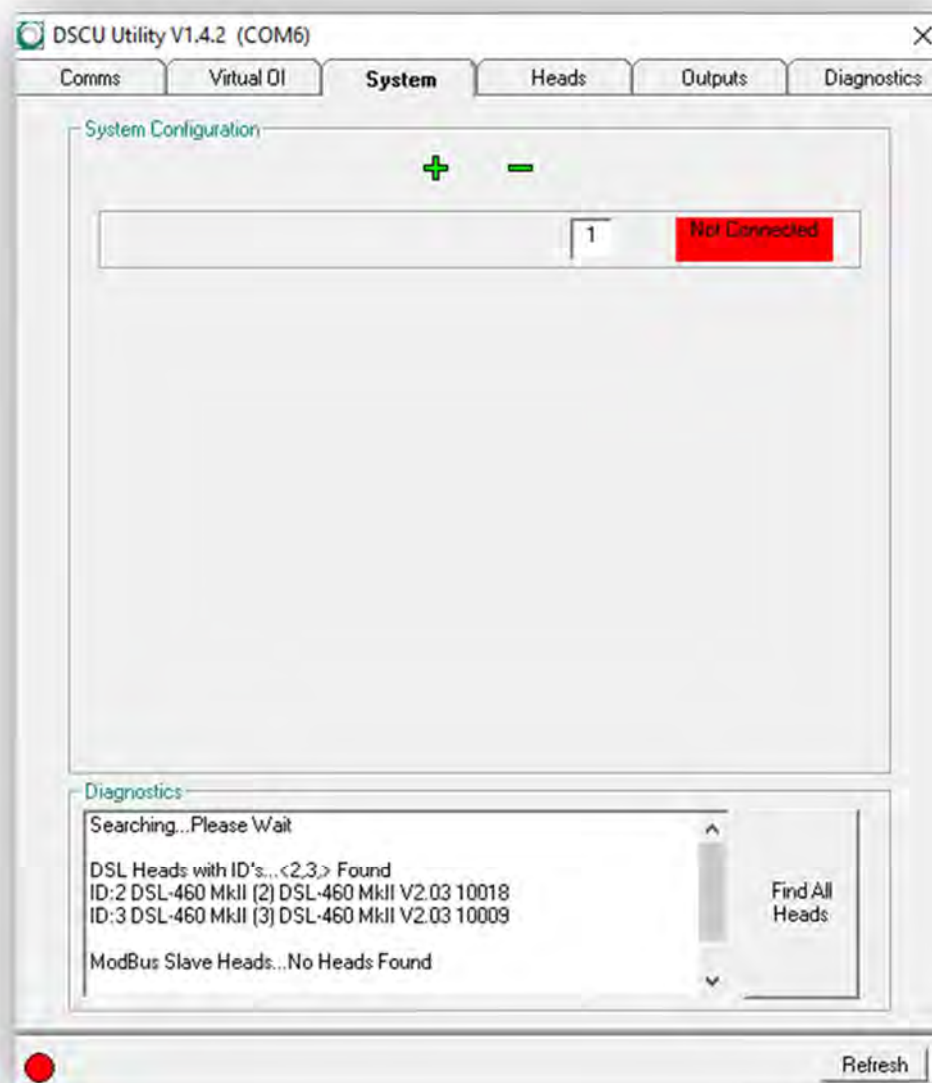
However, if the ID of the attached instrument is not '1' or there are several instruments connected (multi-heading) then the System tab of the DSCU Utility will be required to correctly setup the DSCU.

In the example shown below the DSCU has not found an instrument with ID=1 and so the status is shown in red (Not Connected).

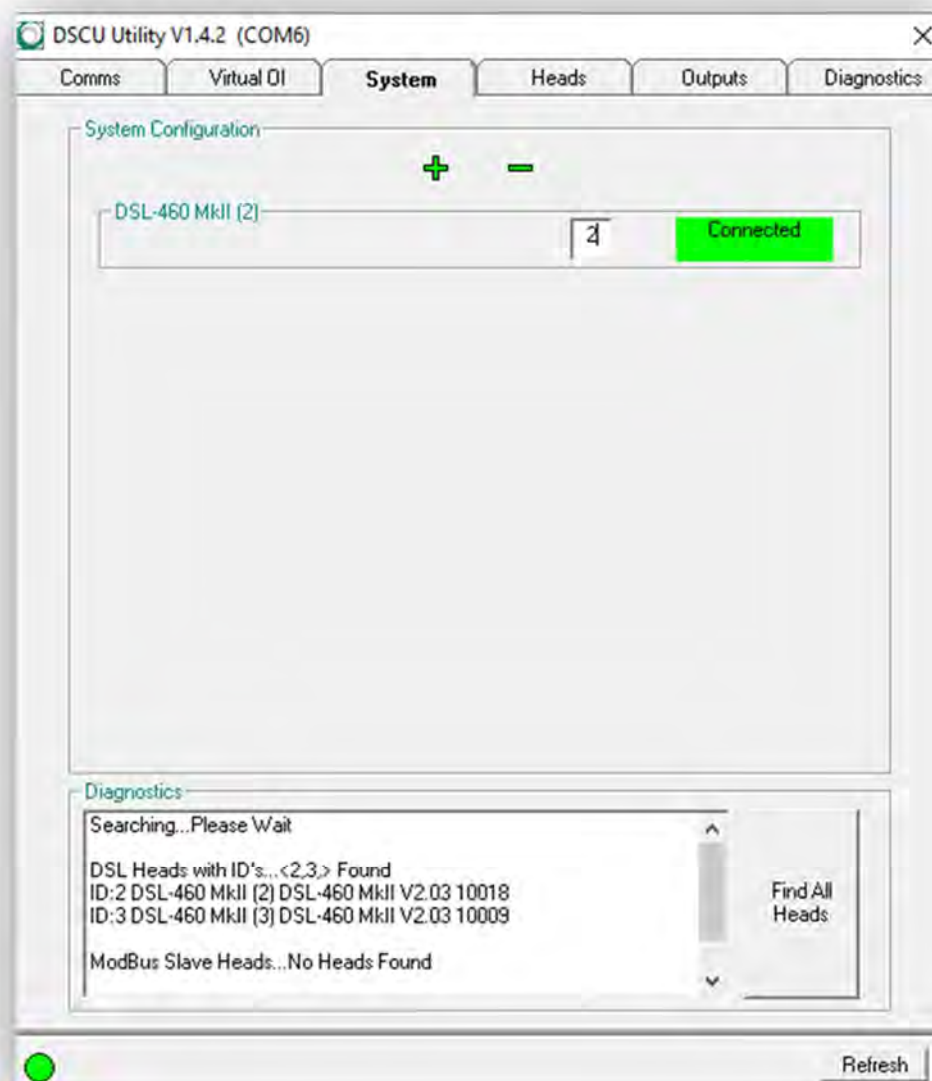


To identify how many heads are connected to the DSCU and what their IDs are, click on the 'Find All Heads' button. In the example below two DSL-460 MkIIs have been found with ID's '2' and '3'.

If the number of heads identified does not equal the actual number physically connected to the RS485 bus it will be necessary to check all of the RS485 connections and also to confirm that all of the instruments have unique ID numbers.

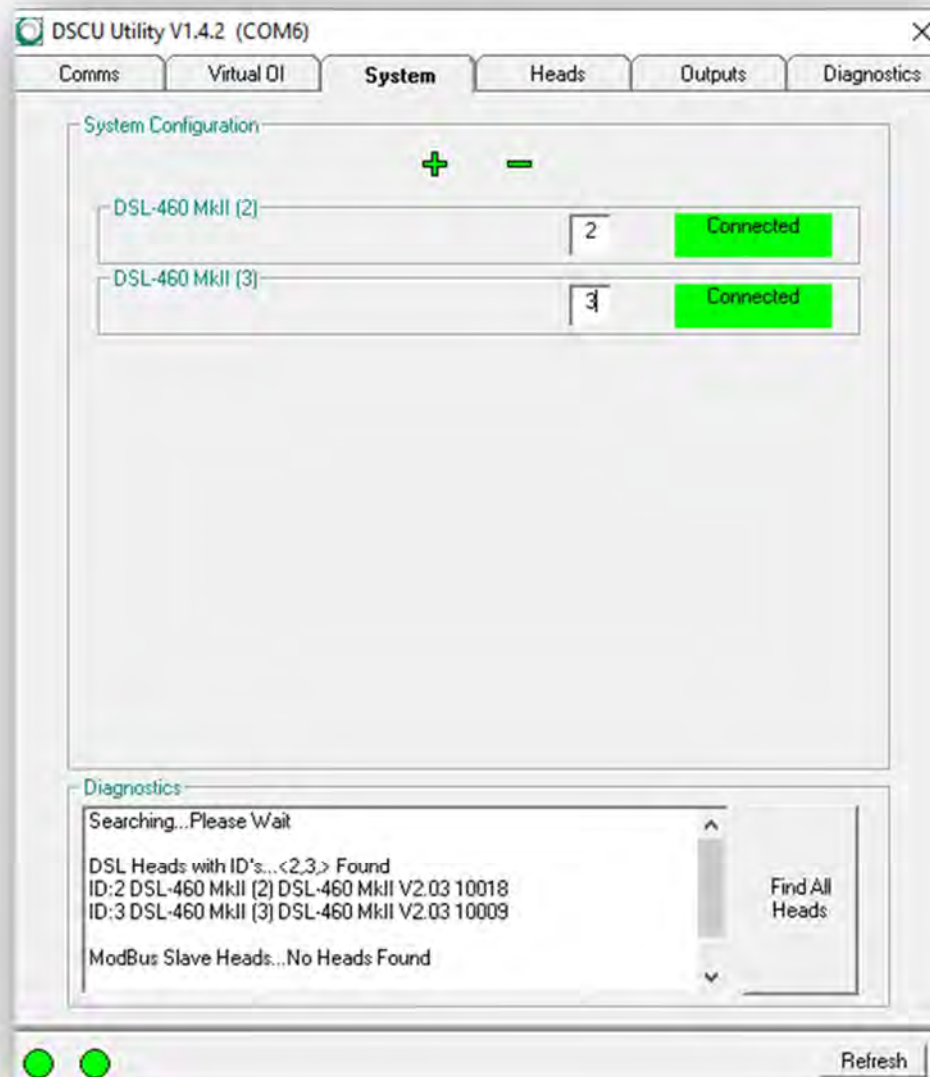


The system configuration can now be edited to match the known number of instruments and IDs. In the example shown below the number in the ID box of the single instrument was changed from '1' to '2'. On pressing enter the DSCU searched and connected to the instrument with ID=2.



The '+' symbol is now used to insert a second connection and '3' was typed into the ID box. On pressing enter the DSCU searched and connected to the DSL-460 MkIII with ID=3.

The DSCU is now correctly set-up to communicate with the two attached instruments.

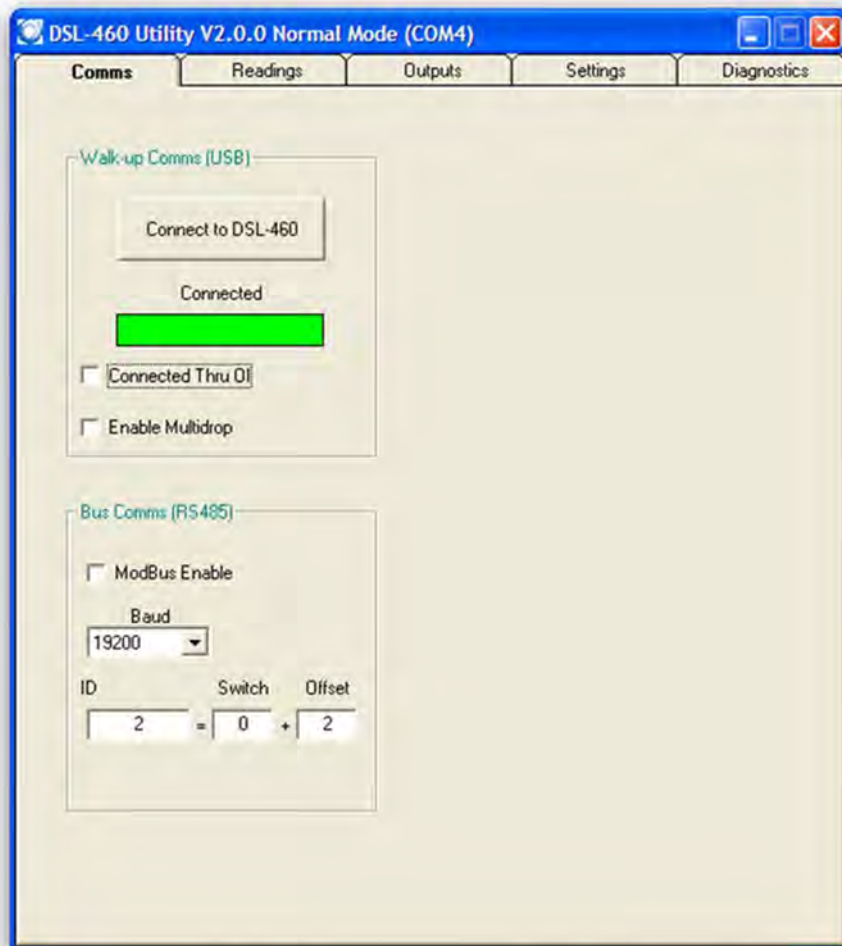


Appendix B – Setting the Instrument Head Address

When setting up a multi-headed system each instrument connected to the DSCU must be assigned a unique head address. This should be carried out when the heads are installed. If the DSCU tries to communicate with two heads that have the same ID the communication will be corrupted and several fault messages will be displayed.

The instrument ID must be set using the instrument Utility connected directly to the walk-up comms of the instrument (i.e., not via the DSCU).

As an example, the screenshot below shows the Utility Comms tab for a DSL-460. At the bottom of the Comms tab is the RS485 ID section.



To set the ID type the required address into the 'Offset' box and press enter.

The instrument ID is displayed in the box under 'ID', as the sum of the Switch and Offset values Make a note of the ID as each instrument must have a unique number.

NOTE: *If one or more instruments are set to the same address conflicts will occur and the system will not work.*

Appendix C: Modbus Configuration

Protocol Version:	Modbus RTU Over RS485
Baud Rate:	9600, 19200 , 28800, 38400, 57600
Data-Parity-Stop Bits:	8-None-1, 8-None-2, 8-Even-1 , 8-Odd-1:
ID:	1 -247 (Via Programmable ID) (Bold Indicates Default Values)

Modbus Information

Unit Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Unit Version	04	10000	String	13	Read Only
Serial No.	04	10013	String	6	Read Only
Code Version	04	10019	String	13	Read Only

Dynoptic Systems Control Unit – DSCU – Installation & Operators Manual

DSCU Output/Alarm Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Output 1	04	10520	Float	2	Read Only Depending on DSCU Variant
Output 2	04	10522	Float	2	Read Only Depending on DSCU Variant
Output 3	04	10524	Float	2	Read Only Depending on DSCU Variant
Output 4	04	10526	Float	2	Read Only Depending on DSCU Variant
Output 5	04	10528	Float	2	Read Only Depending on DSCU Variant
Output 6	04	10530	Float	2	Read Only Depending on DSCU Variant
Output 7	04	10532	Float	2	Read Only Depending on DSCU Variant
Output 8	04	10534	Float	2	Read Only Depending on DSCU Variant
Relay / Alarm Status	04	10580	Word	1	Read Only Bitwise Flags (Alarm: 0=No Alarm, 1= Alarm) (Data Valid: 0=Not Valid, 1=Valid) Bit00–RLY1–ALARM/DV STATUS Bit01–RLY2–ALARM/DV STATUS Bit02–RLY3–ALARM/DV STATUS Bit03–RLY4–ALARM/DV STATUS Bit04–RLY5–ALARM/DV STATUS Bit05–RLY6–ALARM/DV STATUS Bit06–RLY7–ALARM/DV STATUS Bit07–RLY8–ALARM/DV STATUS (0=Relay Open, 1=Relay Closed) Bit08–RELAY1–RELAY STATUS Bit09–RELAY2–RELAY STATUS Bit10–RELAY3–RELAY STATUS Bit11–RELAY4–RELAY STATUS Bit12–RELAY5–RELAY STATUS Bit13–RELAY6–RELAY STATUS Bit14–RELAY7–RELAY STATUS Bit15–RELAY8–RELAY STATUS

Dynoptic Systems Control Unit – DSCU – Installation & Operators Manual

DSCU Fault/Warning Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Unit Status	04	10590	Word	1	Read Only Bitwise Flags Bit00–OVERRIDE ACTIVE Bit01–MEMORY FAULT Bit02–SYSTEM ALARM Bit03–SYSTEM DATA INVALID Bit04–ALL HEADS MISSING Bit05–SOME HEADS MISSING Bit06–COMMS CRC FAULT Bit07–CLOCK FAULT Bit08–SERVICE OVERRIDE ACTIVE Bits09–15–SPARE
Unit Status	04	10591	Word	1	Read Only Bitwise Flags Bits00–15–SPARE

Connected Instrument Information (Information available depends upon connected Instrument)					
Connected Instrument Unit Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Instrument Identifier	04	1x000	String	13	Read Only
Connected Instrument Reading Information					
Reading 1	04	1x500	Float	2	Read Only
Reading 2	04	1x502	Float	2	Read Only
Reading 3	04	1x504	Float	2	Read Only
Reading 4	04	1x506	Float	2	Read Only
Reading 5	04	1x508	Float	2	Read Only
Reading 6	04	1x510	Float	2	Read Only
Reading 7	04	1x512	Float	2	Read Only
Reading 8	04	1x514	Float	2	Read Only

Dynoptic Systems Control Unit – DSCU – Installation & Operators Manual

Connected Instrument Output/Alarm Information					
Output 1	04	1x520	Float	2	Read Only
Output 2	04	1x522	Float	2	Read Only
Output 3	04	1x524	Float	2	Read Only
Output 4	04	1x526	Float	2	Read Only
Output 5	04	1x528	Float	2	Read Only
Output 6	04	1x530	Float	2	Read Only
Output 7	04	1x532	Float	2	Read Only
Output 8	04	1x534	Float	2	Read Only
Relay/Alarm Status	04	1x580	Word	1	Read Only (Alarm: 0=No Alarm, 1= Alarm) (Data Valid: 0=Not Valid, 1=Valid) Bit0-RLY1-ALARM/DV STATUS Bit1-RLY2-ALARM/DV STATUS Bit2-RLY3-ALARM/DV STATUS Bit3-RLY4-ALARM/DV STATUS Bit4-RLY5-ALARM/DV STATUS Bit5-RLY6-ALARM/DV STATUS Bit6-RLY7-ALARM/DV STATUS Bit7-RLY8-ALARM/DV STATUS (0=Relay Open, 1=Relay Closed) Bit8-RLY1-RELAY STATUS Bit9-RLY2-RELAY STATUS Bit10-RLY3-RELAY STATUS Bit11-RLY4-RELAY STATUS Bit12-RLY5-RELAY STATUS Bit13-RLY6-RELAY STATUS Bit14-RLY7-RELAY STATUS Bit15-RLY8-RELAY STATUS
Connected Instrument Fault / Warning Information					
Instrument Status	04	1x590	Word	1	Read Only Bits0-15 – See Instruments Modbus Configuration
Instrument Status	04	1x591	Word	1	Read Only Bits0-15 – See Instruments Modbus Configuration

- The Control Units' data is in the range 10000-10999 (not all addresses are valid please see the map above).
- The Control Unit can be configured to communicate with up to 8 instruments.
- The address structure for each instrument is the same, however the hardware available for each instrument may differ (e.g. Instruments can have different numbers of readings, outputs and relays).
- The Status information is dependent on the type of instrument that is connected. Please see the relevant instrument's manual for more detailed explanation of how the Status information is structured.
- To access the data for each instrument position, use the address ranges listed below

Instrument Position Address Range	
Instrument Position	Address Range
DSCU	10000-10999
1	11000-11999
2	12000-12999
3	13000-13999
4	14000-14999
5	15000-15999
6	16000-16999
7	17000-17999
8	18000-19999

Example:

To read Instrument 4's Reading #3 use address 14504.

To read Instrument 8's Relay/Alarm information use address 18580.

Example

The following examples are for illustrative purposes and the Data received could be different dependent on the type of instrument.

String Read	
Unit Version Read	ModBus Addr:1, Address:10000, Length:13
Master Message (Command)	01 04 27 10 00 0D 3A BE
Instrument Message (Reply)	01 04 1A 56 31 2E 31 37 00 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 00 2A 82 FB
Received String	V1.17*****
Word Read	
Unit Status Read	ModBus Addr:1, Address:10590, Length:1
Master Message (Command)	01 04 29 5E 00 01 58 44
Instrument Message (Reply)	01 04 02 C4 80 EA 50
Received Word	50304 (C480 Hexadecimal)
Float Read	
Reading 1 Read	ModBus Addr:1, Address:10500, Length:2
Master Message (Command)	01 04 29 04 00 02 38 56
Instrument Message (Reply)	01 04 04 41 58 C8 CE B8 3F
Received Word	13.55

Notes

All technical details and specifications are subject to change without notice.

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CUSTOMER:

Ingredion (Thailand) Co., Ltd.

LOCATION:

Kalasin

PROJECT TITLE:

Project SPRINT 18K TPA Spray Dry

CUSTOMER P.O. NO.:

7001730092

PACKAGE:

Dust Analyzer

EPM PROJECT NO:

6626013

DOCUMENT TITLE:

Installation, Operation and Maintenance Manual

2.DSL-800 Installation Manual

Installation Manual V1.5

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Precautionary Statements

Throughout this manual, precautionary statements are included to draw attention to potential hazards.

These statements are always boxed and include the signal words DANGER, WARNING, CAUTION, or NOTE, to indicate the degree of hazard that may be encountered. These words are defined below:

DANGER: *Imminent hazard and failure to avoid hazard will result in death.*

WARNING: *Hazard that may result in injury or death.*

CAUTION: *Hazard that may result in minor or moderate injury.*

NOTE: *Hazard that may result in either, damage to equipment that will require repair or replacement; or a reduction in performance that renders the equipment unable to meet its specification.*

General Precautions

WARNING: *When powered this instrument emits a visible laser beam (Class 3R). Avoid direct eye exposure. Only connect power to the instrument after it has been correctly installed. Always disconnect the power before removing the instrument for repair or maintenance.*

CAUTION: *All electrical work must be carried out with the power to the instrument OFF i.e., it must be isolated from any source of voltage. Failure to do so could result in equipment failure or injury.*

CAUTION: *All installations and connections must be made by suitably qualified and experienced engineers with proper regard to good practice and compliance with applicable legislation.*

CAUTION: *Appropriate Protective Personal Equipment (PPE) must be worn when installing, commissioning or otherwise working on this device.*

CAUTION: *Appropriate access equipment, operated by qualified personnel, must always be used when working at height.*

CAUTION: *Care must be taken when working on a stack. In particular hot surfaces and hot stack gases. Take all necessary precautions.*

NOTE: *Prior to use, this equipment must be stored in its original packaging in a dry and sheltered area that is not exposed to extremes of temperature or humidity.*

NOTE: *This device is particularly vulnerable during transportation and manual handling. Always transport the instrument in its original packaging and handle with utmost care.*

NOTE: *This device contains sensitive optics and electronics and must be protected from physical shock, excessive vibration, and ingress, at all times. Always handle with care and protect from ingress.*

NOTE: *Do not open the instrument or attempt to remove the back panel. All electrical connections to the instrument are made using external connectors.*

NOTE: *Cables connected to this device may carry sensitive electronic signals which can be affected by RFI and EMI. Care must be taken to run cables away from high voltage power cables, high frequency signal cables, or switch gear cabling. Ideally any cables connecting to the device will follow an independent route and will not be installed in bundles or cable ducts with other cabling.*

NOTE: *Protective earth connections are not only essential for safety, but also to maintain the instrument's performance and compliance with European safety directives and CE marking.*

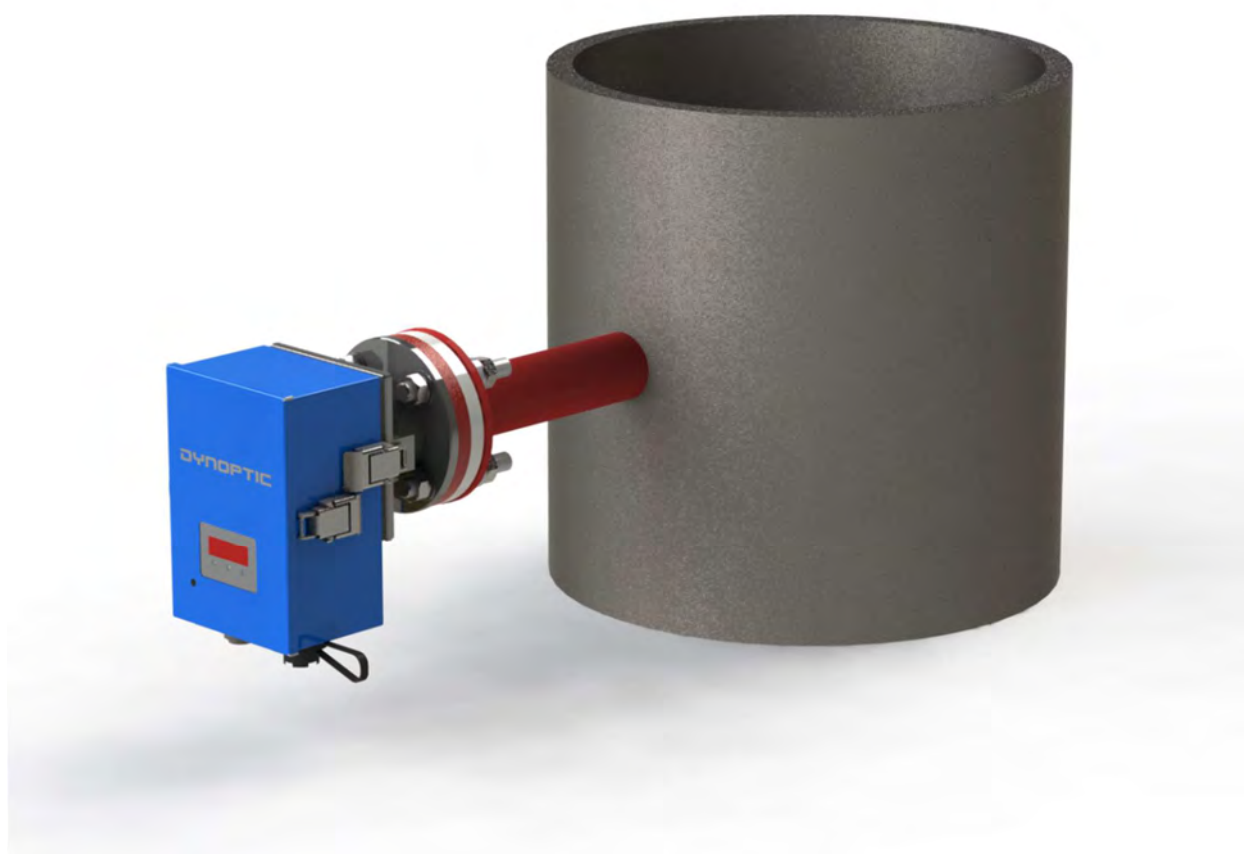
NOTE: *Do not make or break electrical connections whilst power is connected to this device, or any device connected to it. "Hot plugging" as this is known may damage the electronics. Such damage is NOT covered under warranty.*

NOTE: *Applications vary and all installations will involve a degree of application engineering which is the responsibility of the customer, not Dynoptic.*

NOTE: *Read this manual in its entirety before installing, commissioning, or otherwise using this device. If in doubt, contact Dynoptic directly for advice.*

Product Overview

The DSL-800 Particulate Monitor uses laser backscatter to measure the level of suspended particles (e.g., dust and smoke) in a duct, stack, chimney or flue.



The DSL-800 consists of a single TRX head mounted on one side of the stack. The TRX head emits a beam of laser light into the stack and any light backscattered from suspended particles is detected by the TRX. Increased particulate or smoke density in the stack gas produces an increase in the level of the scattered light. When calibrated against a standard reference measurement, the measured backscattered light can be used to continuously monitor the particulate concentration and present a reading in mg/m^3 .

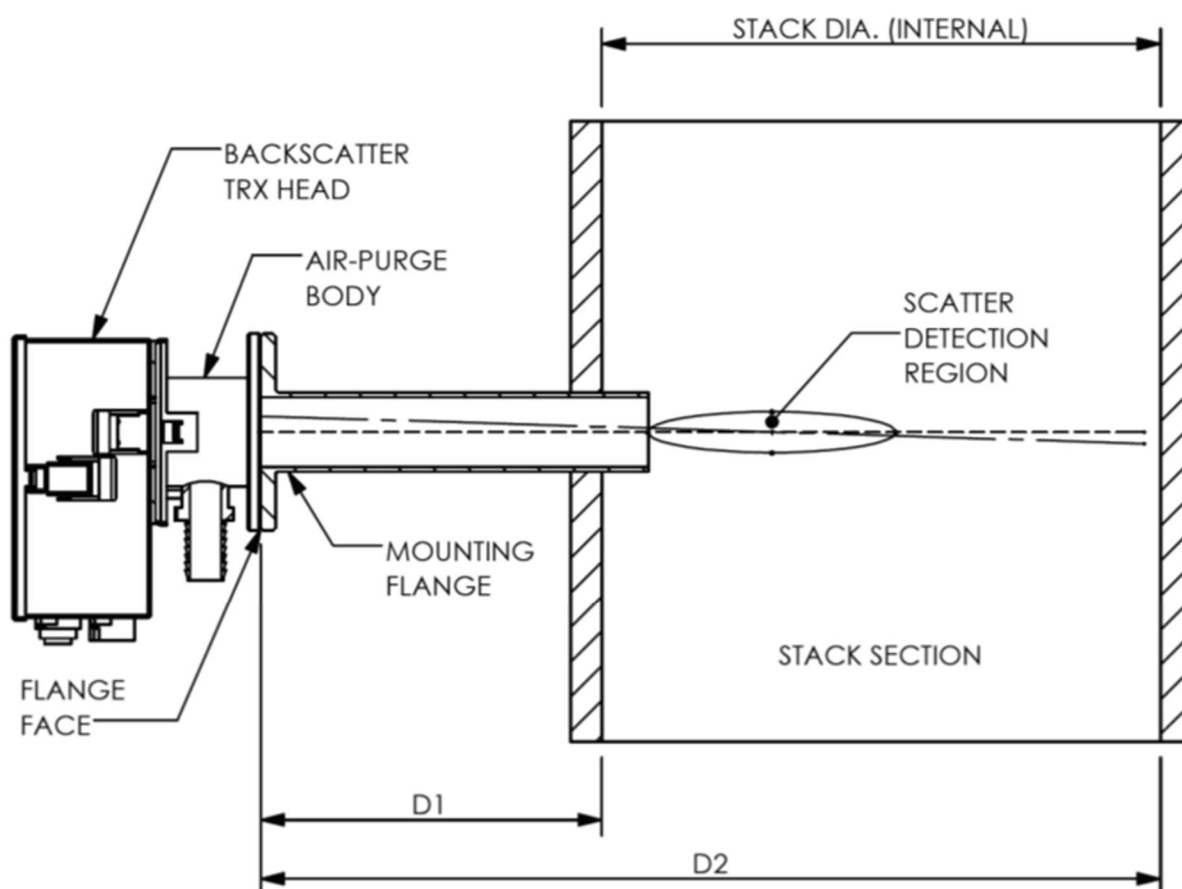
The DSL-800 is particularly suited to small stacks, with internal diameters below 4m. Three variants of the DSL-800 are available: short, medium or long path. The variant to be used depends on the internal diameter of the stack.

Short Path: for internal stack diameters 0.5 to 1.0m

Medium Path: for internal stack diameters 1.0 to 2.0m

Long Path: for internal stack diameters 2.0 to 4.0m

The mounting distance of the TRX head from the inner walls of the stack is also critical for optimum performance. The ideal distances from the face of the mounting flange plate to the inner walls of the stack are illustrated in the following drawing and table.



VARIANT	D1	D2
SHORT	$\leq 0.25\text{m}$	$\geq 0.75\text{m}$
MEDIUM	$\leq 0.25\text{m}$	$\geq 1.25\text{m}$
LONG	$\leq 0.35\text{m}$	$\geq 2.35\text{m}$

A number of checks and measures are employed to maintain continued accuracy of the instrument, including direct monitoring and compensation for the light source intensity. The DSL-800 also has an automatic, internal, zero and span check

capability which can be used to check for any performance drift. This automatic check routine can be configured to run at a regular, pre-defined, time interval set using the supplied Utility software.

The DSL-800 can be supplied for “stand-alone” operation (i.e., TRX head only – no control unit), in which case command and control of the instrument is performed using either the Utility software provided (for use on a laptop PC), or by Modbus serial communications. In the stand-alone configuration, the TRX head has a range of interface outputs including 4-20mA loop, level and service alarm contacts, Modbus and USB, allowing the head to be connected into standard industrial monitoring systems. The DSL-800 also contains a four-digit LED display which shows the live reading, and three status indicator LEDs (Green, Yellow and Blue).

Alternatively, the DSL-800 can be supplied with a DSCU (Dynoptic Systems Control Unit), which is an OI (operator interface) that consists of a numeric / directional keypad, a two-line LCD display and a terminal compartment.

The keypad and display on the DSCU combine to form an interface which can be used to set up and control any connected instrument, with similar functionality to the PC based Utility software supplied with each instruments.

TRX head is supplied with a stainless-steel air-purge body, this not only provides the physical mounting point for the instrument, but also allows users to connect a high-volume, low-pressure air supply to help keep the optics clean and prolong service intervals.

By default, the instrument is a 24Vdc powered device, but it can be optionally supplied with universal input 90 265VAC PSU for AC operation.

Product Specification

	Class 3R visible laser beam. Avoid direct eye exposure.
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Measurement Performance

No.	Parameter	Units	Min	Max	Comment
1	Stack Diameter	m	0.5	4	Internal diameter (three instrument variants)
2	Measuring Range	mg/m ³	0.0	1,000	User selectable ^[1]
3	Accuracy	%	-5	+5	Of measured value
		mg/m ³	-2	+2	Whichever is greater ^[1]
4	Resolution	mg/m ³	0.1	1	Display resolution (Depends on absolute value)
5	Damping	s	1	60	User selectable
6	Drift with Temperature	%	-5	+5	Over the full operating range
7	Operating Wavelength	nm	650	660	Laser (Class 3R)

Power & Air Requirements

8	Voltage	Vdc	+24		Optional 90-260 Vac PSU available
9	Voltage Tolerance	%	-15	+15	
10	Nominal Current Consumption	mA		500	
11	Power Up Current Consumption	mA		200	
12	Air Supply Volume	L/min	50	100	Supplied to the air purge body
13	Air Supply Fitting				1" Hosetail in the air purge body

Interface Options

14	Serial Comms				Modbus RTU via RS485 & External USB
15	Analogue Output (one)	mA	0 / 2 / 4	20	Isolated, User configurable
16	Relay Contacts (Two)	A		3	@30 Vdc User configurable
17	Instrument Displays	Dust Reading (mg/m ³)			4-digit LED display
		Three Status Lights			Green-Power, Yellow-Fault, Blue-Maintenance

Physical

18	Ingress Protection		IP65		For external use
19	Ambient Operating Temperature	°C	-20	+50	Air temperature around the heads
20	Operating Humidity	%		100	Air humidity around the heads
21	Gas Temperature	°C		+400	
22	Materials				AISI/SAE 316L stainless steel
23	Weight	kg		2.5	TRX head plus air-purge body

Compliance & Design

24	Regulatory Compliance				2014/30/EU (Electromagnetic Radiation) 2014/35/EU (Low Voltage)
25	Design Life	Years	20		
26	MTBF	Hours	30,000		Replaceable Laser
27	Warranty	Months	24		Return to base warranty. Extensions available

[1] With default calibration settings

Physical Installation

Selecting a location

Points for consideration include:

- Legislation
- Accessibility
- Gas flow within the stack
- Liquid aerosols
- Temperature
- Cable routes
- Earth connection
- Air-purge tubing

Legislation

Local or national legislation covering the installation may specify or make specific requirements on the positioning of the sensor head. If legislation does affect the head position, priority must be given to this legislation, but do not disregard other considerations discussed here.

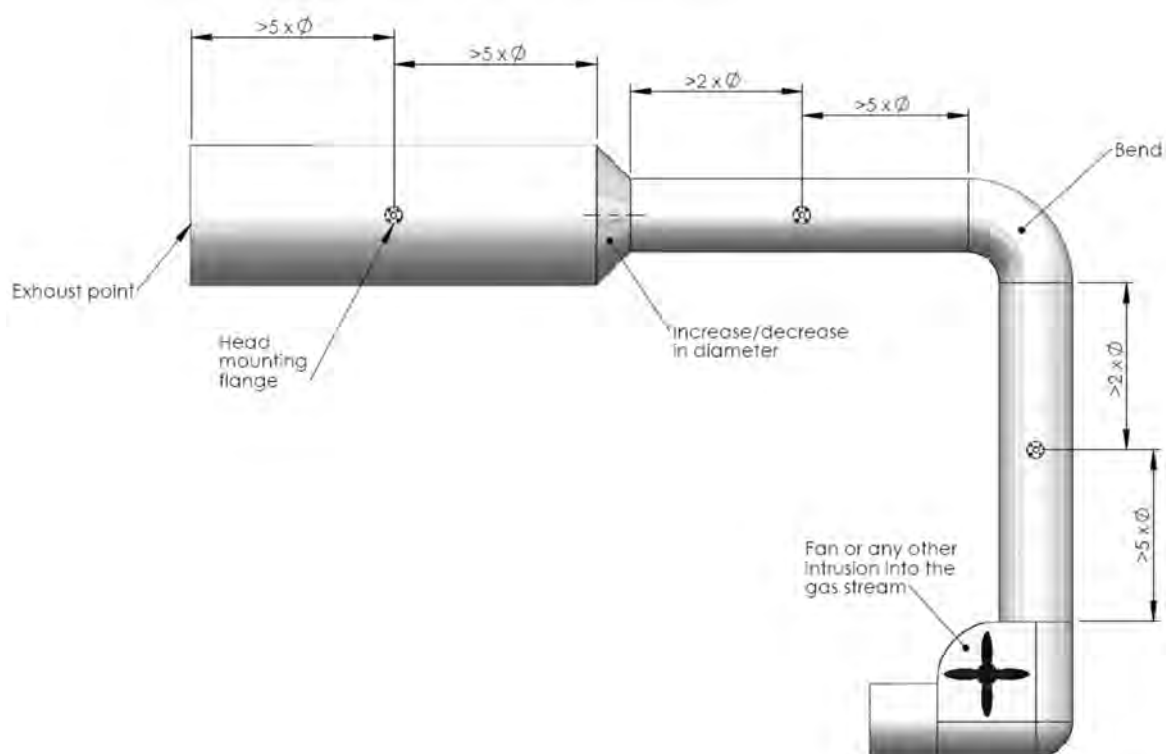
Accessibility

Access to the head will be required for maintenance, performance testing and servicing. Positioning the instrument at the level of a gantry or platform is preferable.

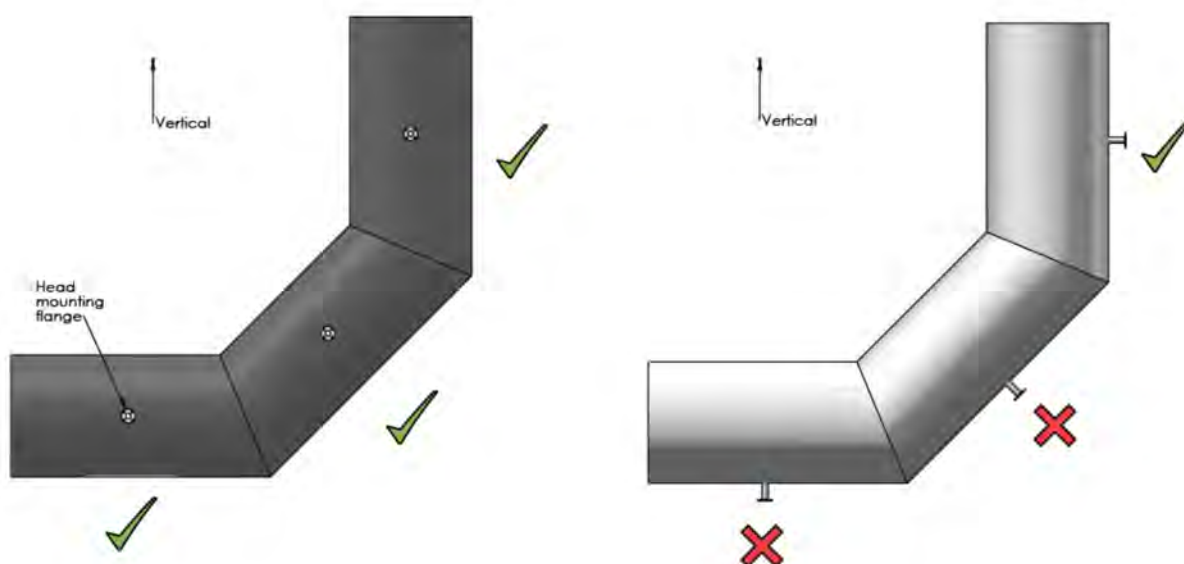
Gas Flow Within the Stack

The stack gas flow characteristics at the measurement point should be considered carefully when locating the head. Ideally the head should be positioned on a straight section of duct with a uniform gas flow. This requires the mounting location to have no abrupt changes in diameter, no bends, no inlets/outlets and no intrusions/protrusions, within at least 5 duct diameters of the head downstream and 2 duct diameters upstream.

Ideally the measurement point should be located at least 5 duct diameters from the exhaust point, see image below.



Where suitable sample planes exist in both vertical and horizontal sections of the duct, the vertical section is preferable

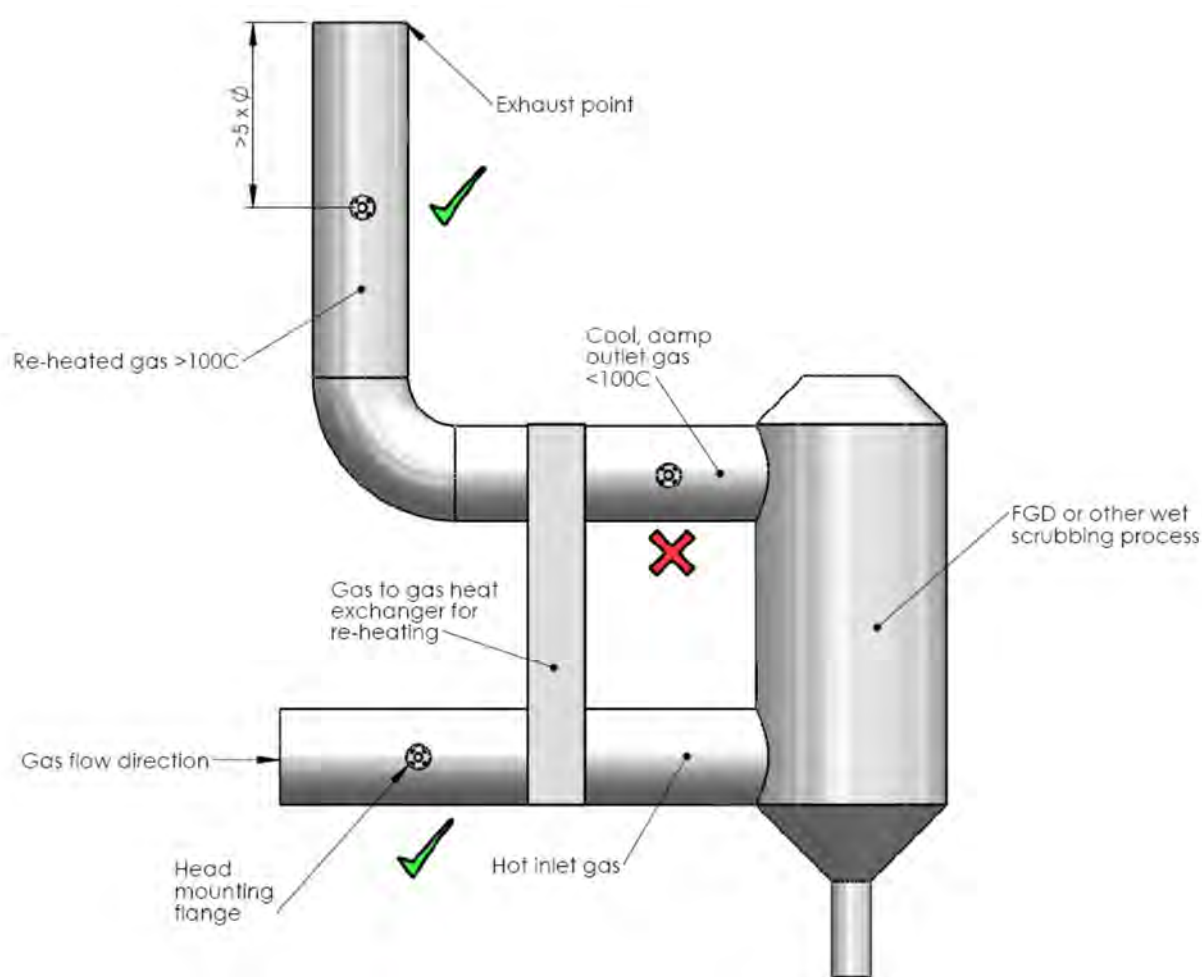


NOTE: The head should never be positioned at the bottom of a horizontal duct or angled duct, as particle matter will fall under gravity into the head and cause the optics to become dirty more quickly.

Liquid Aerosols

Mist, spray, droplets and condensing liquids will scatter the light in a similar way to the dust or smoke particles. In most installations it is undesirable to measure the contribution from liquid aerosols and therefore a mounting location should be chosen that is free from such contaminants.

As a general rule, the head should be located at least 5 stack diameters upstream of the exhaust point, at a point where the exhaust gas is consistently in excess of 100°C, and upstream of any wet scrubber or other process, which may introduce liquid mist or spray (unless fitted with a re-heater or demister).



If the head is mounted on a vertical stack section near to the exhaust point, there is the possibility that heavier droplets (from rain or condensing steam) may fall down into the stack, against the direction of the gas flow. These droplets will also generate a reading, which may be undesirable, and an alternative location should be considered.

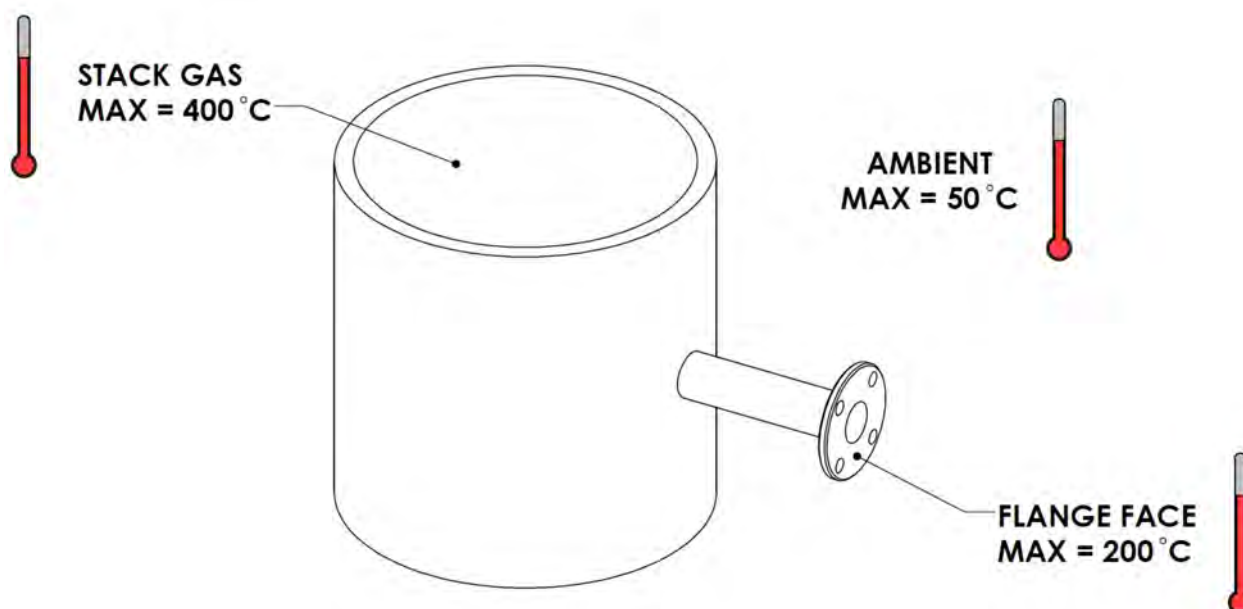
Knowledge of the process gas conditions, engineering logic, and common sense should be employed to choose a location for the head that will not be affected by mist, spray, droplets, or condensing liquids.

Temperature

Temperature is an important consideration when choosing a location for the head, the main considerations are:

- Stack gas temperature
- Mounting flange temperature
- Ambient air temperature around the head
- Radiated heat
- Direct sunlight

The mounting flange face temperature must not exceed 200°C and the standard installation recommendations provided in this manual are for a maximum stack gas temperature of 400°C. Please consult Dynoptic Systems if the application has stack gas temperatures higher than 400°C.



The ambient air temperature around the heads should be in the region -20°C to +50°C. If the ambient air temperature around the head is being affected by radiated heat from the duct, the use of lagging on the duct, or a heat shield in front of the heads, may be necessary.

Where the head is mounted externally it should be shielded from direct sunlight at all times. The head may be mounted under a canopy or shelter, or alternatively an optional weather cover is available from Dynoptic.

It is essential that the instrument is protected from extreme temperature conditions.

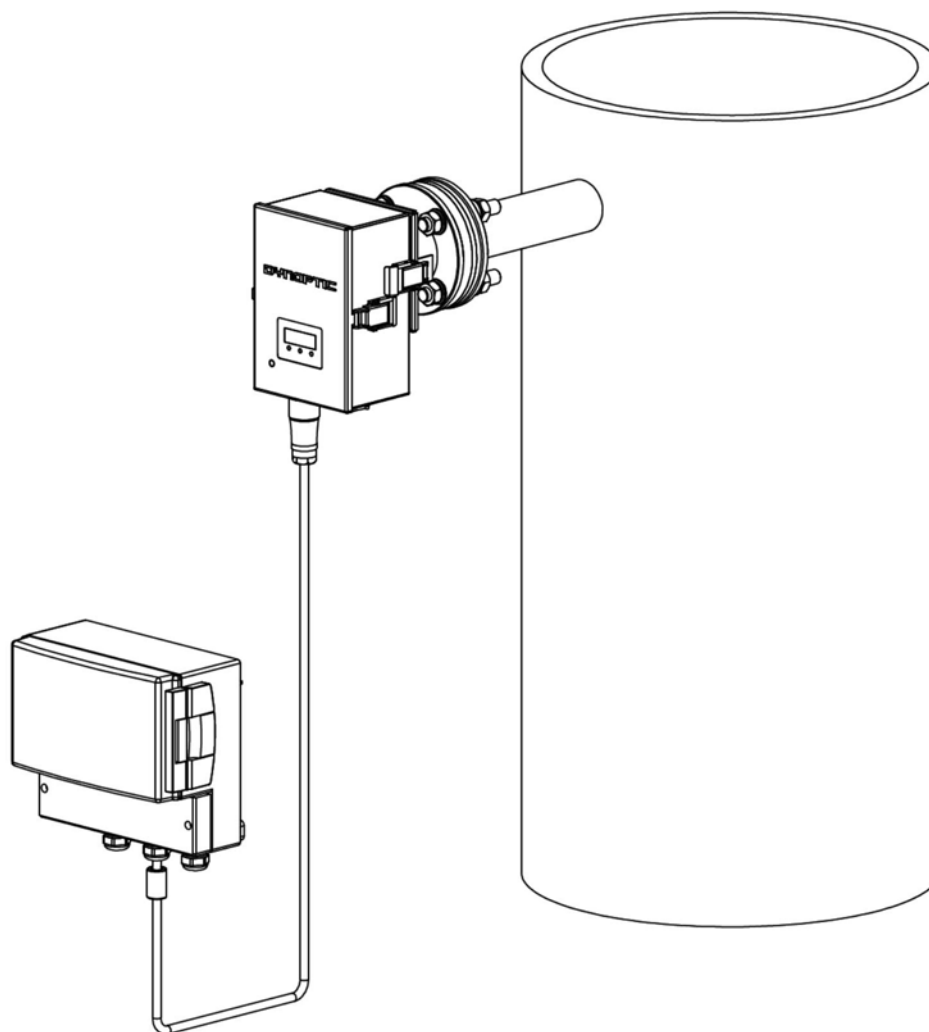
Cable Routes

When choosing a location for the DSL-800 consideration must be given to the route and length of the power and interface cables.

The length of the 24Vdc power cable is limited by the voltage drop likely to occur on the 24Vdc supply carried in this cable. Always calculate the potential voltage drop when specifying the type/length of this cable, taking care to allow for the output tolerance of the power supply. The 24Vdc supply at the TRX must not fall below 20Vdc

The RS485 or analogue signal output cables can be much longer, up to 1km. These cables carry sensitive electronic signals which can be affected by RFI and EMI. These cables must follow an independent route away from any high voltage power cables, high frequency signal cables, or switch gear cabling.

When the instrument is being powered through a DSCU (as illustrated in the following drawing) the connecting cable contains both power (24Vdc) and RS485 communications. The maximum length of this cable is determined by the 24Vdc voltage drop.



Earth Connection

A protective earth connection to the TRX head is essential, not just as a safety precaution, but also to ensure correct performance and to maintain the instrument's compliance with the safety directives and CE marking.

A location for the head must be chosen where a proper electrical earth is available. It is not acceptable to rely on the stack or the mounting flanges as an earth point.

Air-Purge Tubing

If the purge air to the TRX head is being provided by a blower then there is a maximum tube length and height differential that can be tolerated to maintain optimum purge air performance.

Ideally the blower should be located close to the instrument. If this is not practical,

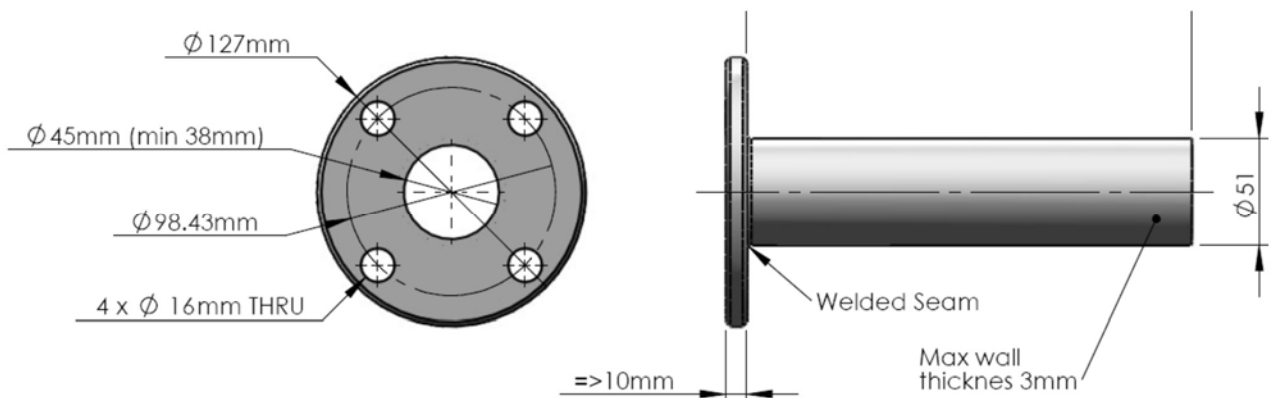
then care must be taken to ensure that the length and position of the tubing does not adversely reduce the air flow. For blowers supplied by Dynoptic the total length of the 25mm diameter tubing should not exceed 50m. In addition, the location of the blower must be less than 20m below the height of the air-purge body.

These limitations do not apply if compressed air is being used as the purge air.

Fixing the Mounting Flange & Head

NOTE: Care must be taken when working on the stack. In particular hot surfaces and hot stack gases. Take all necessary precautions.

The TRX head air-purge body should be bolted to a suitable mounting flange (1.5" ANSI 150 pattern) extended away from the side of the stack on a short extension tube. The drawing below shows the construction details of a suitable mounting flange extension. A pre-fabricated mounting flange extension, with a tube length of 240mm, can be purchased from Dynoptic as an accessory.



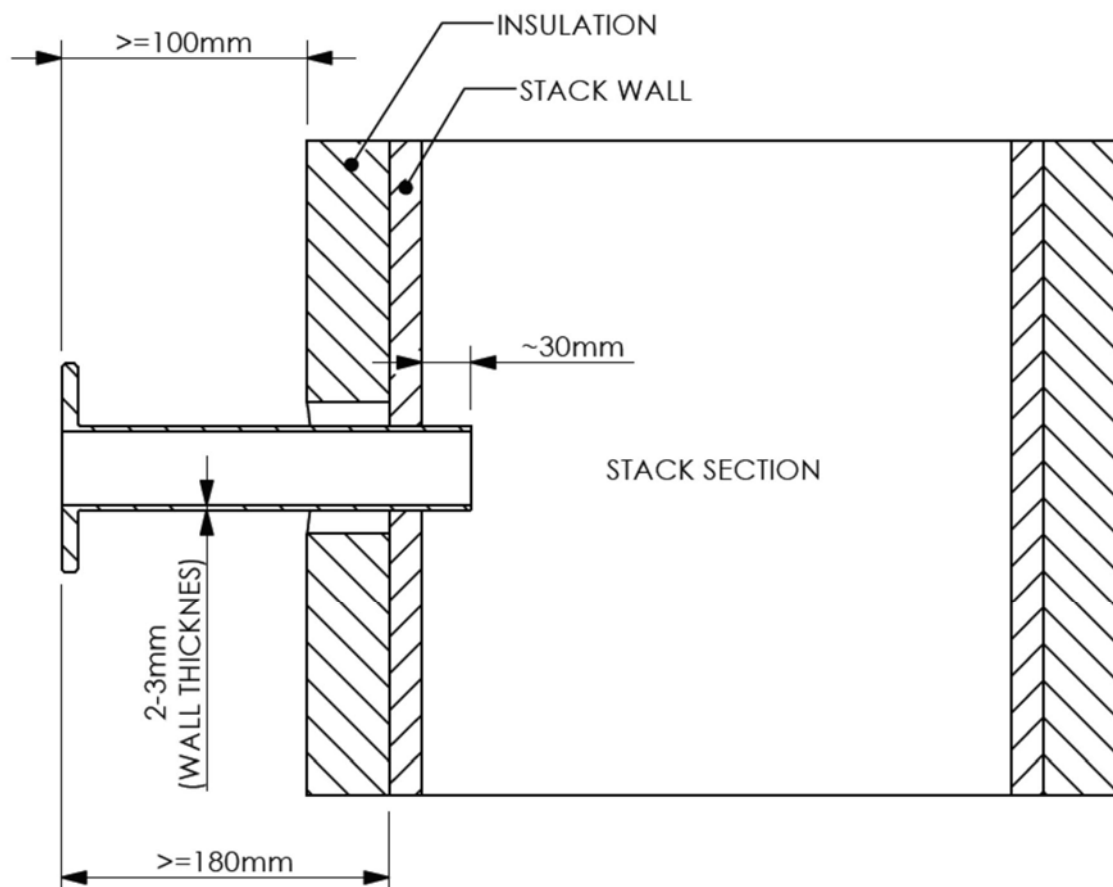
When fitting the mounting flange to the stack the extension tube should pass through the wall of the stack and protrude slightly (~30mm) into the duct. The purpose of the intrusion is to prevent any dirt or slime build-up on the inner wall of the duct falling down the wall and passing through the light path.

NOTE: For correct operation of the purge air the inner diameter of the flange tube must be $\leq 45\text{mm}$ and must protrude into the stack.

A typical installation with a pre-fabricated flange extension is illustrated below. The

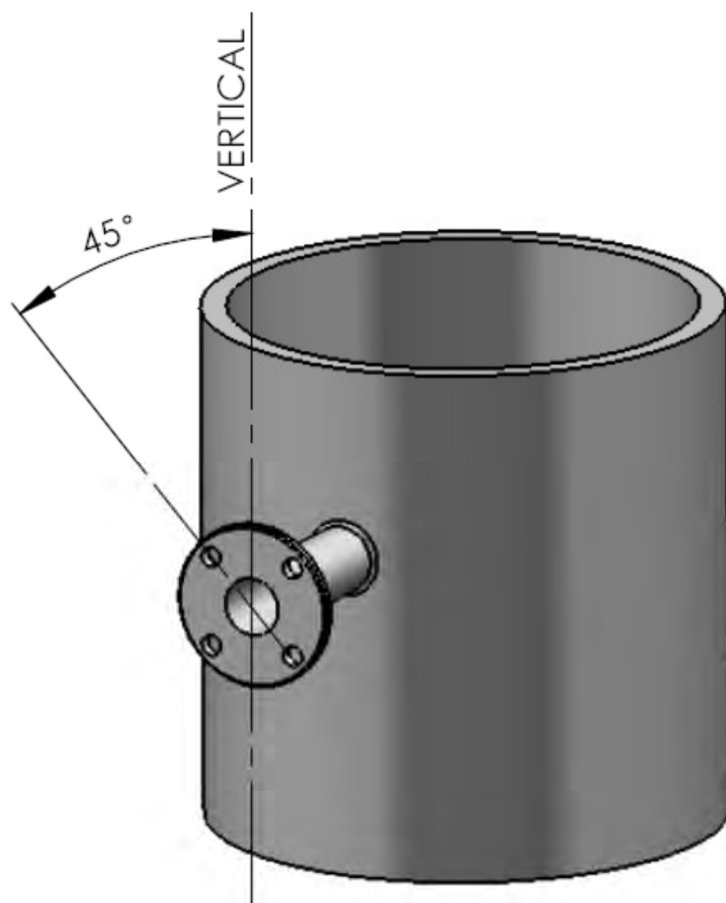
minimum flange plate distances shown in this drawing are for 400degC stack gas temperature. At lower maximum stack gas temperatures, the flange plate can be positioned closer to the stack wall.

The flange extension tube should be horizontal or, if possible, sloping slightly towards the stack. This is to reduce the possibility of any corrosive liquid present in the duct running down the flange extension tube towards the air-purge body.

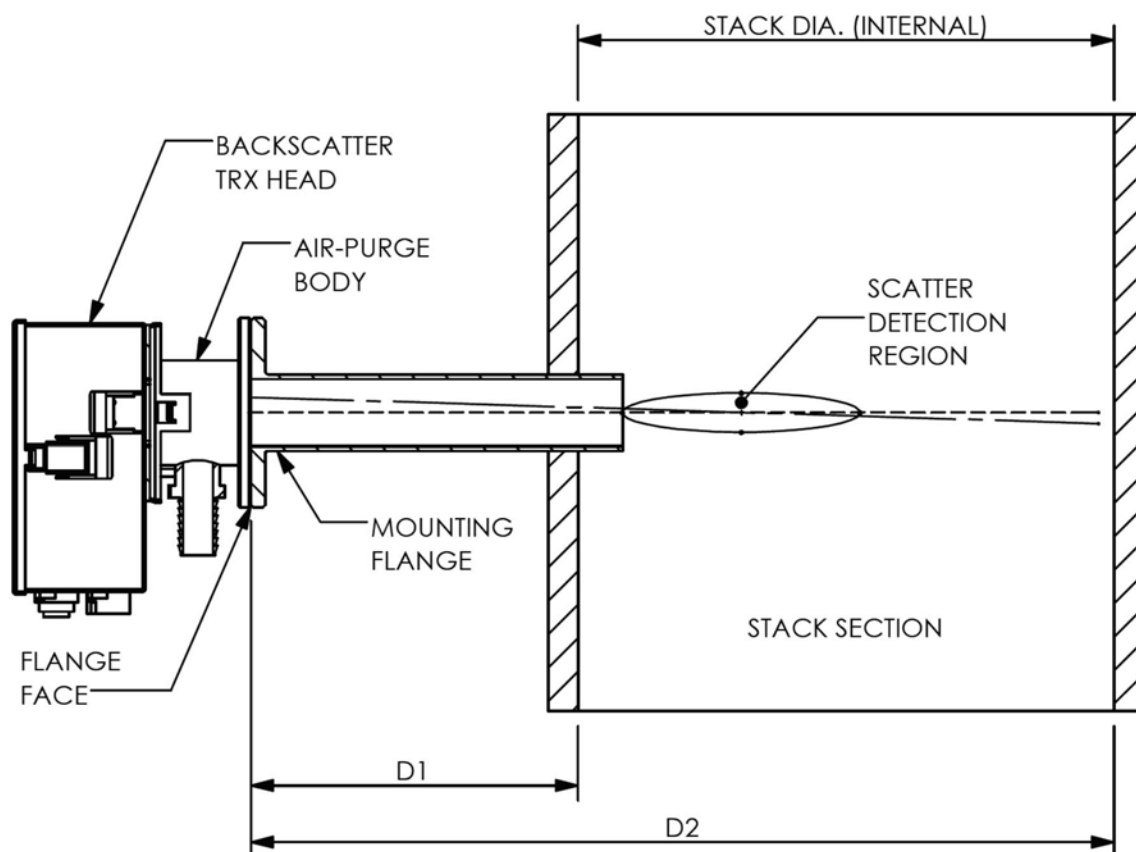


The flange extension tube must be installed with the bolt holes at 45° to the vertical, as shown below.

NOTE: When installed, this instrument must be orientated so that the electrical connectors are positioned on the underside, so that no ingress can be made through the connectors under gravity.



The DSL-800 is a laser backscatter instrument and is designed to measure light scattered from a region at a fixed distance in front of the TRX head. The length of the flange extension tube is critical because it sets the position of this scatter region inside the stack. The following drawing illustrates the maximum distance allowed for the three path length variants. Note the pre-fabricated flange extension, installed as recommended above, would place the flange plate at D1=220mm from the inner wall of the stack.

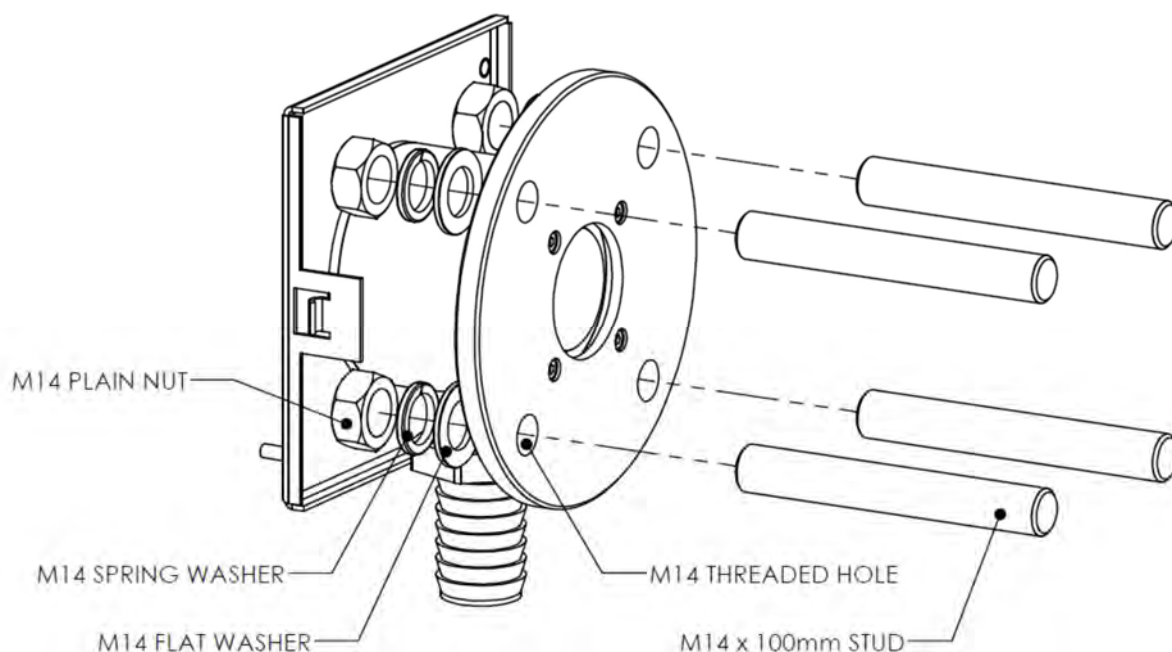


VARIANT	D1	D2
SHORT	≤0.25m	≤0.75m
MEDIUM	≤0.25m	≤1.25m
LONG	≤0.35m	≤2.35m

The DSL-800 comes with a stainless-steel air-purge body. The instrument TRX head clips onto this air-purge body. The installation of this air-purge body onto the mounting flange plate is illustrated in the following drawings.

The air-purge body has four (4) M14 threaded holes on a PCD to match the mounting flange (1.5" ANSI 150 pattern). This air-purge body is installed on the mounting flange by using short lengths of M14 threaded stud with suitable nuts and washers on both ends.

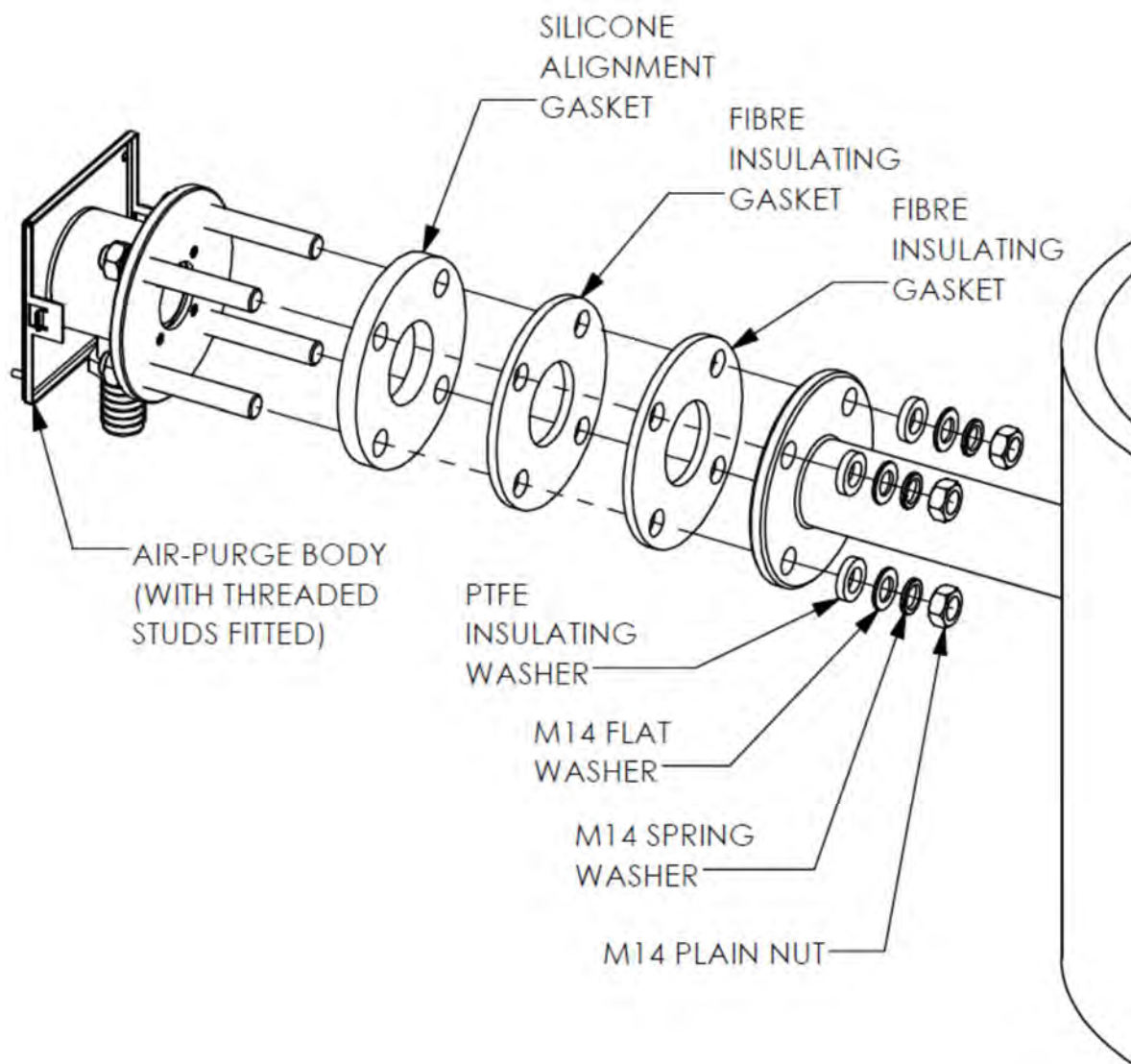
The M14 threaded studs should first be screwed into the threaded holes of the air-purge body until they protrude out of the back of the plate by at least 25mm. These must then be locked in place with an M14 nut, spring washer and flat washer.



The alignment gasket (red) and two insulating gaskets (white) are then placed onto the long threaded studs protruding from the air-purge head. This full assembly is then attached to the flange plate. Each stud is fixed to the flange plate using a PTFE washer (for insulation), flat washer, spring washer and then M14 nut.

The M14 nuts on the flange plate side of the assembly should all be tightened equally to a torque setting of approximately 20Nm.

Select threaded studs of a suitable length and remember to allow for the additional thickness of the rubber alignment gaskets, heat insulating gaskets, washers, and any weather shield that may be fitted as part of the installation. Fixing kits containing suitable M14 studs, washers and nuts are available from Dynoptic.



The instrument is supplied with the two (2) heat insulating gaskets manufactured from a 6mm thick woven ceramic fibre cloth (white in colour). These gaskets must be installed as shown, against the hot mounting flange surface, to protect the red silicone rubber alignment gasket and the TRX head, from conducted heat from the flange face.

The instrument is supplied with the four (4) heat insulating washers manufactured from 5mm thick Teflon (white in colour). These washers must be installed as shown, so that they are fitted between each nut/bolt/washer and the hot mounting flange surface. Their purpose is to prevent heat from the mounting flange conducting through the fixing bolts.

The instrument is supplied with one (1) red silicone rubber gasket which must be fitted as shown, so that it is protected from the heat of the mounting flange by the

heat insulating gaskets. This rubber gasket forms a watertight seal and, if required, will allow for a degree of adjustment to be made in the angular alignment of the TRX head. The rubber gasket must not come into contact with any surface that has a temperature in excess of 150°C.

Earth Connection

A heavy-duty earth wire, or braid, must be connected between the M5 stud on the outside of the TRX head enclosure and a proper electrical earth point. It is not acceptable to rely on the stack or the mounting flange as an earth point.



NOTE: A proper protective earth connection is not only essential for safety, but also to maintain the instrument's performance and compliance with safety directives, CE and UKCA marking.

Air-Purge System

NOTE: *The air-purge system must be in place and operational prior to the installation of the TRX head. The air-purge system must remain operational at all times, even if the instrument itself is not. Failure to maintain an operational air-purge system may result in the head being damaged by hot, corrosive stack gases.*

Overview and Requirements

The DSL-800 uses optical techniques to make the particulate measurement. The head therefore has optical surfaces exposed to the dirty and potentially harmful environment of the stack interior. In order to protect the optical surfaces from contamination by particle depositions and the damaging effects of heat and/or corrosive stack gas, it is necessary to employ an air-purge system to repel the stack gas and prevent it from coming into contact with the head.

The air-purge system achieves this by producing a flow of clean and dry air flowing away from the head and into the stack. This continuous flow of air provides a clean air barrier between the optics and the stack gas.

A complete air-purge system will comprise of:

- **Air-purge body** (supplied with instrument) which serves to duct the purge air away from the instrument head and into the stack.
- **Flange Tube** (not supplied as standard) to produce a uniform flow of purge air into the stack.

NOTE: *For correct operation of the purge air the inner diameter of the flange tube must be $\leq 45\text{mm}$ and must protrude into the stack.*

- **A clean, dry, purge air supply** (not supplied as standard) which could be derived from a suitably filtered air blower or pressure regulated and filtered compressed air.

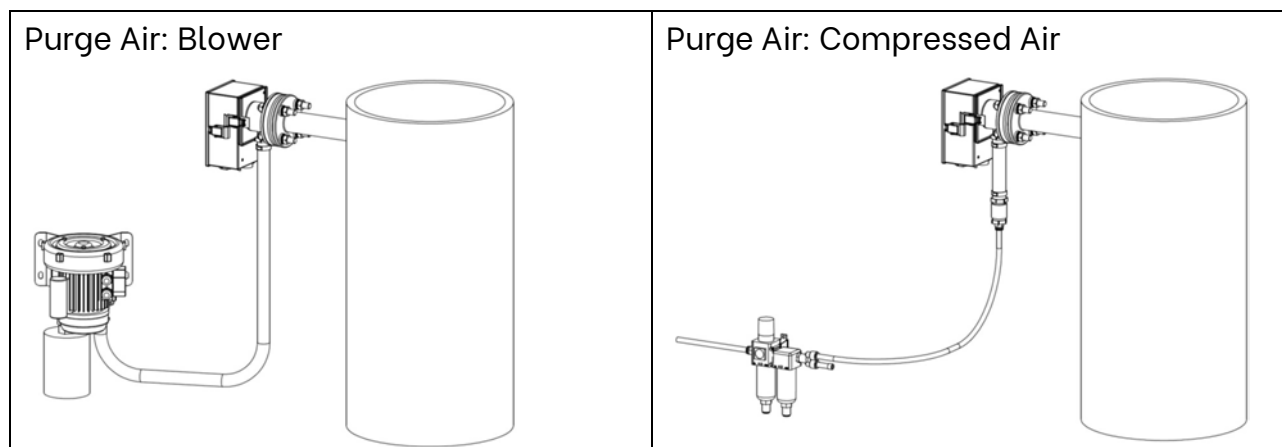
- **Hose and fittings** (not supplied as standard) necessary to connect the air supply to the air-purge body.

The air-purge body is supplied with the instrument; all the other elements are available as accessories. For detailed information on the purge air requirements, please refer to the Dynoptic technical note 'Air Supply for Optical Stack Monitor Purge Systems'.

The air inlet to the air-purge body is a 1" hosetail, and any blower used must be capable of providing a flow rate of 50 to 100 litres/min through the air-purge body.

If compressed air is to be used a flow restrictor must be attached to the inlet of the air-purge body. A suitable part is available as an accessory from Dynoptic (DSL-CAFR). This has a short length of tubing to attach to the air-purge body hosetail and a 3/8" BSP female thread for connecting compressed air fittings.

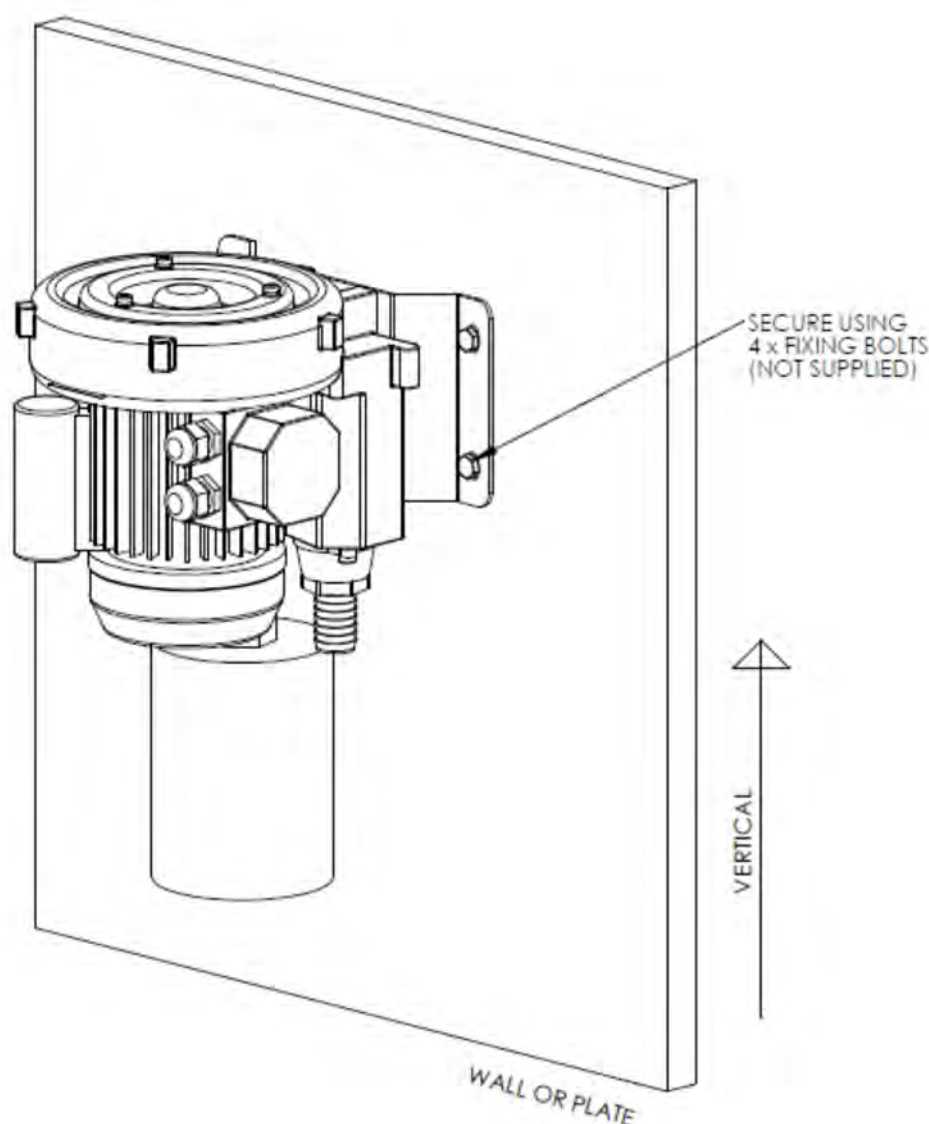
Full details of installing and operating a blower or compressed air system are provided with the optional accessory. The following is a very brief overview.



Blower

The required high volume, low pressure, clean air can be supplied by a filtered blower. The blower inlet must be filtered to remove particulates smaller than 5µm. To provide the required airflow through the air-purge body the blower must have a maximum flow rate (zero pressure drop) of >400 litres/min and a maximum differential pressure (zero flow) of >30 mbar.

A suitable blower and tubing are available from Dynoptic (DSL BK40B). If possible, this blower should be mounted in a clean air environment, vertically as illustrated below. The total length of 25mm ID tubing from the blower to the air-purge body must be less than 50m.

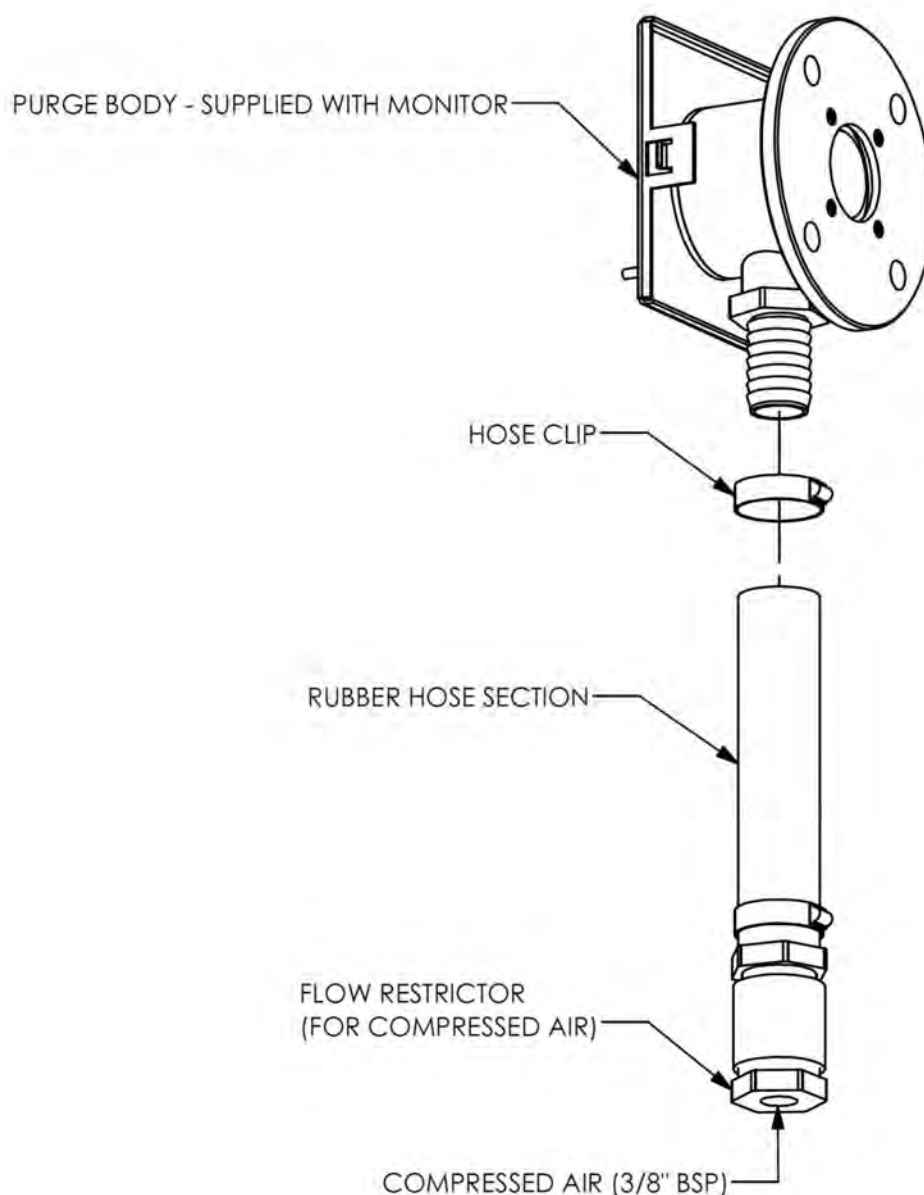


Compressed Air

The **pressure of a standard compressed air line is too high for the air-purge body**. A flow restrictor must be fitted to the inlet of the air-purge body to reduce the pressure. A suitable flow restrictor assembly is available as an optional accessory from Dynoptic (DSL-CAFR).

The following drawing shows how the flow restrictor assembly fixes onto the inlet 1"

hosetail of the air-purge body.



The compressed air going into the flow restrictor must be regulated to ~4 bar and filtered to remove all particulates, water and oil. Dynoptic can provide a compressed air kit (DSL-CAK-1), which contains a suitable regulated filter assembly as well as the flow restrictor assembly.

Electrical Installation

NOTE: *Electrical power must be disconnected from the instrument when making or breaking electrical connections of any kind.*

NOTE: *Do not open the instrument or attempt to remove the back panel. All electrical connections to the instrument are made using external connectors.*

General Description

All electrical connections to the DSL-800 are via two multi-pin sockets on the bottom of the TRX head. The Power/Comms socket and the Outputs socket.

The Power/Comms socket is used to supply the 24Vdc power to the TRX head and it also provides connection to the isolated RS485 communications. If the instrument is connected directly to an optional DSCU then any cable from this connector can go directly to the DSCU (24Vdc and RS485 comms). However, if the power and comms need to be separated a suitable junction box will be required.

The Output socket provides the one analogue output and two relay outputs.

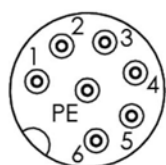
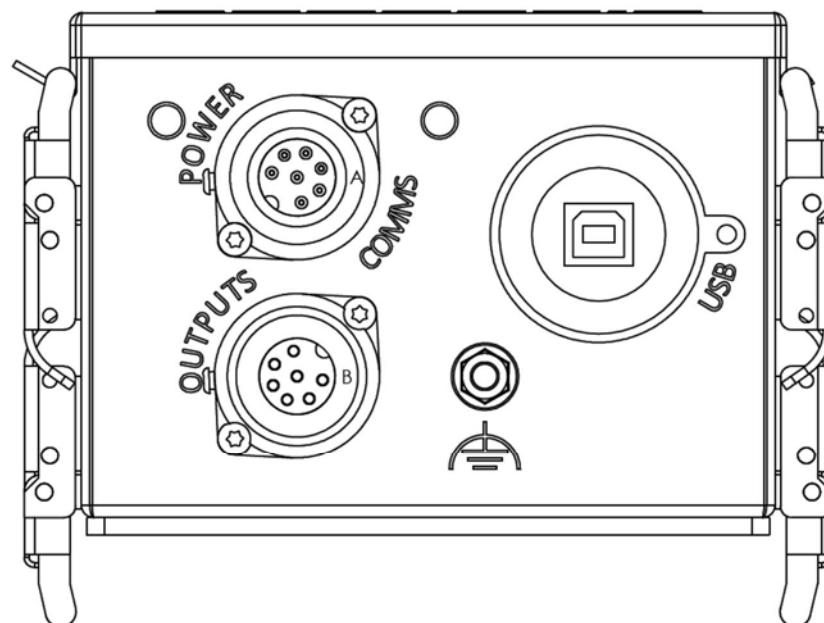
Power / Comms Cable and Outputs Cable

A standard DSL-800 is supplied with a short (2m) power / comms cable, and a short (2m) outputs cable. Both cables include a unique pre-terminated connector at one end only, for connection to the TRX head. The other end of the cable is to be made off, as required, by the customer.

Longer cables are available as optional accessories but should be discussed and ordered separately. Connector bodies are also available as optional accessories. These connector bodies need to be connected to a suitable cable (see later)

The pinout connection of the DSL-800 is shown in the following drawing and instrument wiring diagram.

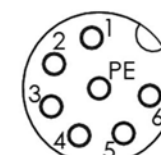
DSL-800 – Laser Backscatter Particulate Monitor – Installation Manual



DETAIL A
SCALE 2 : 1

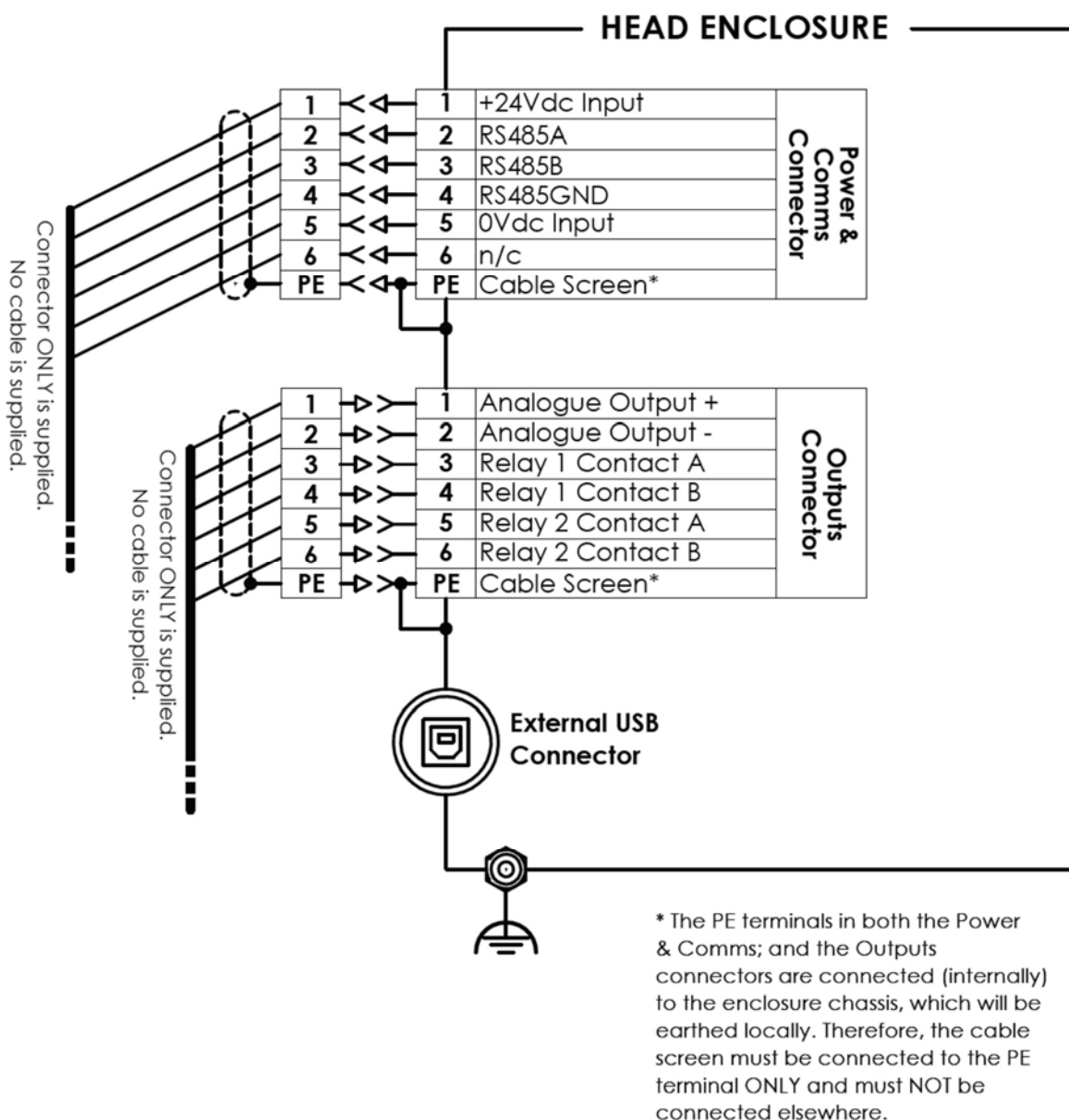
POWER / COMMS	
Pin #	Connection
1	+24Vdc Input
2	RS485A
3	RS485B
4	RS485GND
5	0Vdc Input
6	n/c
PE	Cable Screen*

OUTPUTS	
Pin #	Connection
1	Analogue Output +
2	Analogue Output -
3	Relay 1 Contact A
4	Relay 1 Contact B
5	Relay 2 Contact A
6	Relay 2 Contact B
PE	Cable Screen*



DETAIL B
SCALE 2 : 1

* THE CABLE SCREEN MUST BE CONNECTED HERE, AT THE DSL-800 END OF THE CABLE, BUT IT MUST BE LEFT UNCONNECTED AT THE OTHER END OF THE CABLE.



Each cable assembly supplied includes a cable diagram which details wire colours in relation to pin number. The typical relationship between pin number / pin function / wire colour is defined below. However, the printed cable diagram, supplied with each cable assembly, takes precedence and should always be considered definitive.

POWER / COMMS Cable			OUTPUTS Cable		
Pin No.	Connection	Colour	Pin No.	Connection	Colour
1	+24Vdc Input	Brown	1	Analogue Output +	Brown
2	RS485 A	Pink	2	Analogue Output -	Pink
3	RS485 B	White	3	Relay 1 Contact A	White
4	RS485 GND	Yellow	4	Relay 1 Contact B	Yellow
5	0Vdc Input	Green	5	Relay 2 Contact A	Green
6	n/c	Blue	6	Relay 2 Contact B	Blue
PE	Cable Screen	Screen	PE	Cable Screen	Screen

NOTE: These wire colours are based on the standard cable assemblies. The supplied cables may differ. Please refer to the reference drawing supplied with each cable assembly

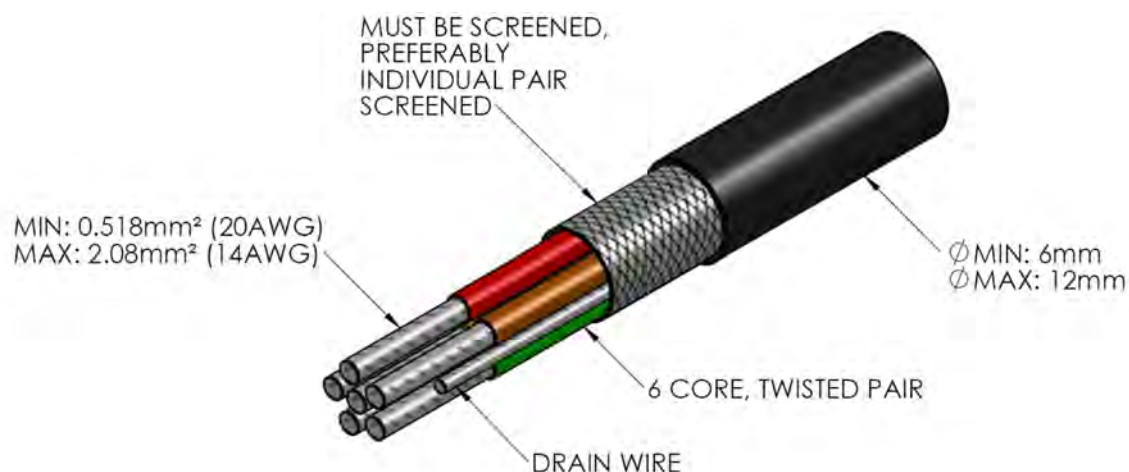
Cable

If the connector body accessories are used, they will need to be connected to a suitable cable. This cable must be sourced locally or can be purchased separately from Dynoptic.

All cables should have an individual core size of between 0.518mm² (20AWG) and 2.08mm² (14AWG).

All cables must be screened, ideally individually screened twisted pairs with a common drain wire, such as Belden 9873 (6 core, screened pair, 20AWG).

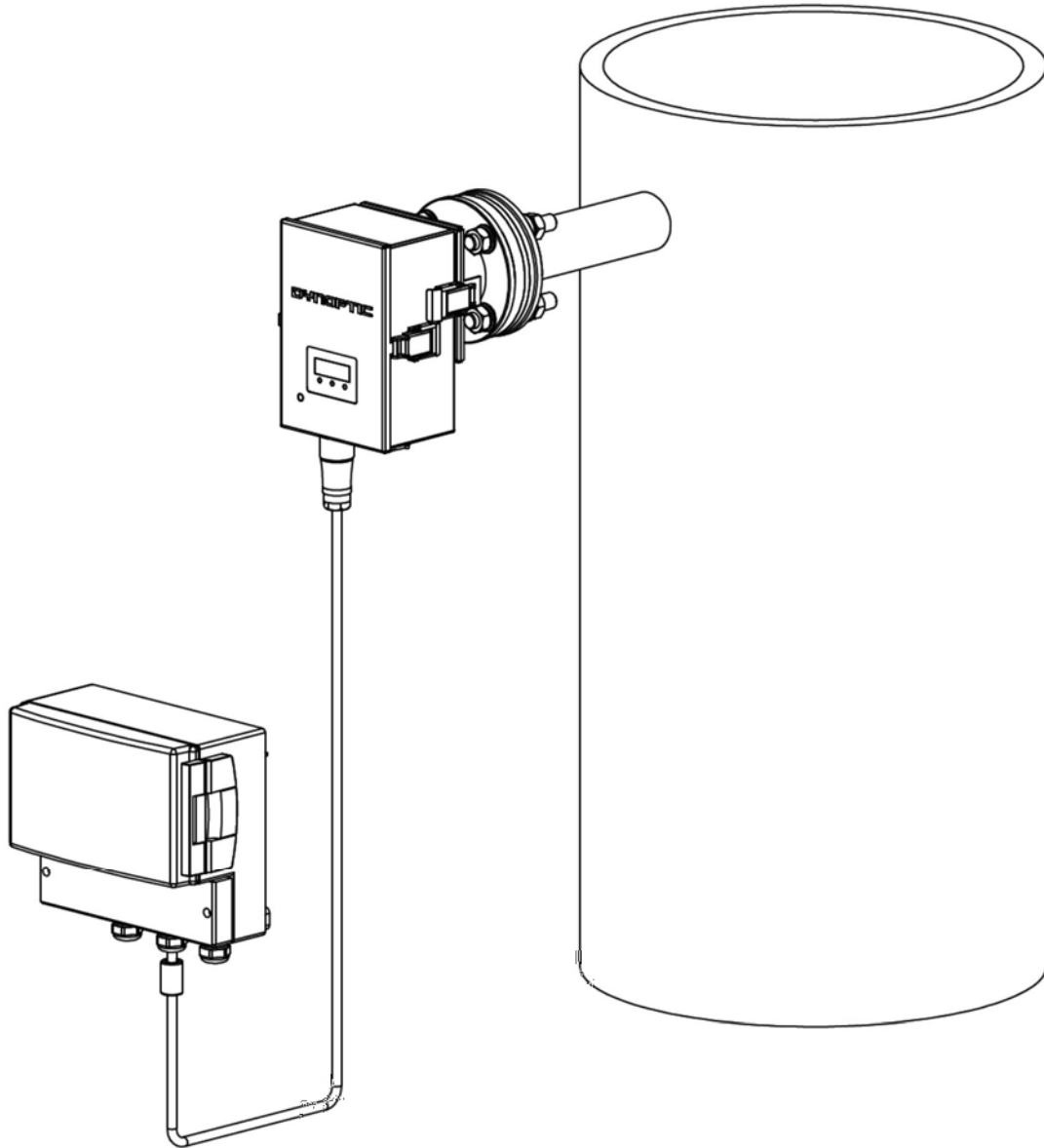
The overall diameter of the cable should be between 6 and 12mm.



Connecting to a DSCU

When an optional DSCU is installed, the cable connecting the DSCU to the TRX Power/Comms socket must have a minimum of 5 cores and meet the previous cable specification. The maximum length of this cable is limited by the voltage drop likely to occur on the 24Vdc supply carried in this cable. Always calculate the potential voltage drop when specifying the type/length of this cable, taking care to allow for the output tolerance of the power supply. The 24Vdc supply to the TRX head must not fall below 20Vdc. This typically limits the cable length from the DSCU to the TRX head to about 20m

The single cable connection from a DSCU to the TRX head is illustrated below.



The detailed wiring to the DSCU is described in the DSCU Installation Manual. Note the DSCU does not use an isolated RS485 and so the RS485 ground connection in the TRX head must be connect to 0Vdc.

If long cable lengths are required between the DSCU and the TRX head then separate, local power supplies should be used to separately power the DSCU and TRX head. With this arrangement the cable from the DSCU to the TRX head contains only the RS485 connections and so the maximum cable length is ~1km.

All cables entering the DSCU must do so through the cable glands provided. All cable glands must be tightened onto the cable to maintain the enclosure's environmental rating.

All unused cable gland entries should be sealed using the rubber bungs supplied with the DSCU. Tighten the gland onto the rubber bung to ensure correct environmental protection.

The ferrite tubes supplied with the DSCU should be fitted on the cables close to where they enter the DSCU enclosure. The ferrites should be secured in place so that they remain fixed adjacent to the cable gland entry.

NOTE: *The cable connecting the TRX head to a DSCU carries sensitive electronic signals which can be affected by RFI and EMI. Care must be taken to run the cable away from high voltage power cables, high frequency signal cables, or switch gear cabling. Ideally the cable will follow an independent route and will not be installed in bundles or cable ducts with other cabling.*

Notes

Disclaimers

All technical details and specifications are subject to change without notice.

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CUSTOMER:

Ingredion (Thailand) Co., Ltd.

LOCATION:

Kalasin

PROJECT TITLE:

Project SPRINT 18K TPA Spray Dry

CUSTOMER P.O. NO.:

7001730092

PACKAGE:

Dust Analyzer

EPM PROJECT NO:

6626013

DOCUMENT TITLE:

Installation, Operation and Maintenance Manual

3.DSL-800 Operators Manual



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Precautionary Statements

Throughout this manual, precautionary statements are included to draw attention to potential hazards.

These statements are always boxed and include the signal words DANGER, WARNING, CAUTION, or NOTE, to indicate the degree of hazard that may be encountered. These words are defined below:

DANGER: *Imminent hazard and failure to avoid hazard will result in death.*

WARNING: *Hazard that may result in injury or death.*

CAUTION: *Hazard that may result in minor or moderate injury.*

NOTE: *Hazard that may result in either, damage to equipment that will require repair or replacement; or a reduction in performance that renders the equipment unable to meet its specification.*

General Precautions

WARNING: *When powered this instrument emits a visible laser beam (Class 3R). Avoid direct eye exposure. Always disconnect the power (or turn on Maintenance Mode) before removing the instrument for repair or maintenance.*

CAUTION: *Appropriate Protective Personal Equipment (PPE) must be worn when installing, commissioning or otherwise working on this device.*

CAUTION: *Appropriate access equipment, operated by qualified personnel, must always be used when working at height.*

CAUTION: *Care must be taken when working on a stack. In particular hot surfaces and hot stack gases. Take all necessary precautions.*

NOTE: *This device contains sensitive optics and electronics and must be protected from physical shock, excessive vibration, and ingress, at all times. Always handle with care and protect from ingress.*

NOTE: *Do not open the instrument or attempt to remove the back panel. All electrical connections to the instrument are made using external connectors.*

NOTE: *Cables connected to this device may carry sensitive electronic signals which can be affected by RFI and EMI. Care must be taken to run cables away from high voltage power cables, high frequency signal cables, or switch gear cabling. Ideally any cables connecting to the device will follow an independent route and will not be installed in bundles or cable ducts with other cabling.*

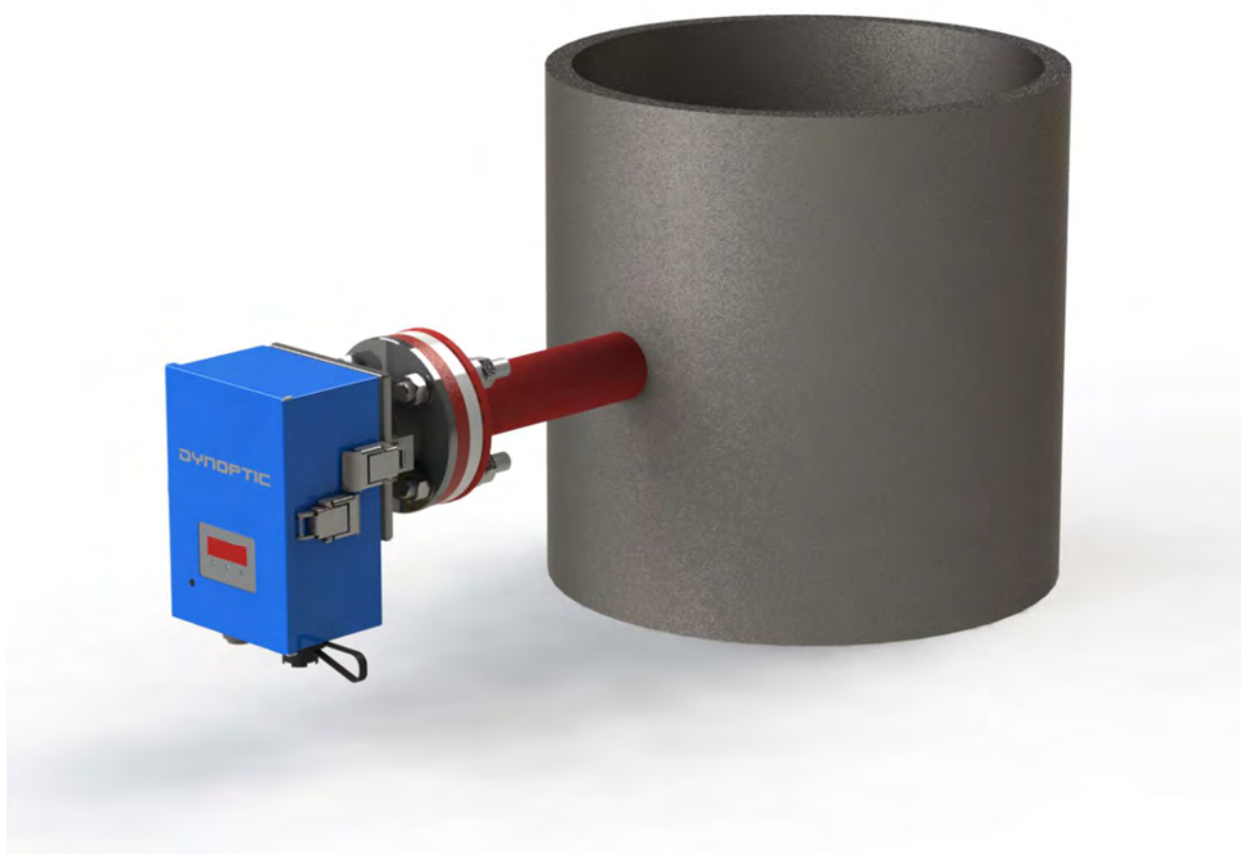
NOTE: *Do not make or break electrical connections whilst power is connected to this device, or any device connected to it. “Hot plugging” as this is known, may damage the electronics. Such damage is NOT covered under warranty.*

NOTE: *Protective earth connections are not only essential for safety, but also to maintain the instrument’s performance and compliance with safety directives, CE and UKCA marking.*

NOTE: *Read this manual in its entirety before commissioning, or otherwise using this device. If in doubt, contact Dynoptic directly for advice.*

Product Overview

The DSL-800 Particulate Monitor uses laser backscatter to measure the level of suspended particles (e.g., dust and smoke) in a duct, stack, chimney, or flue.



The DSL-800 consists of a single TRX head mounted on one side of the stack. The TRX head emits a beam of laser light into the stack and any light backscattered from suspended particles is detected by the TRX. Increased particulate or smoke density in the stack gas produces an increase in the level of the scattered light. When calibrated against a standard reference measurement, the measured backscattered light can be used to continuously monitor the particulate concentration and present a reading in mg/m^3 .

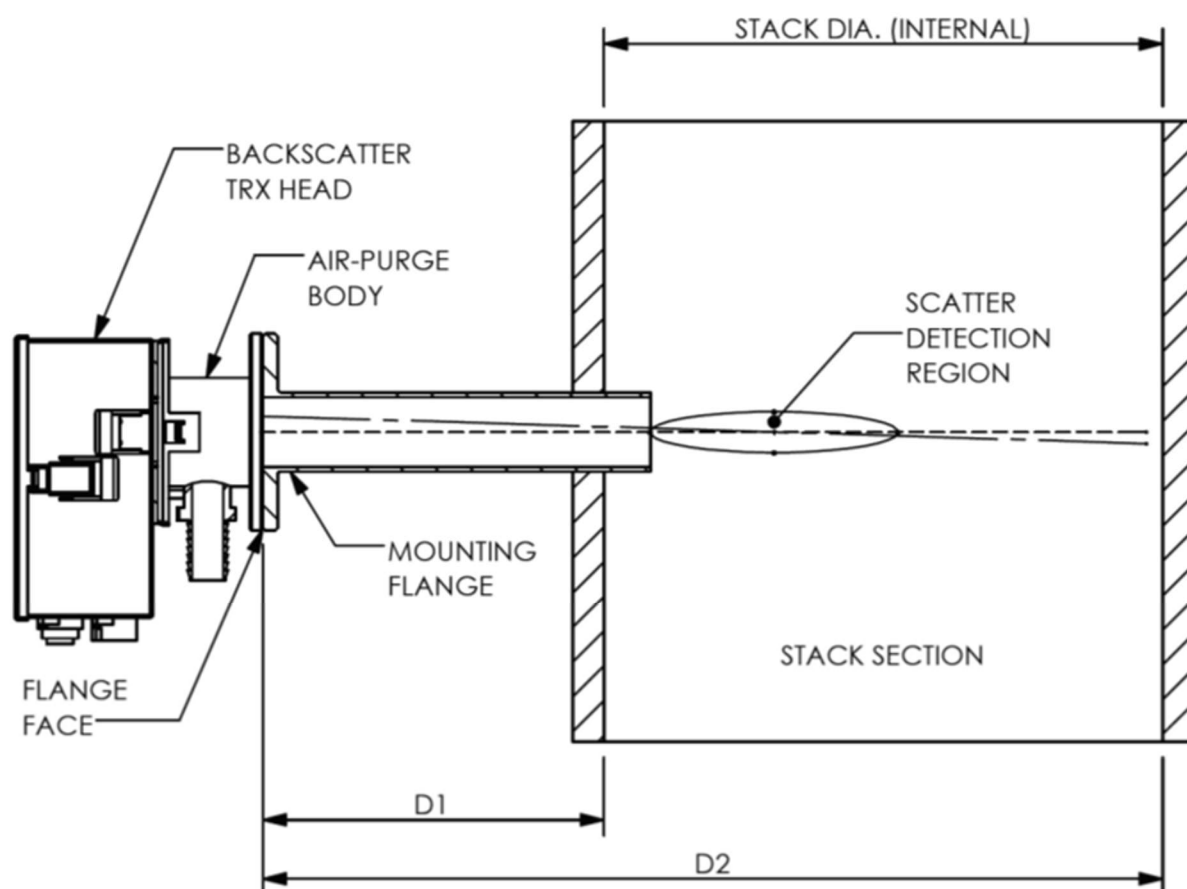
The DSL-800 is particularly suited to small stacks, with internal diameters below 4m. Three variants of the DSL-800 are available: short, medium or long path. The variant to be used depends on the internal diameter of the stack.

Short Path: for internal stack diameters 0.5 to 1.0m

Medium Path: for internal stack diameters 1.0 to 2.0m

Long Path: for internal stack diameters 2.0 to 4.0m

The mounting distance of the TRX head from the inner walls of the stack is also critical for optimum performance. The ideal distances from the face of the mounting flange plate to the inner walls of the stack are illustrated in the following drawing and table.



VARIANT	D1	D2
SHORT	$\leq 0.25\text{m}$	$\geq 0.75\text{m}$
MEDIUM	$\leq 0.25\text{m}$	$\geq 1.25\text{m}$
LONG	$\leq 0.35\text{m}$	$\geq 2.35\text{m}$

A number of checks and measures are employed to maintain continued accuracy of the instrument, including direct monitoring and compensation for the light source intensity. The DSL-800 also has an automatic, internal, zero and span check capability which can be used to check for any performance drift. This automatic

check routine can be configured to run at a regular, pre-defined, time interval set using the supplied Utility software.

The DSL-800 can be supplied for “stand-alone” operation (i.e., TRX head only – no control unit), in which case command and control of the instrument is performed using either the Utility software provided (for use on a laptop PC), or by Modbus serial communications. In the stand-alone configuration, the TRX head has a range of interface outputs including 4-20mA loop, level and service alarm contacts, Modbus and USB, allowing the head to be connected into standard industrial monitoring systems. The DSL-800 also contains a four-digit LED display which shows the live reading, and three status indicator LEDs (Green, Yellow and Blue).

Alternatively, the DSL-800 can be supplied with a DSCU (Dynoptic Systems Control Unit), which is an OI (operator interface) that consists of a numeric / directional keypad, a two-line LCD display and a terminal compartment.

The keypad and display on the DSCU combine to form an interface which can be used to set up and control any connected instrument, with similar functionality to the PC based Utility software supplied with each instruments.

The TRX head is supplied with a stainless-steel air-purge body, this not only provides the physical mounting point for the instrument, but also allows users to connect a high-volume, low-pressure air supply to help keep the optics clean and prolong service intervals.

By default, the instrument is a 24Vdc powered device, but it can be optionally supplied with universal input 90 265VAC PSU for AC operation.

Product Specification

	Class 3R visible laser beam. Avoid direct eye exposure.
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Measurement Performance

No.	Parameter	Units	Min	Max	Comment
1	Stack Diameter	m	0.5	4	Internal diameter (three instrument variants)
2	Measuring Range	mg/m ³	0.0	1,000	User selectable (1)
3	Accuracy	%	-5	+5	Of measured value
		mg/m ³	-2	+2	Whichever is greater (1)
4	Resolution	mg/m ³	0.1	1	Display resolution (Depends on absolute value)
5	Damping	s	1	600	User selectable
6	Drift with Temperature	%	-5	+5	Over the full operating range
7	Operating Wavelength	nm	650	660	Laser (Class 3R)

Power & Air Requirements

8	Voltage	Vdc	+24		Optional 90-260 Vac PSU available
9	Voltage Tolerance	%	-15	+15	
10	Nominal Current Consumption	mA		500	
11	Power Up Current Consumption	mA		200	
12	Air Supply Volume	L/min	50	100	Supplied to the air purge body
13	Air Supply Fitting				1" Hosetail in the air purge body

Interface Options

14	Serial Comms				Modbus RTU via RS485 & External USB
15	Analogue Output (one)	mA	0 / 2 / 4	20	Isolated, User configurable
16	Relay Contacts (Two)	A		3	@30 Vdc User configurable
17	Instrument Displays	Dust Reading (mg/m ³)			4-digit LED display
		Three Status Lights			Green-Power, Yellow-Fault, Blue-Maintenance

Physical

18	Ingress Protection		IP65		For external use
19	Ambient Operating Temperature	°C	-20	+50	Air temperature around the heads
20	Operating Humidity	%		100	Air humidity around the heads
21	Gas Temperature	°C		+400	
22	Materials				AISI/SAE 316L stainless steel
23	Weight	kg		2.5	TRX head plus air-purge body

Compliance & Design

24	Regulatory Compliance				2014/30/EU (Electromagnetic Radiation) 2014/35/EU (Low Voltage)
25	Design Life	Years	20		
26	MTBF	Hours	30,000		Replaceable Laser
27	Warranty	Months	24		Return to base warranty. Extensions available

[1] With default calibration settings

Installing the Utility Software

NOTE: *The Utility Software and the USB Driver must be installed on the PC before connecting to the DSL-800. Failure to comply with this requirement is likely to result in a failed installation.*

Minimum Hardware/Software Specification for Host PC

The Utility software supplied with the DSL-800 is not a large or particularly complex program, and therefore has no requirement for a high-performance host PC. Any reasonably modern PC/laptop will be suitable. The only “must have” requirements are:

- Windows 7, or newer, operating system.
- All recent operating system service packs and upgrades (visit the Microsoft website) must have been installed.
- At least one free USB slot (assigned to a COM port between COM1 and COM32 – see later section on ‘Assigning a Valid COM Port’) for direct connection to the DSL-800 (or to an optional DSCU).

Installing the USB Driver

The USB Driver software is supplied in electronic format (CD or flash drive) with each unit. A link to a download file can be provided by Dynoptic on request.

1. Ensure that the PC is logged on to a user account with administrative rights, so that necessary permissions are granted to install software.
2. Copy the USB Driver “CDMxxxxx.exe” file to any suitable location on the host PC (“My Documents” is a good location).
3. Double click on the .exe file and follow the on-screen installation prompts.

Installing the Utility Software

The instrument uses the “DSL-800 Utility” software which is supplied in electronic format (CD or flash drive) with the device. A link to a download file can be provided by Dynoptic on request.

1. Ensure that the PC is logged on to a user account with administrative rights, so that the necessary permissions are granted to install the software.
2. Copy the file “DSL-800 Utility install xxx.exe” to any suitable location on the host PC (“My Documents” is a good location).
3. Double click on the .exe file and follow the on-screen installation prompts.

The software should now be installed and should appear in the “All Programs” list in the folder “DSL-800 Utility”. The software is called “DSL-800 Utility.exe”.

Updating the Utility Software

From time-to-time Dynoptic may release updated versions of the Utility software. If you receive an updated version, it is essential to remove any existing version from the host PC before installing the new one.

Assigning a Valid COM Port

When a PC is first connected to the DSL-800, it is possible that the PC will assign the USB connection to a COM Port number outside the acceptable range (COM1 to COM32), in which case the Utility software will be unable to communicate with the instrument.

It is possible to check which COM Port your PC has assigned to the DSL-800 by opening Windows “Device Manager” before connecting your instrument. When the “Device Manager” window is open scroll down to “Ports (COM & LPT)”, then click the “>” symbol to look at the list of COM Ports.

With the list of current COM Ports visible on the PC screen, connect the USB to the DSL-800. The PC will indicate that it has “Found New Hardware” and a new COM Port number should pop up in the “Ports (COM & LPT)” list.

If the new COM Port is between COM1 and COM32 then your PC has allocated a valid COM Port and no further action is required.

However, if the COM Port number is outside that range, then it must be changed, following the procedure below:

1. Right click on the COM Port (in the “Ports (COM & LPT)” list) that appeared when the DSL-800 was connected.
2. Click “Properties”.
3. Select the “Port Settings” tab.
4. Click on the “Advanced” button.
5. Click the “COM Port” and select a new “COM Port Number” between COM1 and COM32 (avoiding any COM Ports which are marked as in use) from the drop-down list.

Once a valid COM Port number has been assigned to the USB connection, the PC will save this COM Port assignment. There will be no need to repeat the process again, when connecting to a DSL-800 in the future, provided that the connection is made using the same PC and the same USB slot.

NOTE: *If an alternative computer is used it will be necessary to repeat the process above to assign a valid COM Port.*

NOTE: *Always connect to the instrument using the same USB slot on the PC. If an alternative USB slot is used, it may be necessary to repeat the process above and re-assign a valid COM Port, because the PC identifies each USB slot as a separate entity.*

Using the Utility Software

NOTE: *The Utility software and the USB Driver must be installed on the PC before connecting the DSL-800. Failure to comply with this requirement is likely to result in a failed installation.*

Connecting a PC to the DSL-800

A weatherproof USB connector can be found on the bottom of the TRX head. It is necessary to unscrew the protective cover, before gaining access to the USB connection itself.

Connect the PC directly to the DSL-800 using a standard USB cable (Type A to Type B).

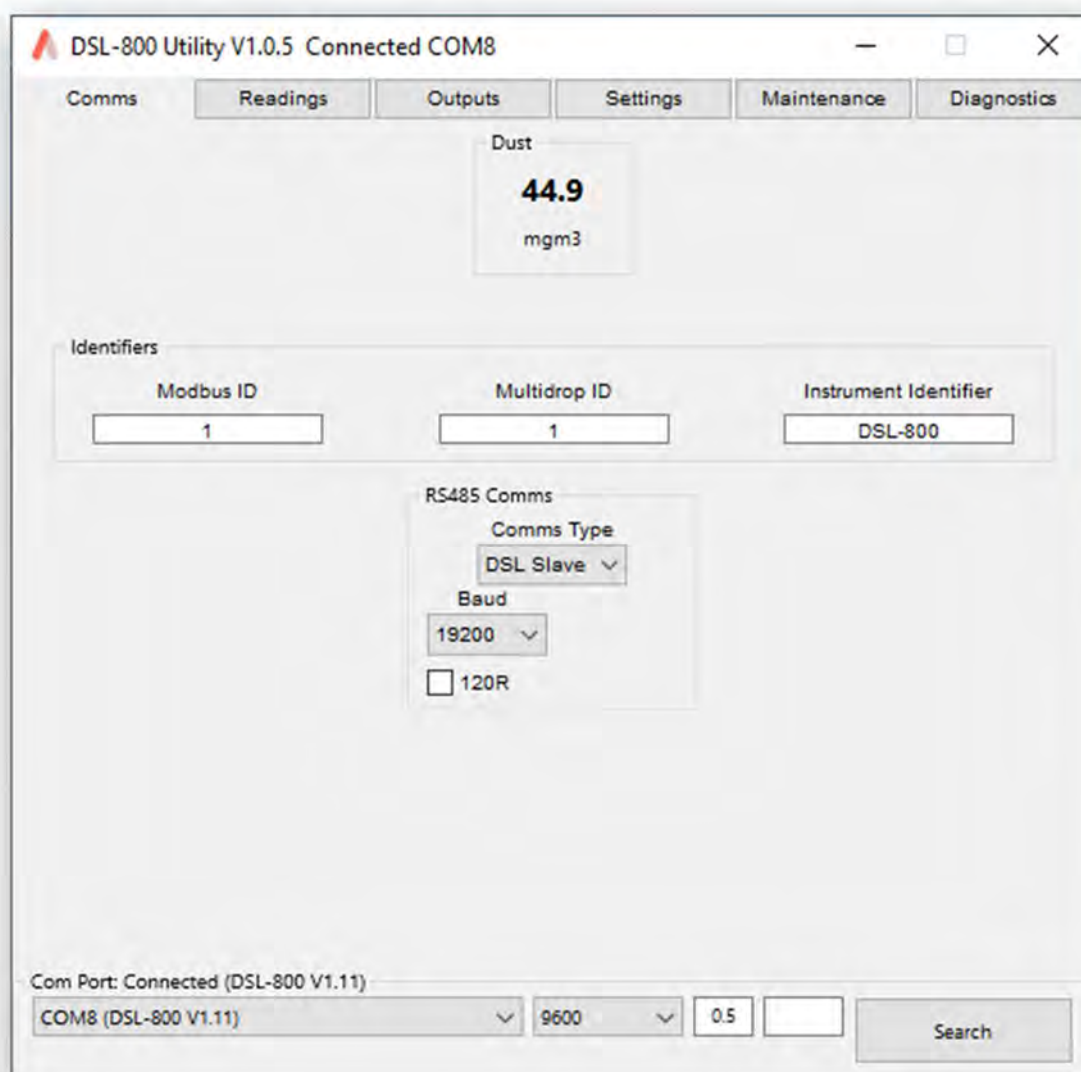
NOTE: *Always replace the weatherproof cap when not connected to the USB. Protect the USB connector from ingress at all times.*

Alternatively, it is possible to connect the PC to the optional DSCU (operator interface). The USB connection for the DSCU can be found inside the enclosure. Although this is an indirect connection to the DSL-800, the functionality allowed is exactly the same as the direct connection to the TRX head.

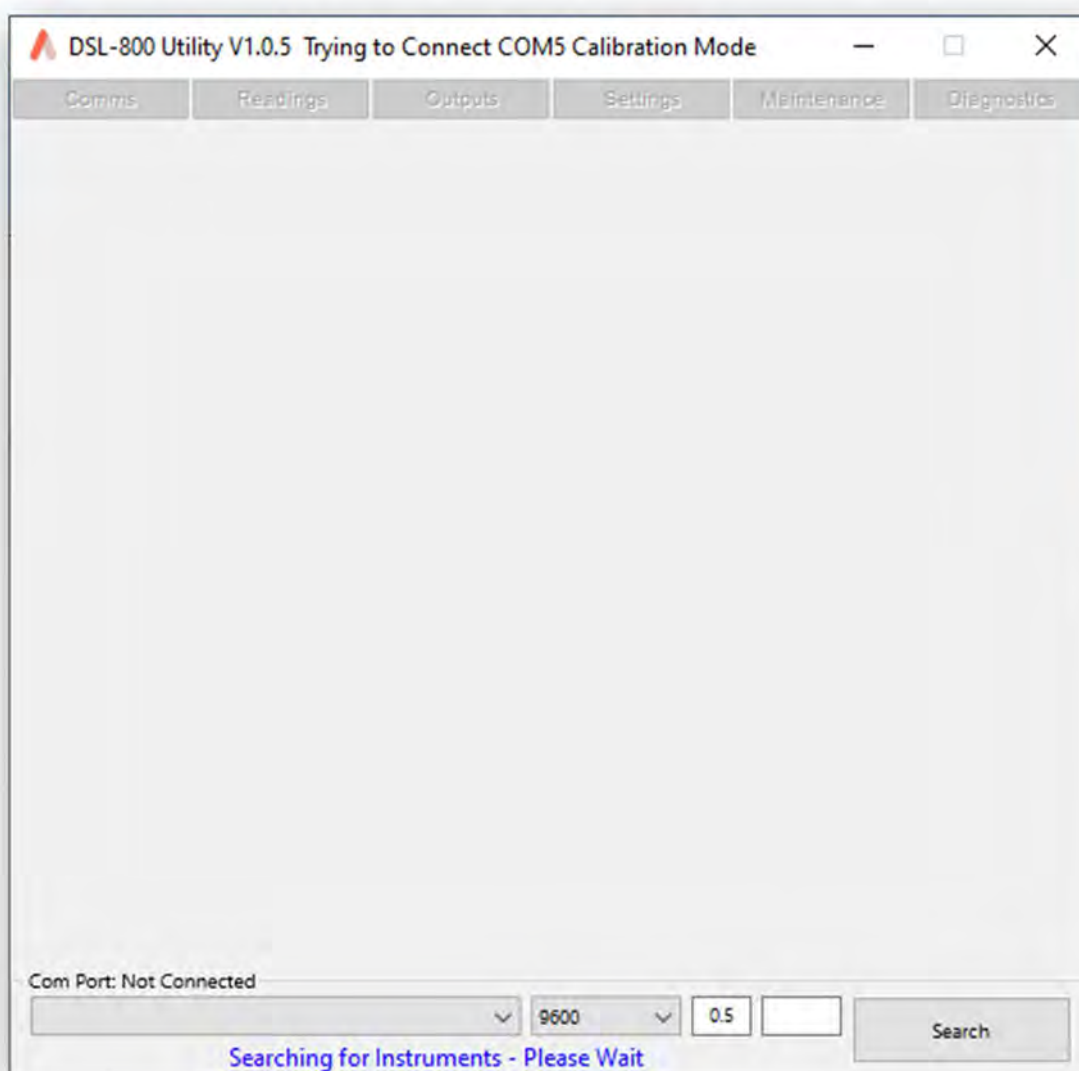
The Comms tab

On running the Utility software, it will open on the Comms tab. The software automatically scans through all of the COM ports (1 to 32) and will connect to any DSL-800 found. A connected Comms tab is shown below.

When a comms connection is established, additional settings become available on the Comms tab and all of the other Utility tabs become accessible.



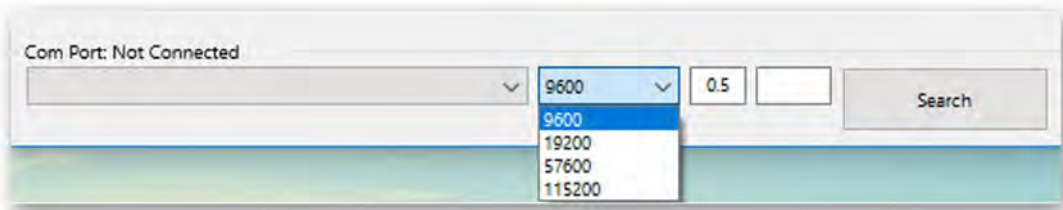
If the software fails to find a DSL-800 instrument the Comms tab will remain blank, as shown below.



If the Utility is not connected to an instrument, every 10 seconds it will automatically scan the COM ports searching for a DSL-800. It is also possible to manually set the software to scan for an instrument by clicking the 'Search' button.

If the Utility fails to find a connected instrument check the following

- The USB cable is working and that it is plugged into the DSL-800.
 - Unplugging and then reconnecting the USB sometimes helps.
- The DSL-800 is powered (green power light is on).
- The PC has assigned the USB connection to a comm port between 1 and 32 on the PC (see "Assigning a valid COM port").
- The Baud Rate is set to 9600 (see below).



The following describes each of the features displayed on the connected Comms tab.

Dust

The box at the top of the tab displays the current measurement reading and units. This appears on all tabs.

Identifiers

This section allows the user to set the instrument identifiers that will be used over the RS485 serial bus. The default IDs are always '1'.

Modbus ID

For use when communicating to the instrument using Modbus (i.e., the RS485 'Comms Type' has been set to 'MB Slave'). The default ID is always '1', it can be changed by entering a new value in this box. The allowed range is 1 to 247. Please consult Dynoptic if you wish to use Modbus.

Multidrop ID

This is used to define the instrument ID when it is connected, via RS485, to a DSCU (i.e., the RS485 'Comms Type' has been set to 'DSL Slave'). The default ID is always '1'. This ID can be changed if more than one Dynoptic instrument is connected to the same DSCU.

Instrument Identifier

This box allows the user to set a unique name for the instrument identifier. This can be useful if a serial comms network is being used with several instruments connected. The default identifier is the instrument type.

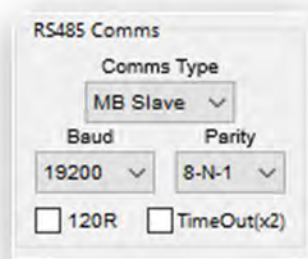
RS485 Comms

This section is used to configure the instrument to operate, as required, over the RS485 serial bus.

Comms Type

This drop-down menu allows the user to select the DSL-800 to act as an RS485 slave with either DSL ('DSL Slave') or Modbus ('MB Slave') protocol. The default setting is 'DSL Slave' which is required for the instrument to communicate with a DSCU.

Selecting Modbus ('MB Slave') modifies the screen as shown below.



For further information on using Modbus to communicate with this instrument please contact Dynoptic.

NOTE: Selecting Modbus ('MB Slave') will prevent any connected DSCU from communicating with the instrument.

Baud

This sets the Baud rate for the RS485 comms, by default it is '19200', this is the rate required for the instrument to communicate with a DSCU. For further information please contact Dynoptic.

Parity

This box is only visible when Modbus has been selected. A drop-down menu allows the user to select the data format as '8-N-1', '8-N-2', '8-E-1' or '8-O-1'. For further information please contact Dynoptic.

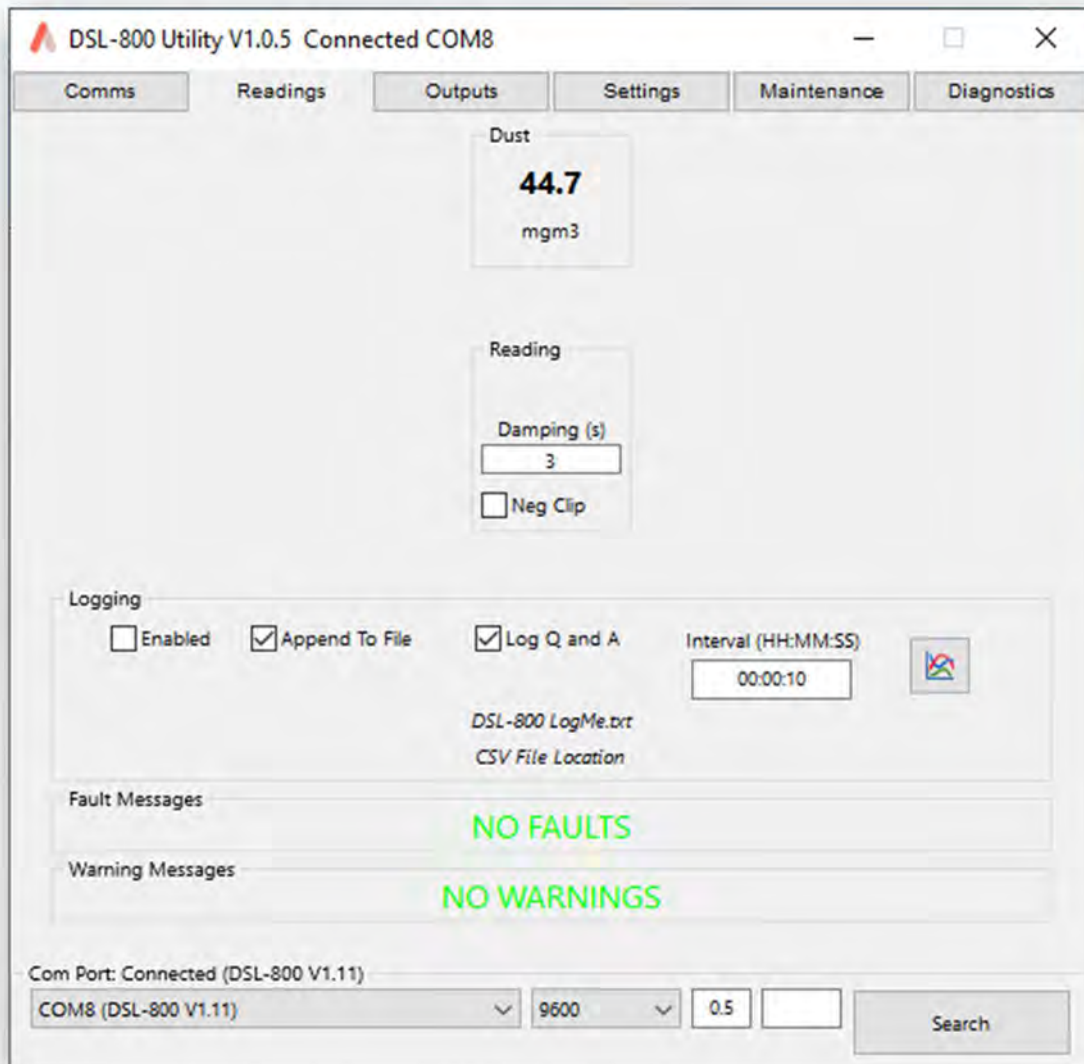
120R

Selecting this box terminates the RS485 line inside the instrument with 120 Ohms. The default setting is unselected, in this case the RS485 line is unterminated. With a long RS485 bus it is advantageous to terminate the furthest instrument on the bus with 120 Ohms. For further information please contact Dynoptic.

NOTE: *Further information on using the Modbus comms is provided in “Appendix B: Modbus Configuration” .*

The Readings tab

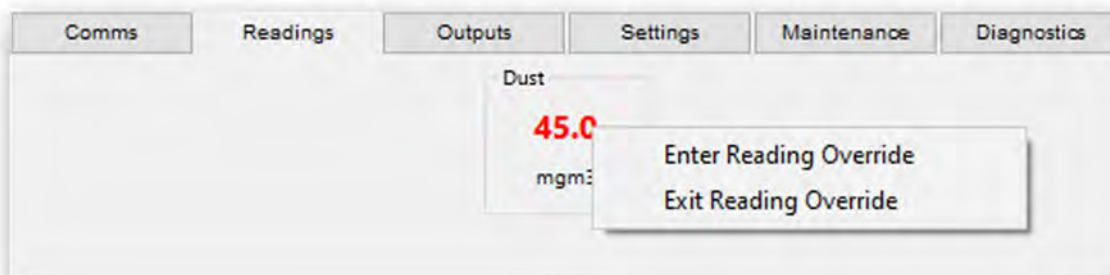
The Readings tab allows the user to view and alter the damping applied to the measurement reading and to select negative clipping. It also provides checkboxes for activating and controlling automatic data logging (which saves log files directly to the PC hard drive) and displays any current fault or warning messages.



Reading Override Mode

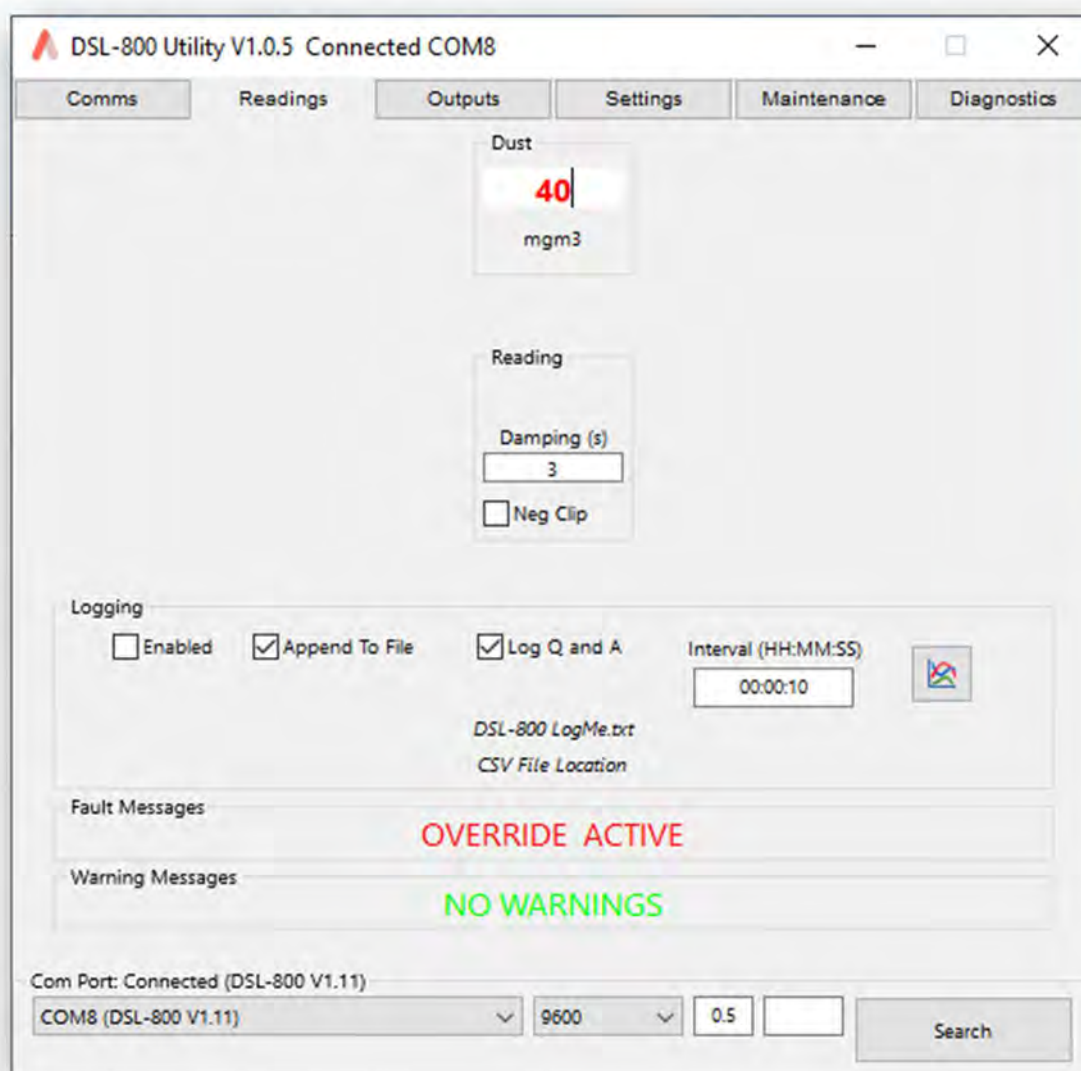
Right clicking on the measurement reading box produces a popup window (see screenshot). From this window select 'Enter Reading Override', this sets the instrument into the Reading Override Mode.

NOTE: Entering the Reading Override Mode activates the 'DATA NOT VALID' and 'OVERRIDE ACTIVE' fault messages.



When in this mode the measurement reading text box is highlighted in white (see below), the user can now manually select and edit the reading value.

Typing a value into the measurement reading box and then pressing enter forces the reading to the value entered. The screenshot below shows a manually entered reading of 40mg/m³.



This override function enables the user to check that the instrument outputs (e.g., Modbus or analogues and relays) are set up and responding as expected at the data acquisition system.

To exit the Reading Override Mode right click on the highlighted measurement reading and select 'Exit Reading Override' from the popup window. Alternatively, the instrument will automatically return to normal readings after 5 minutes of inactivity.

NOTE: Exiting the Reading Override Mode activates the 'DATA NOT VALID' and 'STABILISATION' fault messages for 1 minute.

Damping (s)

The value (in seconds) displays the level of damping (rolling average) applied to the reading. The default value is 3 seconds. Entering a new value in this box will change the damping applied, the allowed values are from 1 to 600 sec.

The response time of the instrument (T90%) to a step change in the reading is approximately three times the damping.

Neg Clip

If this box is UN-CHECKED (default) the instrument will display the true measurement reading, even if that reading is negative. Negative readings are possible during set-up, when the instrument is not calibrated, during stabilisation and during operation due to noise fluctuations.

If this box is CHECKED the instrument will clip the current measurement reading and any related analogue outputs at zero. Positive readings will show as normal but when the instrument calculates a negative reading, it will hold the display and the outputs at zero (0.0mg/m³).

Logging

This area allows the user to log the readings in real time onto the PC running the Utility and, if required, display the readings on a graph. This feature is useful when performing audit or calibration checks.

Logging - Enabled

When this box is CHECKED the measured values (defined by the 'LogMe.txt' file) will be logged in a CSV file. The files are stored in a folder called 'DSL-800', the location of this folder on the PC is given on the Diagnostics tab. The user must be logged onto the PC as an administrator for the logs to be stored and the logs will be created at intervals defined by the interval setting.

The default value logged in the CSV file is the reading in the selected units (mg/m³ or mg/Nm³). Contact Dynoptic to log other parameters.

Logging - Append to File

This box is CHECKED by default. When CHECKED the log file created will be a single

file using the electronic serial number of the instrument as a reference in the filename and each new log is appended within that file. A new file is created each day with the date forming part of the filename.

When the 'Append To File' box is UN-CHECKED, the logging will generate a new file at each logging interval, using the time, date and electronic serial number of the instrument as the file name.

The log file(s) can be opened in a text editor or spreadsheet and can be used for data logging, trend graphing or for diagnostics.

Log Reading – Log Q and A

This box is CHECKED by default. The logging function uses a “question and answer” system to retrieve data for the log. In this system the Utility software sends a question (an ASCII text string) to the instrument, which responds with an answer (usually a number).

When the 'Log Q and A' checkbox is UN-CHECKED, the log file only includes the answer, but when this box is CHECKED (default setting) the log file includes both the question and answer.

By default, the question that is asked (VAL?1) relates to the measurement reading only. However, with advice from Dynoptic, advanced users may modify the question so that the log file records additional or alternative data, for example, relay thresholds or analogue output scaling values. In these circumstances it is beneficial to log both the question and the answer so that the log file is more easily interpreted.

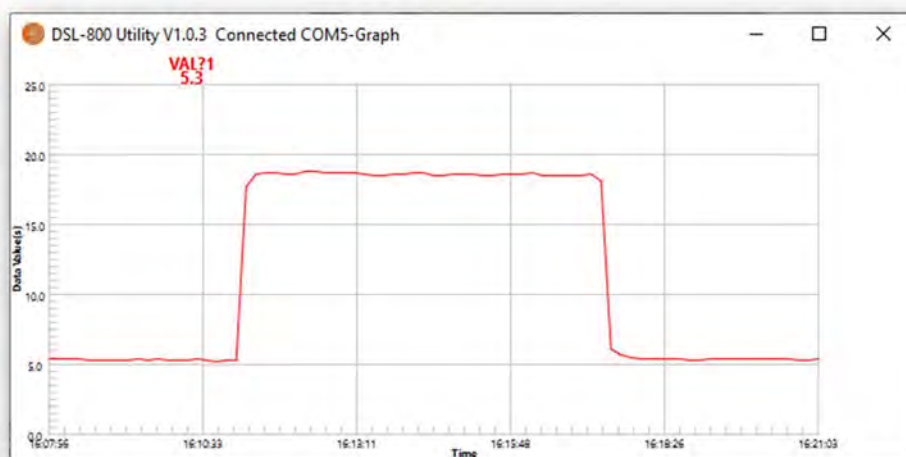
The file that is edited to modify the question is 'DSL-800 LogMe.txt'. Left clicking on the filename opens the current file in a text editor. Right clicking on the filename opens the location folder and allows the users to browse the PC to select a different 'LogMe' file. Please seek advice from Dynoptic before making any changes to this file.

Log Reading – Interval

The time interval between the creation/update of log files is determined here. The default setting is 10sec.

Graph Button

As well as a log file being created, the data can be viewed using the graphing feature. When logging is enabled click on the 'Graph Icon' a graph window will appear (see example screenshot below).



The default file 'DSL-800 LogMe.txt' asks the question 'VAL?1', the response stored in the CSV file is single number at each time interval and this is shown on the graph. If the 'DSL-800 LogMe.txt' file has been changed to log several parameters, only the first four parameters can be displayed on the graph.

To exit the graphing feature, close the graph window using the cross at the top right-hand corner.

To edit the graph (e.g., manually set the y-scale) right click on the graph and use the drop-down menu.

Fault Messages

Under normal operation the message that is displayed is 'NO FAULTS' in green. However, if the instrument detects a fault, the displayed message will switch between 'DATA NOT VALID' and a brief description of the fault, in red. If there is more than one fault, then the messages will switch through each fault description in turn.

Some fault messages will continue to be displayed on this tab until an operator clicks on the message to acknowledge it. If the fault is no longer active the message will disappear, but if the fault status is still active the message will re-appear immediately. See the section on "Fault and Warning Messages" later in this manual for more information.

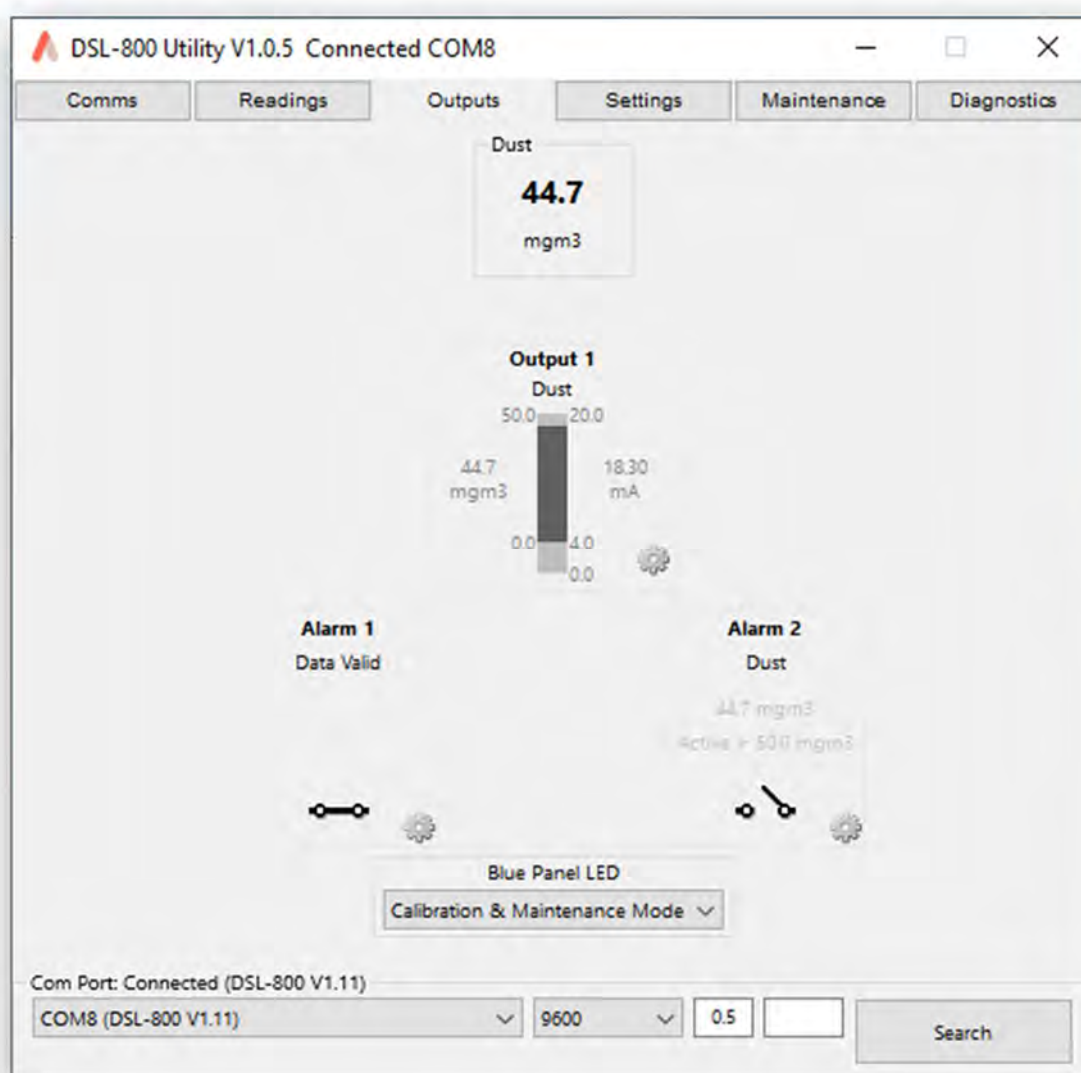
Warning Messages

Under normal operation the message that is displayed is 'NO WARNINGS' in green. If the instrument detects a parameter that is close to or out of range that needs investigating, a brief warning message will be displayed describing the problem, in red. The warning message will continue to be displayed on this tab until an operator clicks on the message to acknowledge it. If the warning is no longer active the message will disappear, but if the problem causing the warning is still active the message will re-appear immediately.

See the section on “Fault and Warning Messages” later in this manual for more information.

The Outputs tab

The Outputs tab allows the user to configure the electrical interface outputs (such as the analogue outputs and alarm relay contacts) and also shows the effects of those settings.

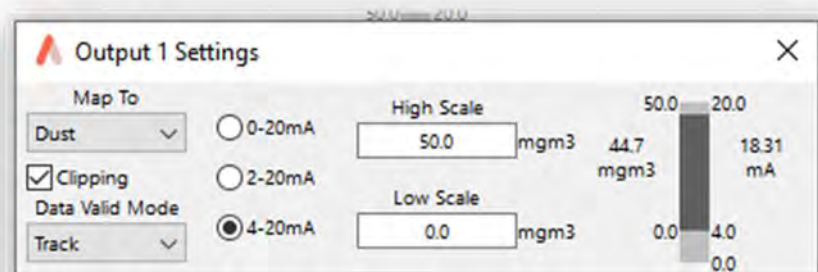


Output 1

The DSL-800 has one analogue output, and it is fully configurable.

The graphic in the 'Output 1' box provides a summary of the measurement parameter (e.g., Dust), the analogue range (e.g., 0 to 50 mg/m³), the reading (e.g., 44.7 mg/m³) and the corresponding analogue output current (e.g., 18.30 mA).

The analogue output settings are configured by selecting the ‘settings’ icon at the bottom right-hand corner of the ‘Output 1’ box. Selecting this icon opens up the following settings window.



Map To

Select from the drop-down menu the measurement you would like the analogue output to act on. The only options available in a DSL-800 are ‘Dust’ and ‘Disabled’ (switches the output off). The default setting is ‘Dust’.

NOTE: Any analogue output not being used should be set to ‘Disabled’.

Clipping

By default, this box is CHECKED. In this mode the analogue output will not respond to readings below the value set as ‘Low Scale’ but will send an analogue output equivalent to the lowest current for the scale selected.

If the box is UN-CHECKED then the analogue output will track the measured signal even if it is below the value set as ‘Low Scale’, but only as low as 0mA.

Data Valid Mode

This drop-down menu is used to select what happens to the analogue output in the event of the data being not valid. Both the actual and calculated analogue outputs are acted on. The available options are: -

Track – In the event of a DATA NOT VALID fault, the analogue output will continue to track the reading. This is the default setting.

Low – In the event of a DATA NOT VALID fault, the analogue output will be forced to the low scale value.

The analogue output will continue to be held at this value for the duration of the data not valid period.

When the data valid status is returned, the output will return to tracking the current measurement reading.

High – In the event of a DATA NOT VALID fault, the analogue output will be forced to the high scale value.

The analogue output will continue to be held at this value for the duration of the data not valid period.

When the data valid status is returned, the output will return to tracking the current measurement reading.

Hold – In the event of a DATA NOT VALID fault, the analogue output will be held at the last known output current.

The analogue output will continue to be held at this value for the duration of the data not valid period.

When the data valid status is returned, the output will return to tracking the current measurement reading.

0-20mA / 2-20mA / 4-20mA

The current loop analogue output can be configured to operate across any one of three scales: 0-20mA, 2-20mA or 4-20mA. Select the required scale using these radio buttons. The default setting is 4-20mA.

High Scale and Low Scale

These two parameters define the upper and lower scaling points of the 0/2/4-20mA output.

Set the low scale value to the reading at which the analogue output should generate 0mA, 2mA or 4mA (depending on selected scale point). This is set to 0.0 mg/m³ by

default.

Set the high scale value to the reading at which the analogue output should generate 20mA. This is set to 50.0 mg/m³ by default.

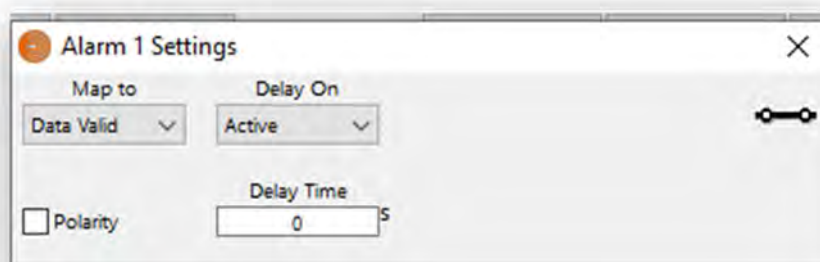
Alarms (Relays)

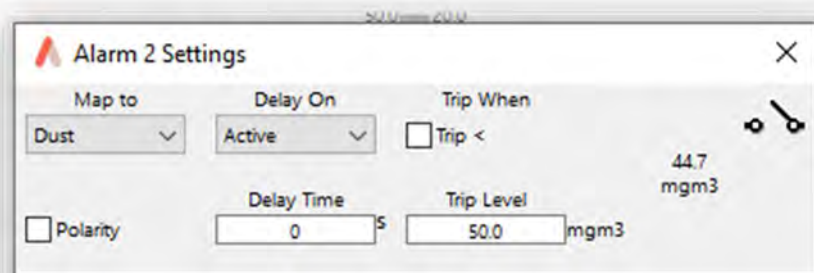
The DSL-800 has two relay alarm outputs, both are fully configurable.

In the below graphic Alarm 1 is set by default as a Data Valid relay in fail safe configuration (i.e., open relay means DATA NOT VALID). Alarm 2 is set as a threshold relay for Dust and the relay will close when the reading goes above 50 mg/m³.



The relay output settings are configured by selecting the 'settings' icon at the bottom right-hand corner of the 'Alarm' box. Selecting this icon opens up the settings window. The default settings windows for the two alarm relays are shown below.





Map To

Select from the drop-down menu the measurement you would like the relay to act on. The options available are 'Dust', 'Data Valid', 'Warnings', 'Maintenance Mode' and 'Disabled' (switches the relay off).

NOTE: Any relay not being used should be set to 'Disabled'.

The Alarm 1 settings window shows how it looks when the alarm relay is set to 'Data Valid'. The settings screen will be the same if 'Warnings' or 'Maintenance Mode' was selected. This relay will trip open (failsafe) when any fault is detected by the instrument.

The Alarm 2 settings window shows how it looks when the alarm relay is set to 'Dust'. This shows the threshold alarm settings screen. This relay will activate when the measured reading passes the set 'Trip Level'.

Polarity

The relay polarity boxes are UN-CHECKED by default.

When UN-CHECKED any Data Valid or Warning relay will be energised when the data is valid, and the relay status indicator will show a closed contact. In this polarity the relay will de-energise when the data becomes not valid (i.e., a fault). This condition represents the recommended failsafe operation.

When the checkbox is UN-CHECKED any Level alarm relay will be de-energised in a below threshold condition and the relay status indicator will show an open. In this polarity the relay will be energised when the reading passes the set 'Trip Level'.

With the checkbox CHECKED, the associated relay polarity will be reversed.

Delay On

The activation of an alarm relay can be delayed before triggering. The following options are available: –

Neither – The triggering of the alarm condition is not delayed at the start, or at the end, of the alarm condition.

Clear – The triggering of the alarm condition is not delayed at the start of the alarm condition but is delayed at the end i.e., when the alarm condition is clear.

Active (default) – The triggering of the alarm condition is delayed by the time set in the ‘Delay’ box. At the end of the alarm condition, the relay returns immediately to its previous state. This is the default setting.

Both – The triggering of the alarm condition is delayed at the start and end of the alarm condition.

Delay Time

This value is the continuous length of time (in seconds) that the relay is delayed by. The exact function is determined by the ‘Delay On’ condition.

Delaying the activation of the relay can be used to prevent borderline level threshold changes from “dithering” the relay state. Only genuine, sustained readings in excess of the threshold will actually trigger the alarm.

Trip When (<)

This tick box only appears when the relay is being used as a reading threshold alarm. By default, it is UN-CHECKED which means the relay is activated when the measurement reading is greater than the set ‘Trip Level’.

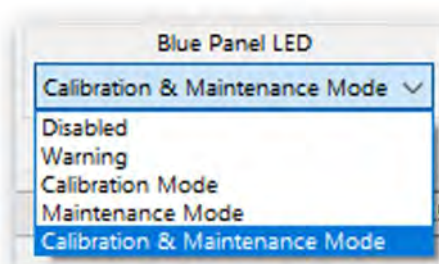
When the box is CHECKED the relay is activated when the measurement reading is less than the ‘Trip Level’.

Trip Level

This parameter defines the measurement reading at which the alarm relay will trip or activate. The default setting is 50.0 mg/m³.

Blue Panel LED

This drop-down menu is used to define the operation of the Blue LED on the rear panel of the instrument. The Green LED will illuminate when instrument is powered. The Yellow LED will illuminate if there is a fault (Data Not Valid). The operation of the Blue LED can be configured by the user using this drop-down menu.



Disabled – The Blue Led is always turned off

Warning – The Blue LED will turn on if the instrument has identified a Warning. See the section on “Fault and Warning Messages” later in this manual.

Calibration Mode – The Blue LED will turn on when the instrument is in ‘Calibration Mode’.

Maintenance Mode – The Blue LED will turn on when the instrument is in ‘Maintenance Mode’.

Calibration & Maintenance Mode – The Blue LED will turn on when the instrument is either in ‘Calibration Mode’ or ‘Maintenance Mode’. This is the default setting.

The Settings tab

NOTE: The Settings tab should only be used by a trained instrument engineer who has read and understood the later section on “Set Up and Commissioning”. The settings and controls on this tab determine the calibration of the instrument and the measurement reading that it generates.

The Settings tab provides the settings and controls for the initial set up of the instrument.

The screenshot shows the 'DSL-800 Utility V1.0.5 Connected COM8' window with the 'Settings' tab selected. The interface includes several sections for configuration:

- Comms:** A dropdown menu showing 'COM8 (DSL-800 V1.11)'.
- Readings:** A display showing 'Dust' with a value of '44.0' and units 'mgm3'.
- Outputs:** A section for 'Signal Strength' with input fields for 'Signal (V)' (0.198) and 'Reference Signal (V)' (4.195).
- Set Up:** A section for 'mg/m3 Units' with radio buttons for 'mg/m3' (selected) and 'mg/Nm3'.
- Linearisation:** A section with radio buttons for 'Linear' (selected), 'Logarithmic', 'Exponential', 'Polynomial', and 'Power'. It also includes a formula $Y = a0 + (a1 * X)$ and input fields for 'a0' (0.0) and 'a1' (1.00).
- Bottom Bar:** A status bar indicating 'Com Port: Connected (DSL-800 V1.11)' and a 'Search' button.

Signal Strength – Signal (V)

This box displays the strength of the backscatter signal measured by the instrument. This reading is provided as a diagnostic tool.

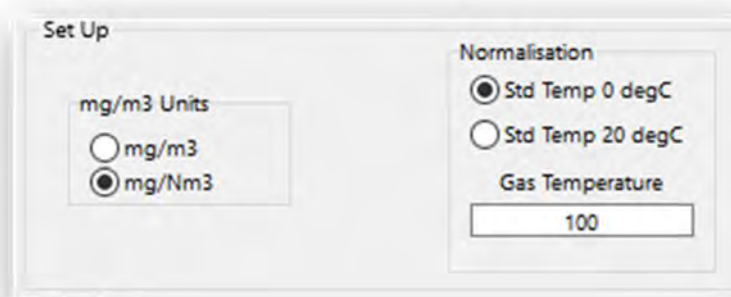
Signal Strength – Reference Signal (V)

This box displays the strength of the laser reference signal and is a measure of the brightness of the source laser. This reading is provided as a diagnostic tool.

Set Up – mg/m³ Units

These tick boxes allow the user to select the units to be used to display the measured particulate concentration. The options are milligrams per meter cubed (mg/m³) or milligrams per normalised meter cubed (mg/Nm³). The default setting is mg/m³.

If mg/Nm³ is selected some new options appear in the window, as shown below. These allow the user to select the reference or standard gas temperature to be used for the normalisation (either 0 degC or 20 degC) and to manually input a typical value for the actual gas temperature (in degC) in the stack (default value is 100 degC).



Linearisation

NOTE: This section should only be used by a trained instrument engineer who has read and understood the later section on “Set Up and Commissioning”. The settings in this section determine the calibration of the instrument and the measurement reading that it generates.

This section allows different linearisation functions to be applied to the reading. In most applications the ‘Linear’ function will be used, and this is the default setting.

However, if a standard reference method has been used to perform a multi-point correlation test, then an alternative linearisation functions could be used to provide a best fit to the measured correlation.

NOTE: The simple linear function provides the best correlation in most applications. Non-linear functions are only required at very high dust levels.

When a linearisation function is selected the equation used is shown on the Utility and the values of the coefficients (a_0 , a_1 and/or a_2) are displayed in boxes for editing. Note if a coefficient is changed in one linearisation function it will keep the same value even if the linearisation function is changed.

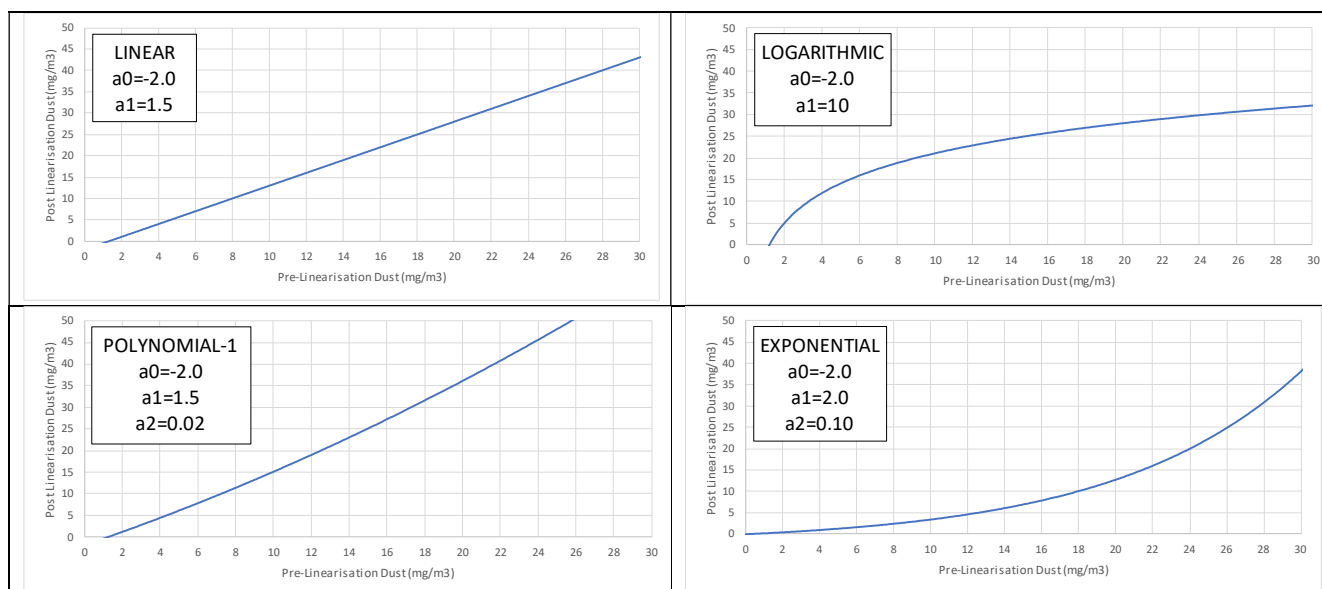
The default values for the linearisation coefficients are

$a_0=0.0$

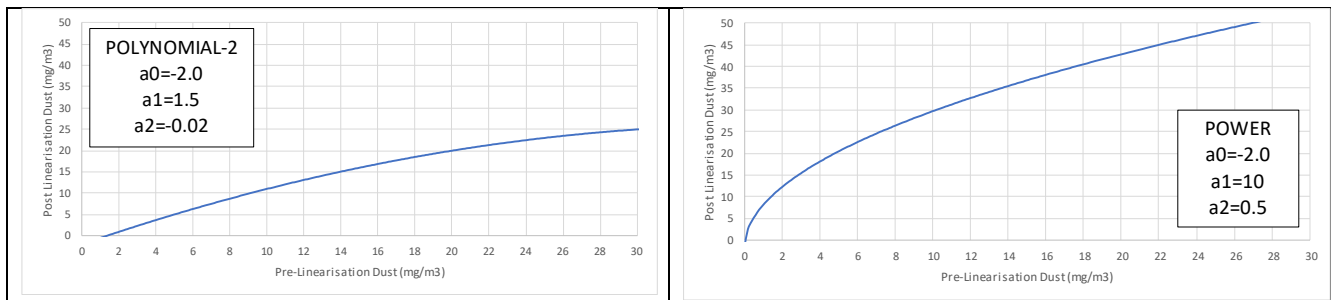
$a_1=1.00$

$a_2=0.00$

The following plots illustrate the effect of applying the various linearisation functions to the measured dust level. In these plots the x-axis is the measured or pre-linearised dust level (range shown 0 to 30 mg/m³) and the y-axis is the output or post-linearised dust level (range shown 0 to 50 mg/m³).



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The Maintenance tab

NOTE: The Maintenance tab should only be used by a trained instrument engineer who has read and understood the later section on “General Maintenance”. The settings and controls on this tab determine the calibration of the instrument and the measurement reading that it generates.

The Maintenance tab provides the settings, functions and controls required for routine maintenance and calibration checking of the instrument.

The screenshot shows the 'DSL-800 Utility V1.0.5 Connected COM8' window with the 'Maintenance' tab selected. The interface includes several sections for instrument configuration and monitoring.

Top Navigation: Comms, Readings, Outputs, Settings, Maintenance (selected), Diagnostics.

Dust Reading: A large display shows '43.7 mgm3'.

Mode Selection: Two checkboxes are present: 'Maintenance Mode' and 'Calibration Mode', both currently unchecked.

Stabilising Time (Seconds): A numeric input field is set to '0'.

Backup to PC: A button labeled 'Backup to PC' is located on the right side.

Zero and Span Check Section:

- Span Value:** A numeric input field is set to '50.0'.
- Manual Check:** Two buttons, 'Zero' and 'Span', are provided for manual calibration.
- Automatic Check:**
 - An 'Enable Auto' checkbox is unchecked.
 - An 'Interval (Hrs)' numeric input field is set to '24'.
 - A 'Disabled' status indicator is shown below the interval field.

Drift Section:

- Zero Drift (mg/m3):** A numeric input field is set to '0.0'.
- Span Drift (%):** A numeric input field is set to '0.0'.

Output Function: A dropdown menu is set to 'Hold'.

Bottom Section:

- Com Port:** A status indicator shows 'Connected (DSL-800 V1.11)'.
- Port Selection:** A dropdown menu is set to 'COM8 (DSL-800 V1.11)'.
- Baud Rate:** A dropdown menu is set to '9600'.
- Parity:** A dropdown menu is set to '0.5'.
- Search:** A button labeled 'Search' is located at the bottom right.

Maintenance Mode

Selecting this tick box will set the instrument into maintenance or safe mode.

NOTE: *Entering the Maintenance Mode activates the ‘DATA NOT VALID’ and ‘MAINTENANCE MODE’ fault messages.*

This mode turns off the laser and should be used when any maintenance is performed on the instrument such as cleaning. See later section on “General Maintenance”.

The box must be UN-CHECKED to turn the instrument back into normal reading mode. After unchecking this box, the instrument will activate the ‘DATA NOT VALID’ and ‘STABILISATION MODE’ faults for 1 minute. If the instrument is power cycled Maintenance Mode will be automatically turned off.

NOTE: *Exiting the Maintenance Mode activates the ‘DATA NOT VALID’ and ‘STABILISATION’ fault messages for 1 minute.*

NOTE: *If the Utility is disconnected from the instrument without exiting Maintenance Mode the instrument will automatically exit Maintenance Mode after 1 hour.*

Calibration Mode

Selecting this tick box will set the instrument into Calibration Mode.

NOTE: *Entering the Calibration Mode activates the ‘DATA NOT VALID’ and ‘CALIBRATION MODE’ fault messages.*

This mode should be used when the audit filter kit is used to perform a calibration check on the instrument. See “Appendix A Performing and Audit Check”.

The box must be UN-CHECKED to turn the instrument back into normal reading mode. After unchecking this box, the instrument will activate the ‘DATA NOT VALID’ and ‘STABILISATION MODE’ faults for 1 minute. If the instrument is power cycled Calibration Mode will be automatically turned off.

NOTE: *Exiting the Calibration Mode activates the ‘DATA NOT VALID’ and ‘STABILISATION’ fault messages for 1 minute.*

NOTE: *If the Utility is disconnected from the instrument without exiting Calibration Mode the instrument will automatically exit Calibration Mode after 1 hour.*

Stabilising Time (Seconds)

This number displays the remaining stabilisation time of the instrument in seconds. When the instrument is first powered up, or it is switched out of Maintenance Mode, or Calibration Mode or Reading Override Mode, the instrument goes into Stabilisation Mode for a period of time ranging from 1 to 20 minutes. During stabilisation it is not possible to perform some of the functions on the Maintenance tab. This counter tells you how long to wait (in seconds) before the instrument stabilisation is complete.

On power up the default stabilisation time is 20 minutes. However, if the power off time has been very short (e.g., less than 10 minutes) then the stabilisation will actually turn off after only 5 minutes.

Backup to PC

Clicking this button will cause the current instrument settings to be saved in a backup text file. The location of this file on the PC is shown in the 'folder location' box on the Diagnostics tab. The filename of the backup file will incorporate the electronic serial number of the interface board, the date, and the time of the backup.

It is highly recommended that this button is used to take a backup of the instrument settings after successfully completing the set up and commissioning, before and after any maintenance and if there is an instrument fault. The backup file can be used to restore the instrument settings, should the live settings ever be lost or become corrupted, they can also be used to help diagnose faults or installation issues.

NOTE: *A backup file should be downloaded and archived immediately after the successful set up and commissioning of this device, or whenever the settings and/or parameters are permanently altered.*

Zero and Span Check

The DSL-800 has the ability to perform an internal, electronic, zero and span check. The zero check involves turning off the laser and the span check injects a fixed, known amount of light into the detection circuit. These tests check the instrument for any internal long-term drifting but does not measure any effects due to dirt build up on the optics. The external instrument performance can be checked using the

manual external audit check, which requires the optional calibration head and audit filters (see Appendix A).

Span Value

This box shows the set span value used in the internal span check. The default setting is 50 mg/m³. This span value can be changed by manually entering the required value into this box.

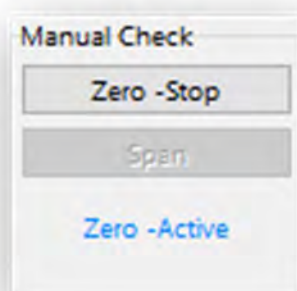
The maximum and minimum allowed setting for this span value will depend on the linearisation settings. If a span value entered that is outside of the allowed range the value will change to the closest allowed value.

NOTE: Do not change the Span Value if the instrument is performing a manual span check or an automatic zero and span check.

Manual Check

It is possible to perform a manual zero check or a manual span check by selecting one of these buttons. If the button is greyed out, then the function is inactive (e.g., during stabilisation).

Selecting the 'Zero' button turns on the manual zero check. A 'Zero-Active' message appears under the buttons, the 'Span' button is greyed out and the 'Zero' button changes to 'Zero-Stop'.

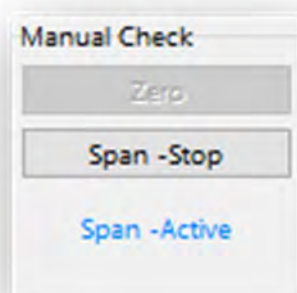


To exit the manual zero check, press the 'Zero-Stop' button. It may take a few seconds for the zero routine to finish.

NOTE: Selecting the Zero Manual Check activates the 'DATA NOT VALID' and 'ZERO CHECK ACTIVE' fault messages.

NOTE: Exiting the Zero Manual Check will briefly activate the 'STABILISATION' fault message.

Selecting the 'Span' button turns on the manual span check. A 'Span-Active' message appears under the buttons, the 'Zero' button is greyed out and the 'Span' button changes to 'Span-Stop'.



To exit the manual span check, press the 'Span-Stop' button. It may take a few seconds for the span routine to finish.

NOTE: Selecting the Span Manual Check activates the 'DATA NOT VALID' and 'SPAN CHECK ACTIVE' fault messages.

NOTE: Exiting the Span Manual Check will briefly activate the 'STABILISATION' fault message.

Output Function

This drop-down menu can be used to set the instrument reading to either 'Hold' or 'Track' during a zero or span check.

If 'Hold' is selected then during the zero and span check the instrument output will be held at the last reading before the routine started. This is the default setting.

If 'Track' is selected then the outputs and displays from the instrument will track the readings that occur during the zero and span check.

Automatic Check

The DSL-800 can be set to perform an automatic, internal, zero and span check. This

check takes about 50 seconds, and it can be set to automatically repeat at a fixed time interval. At the end of the automatic check the measured zero drift (mg/m³) and span drift (%) are displayed.

Interval (Hrs)

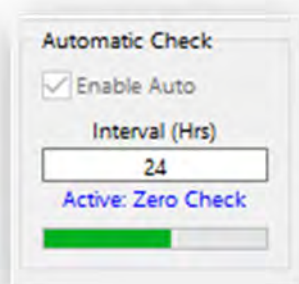
Sets the frequency of the automatic calibration checks in hours (1 to 1000), the default setting is 24 hours. Only change this value when the automatic zero and span check is disabled.

NOTE: The accuracy of the internal clock means that the actual time interval can drift by up to ± 30 minutes / year.

Enable Auto

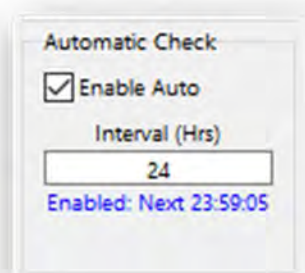
Selecting the 'Enable Auto' tick box starts the automatic zero and span calibration check routine. Please make sure you have selected the correct 'Span Value', set the required 'Output Function' and set the correct 'Interval (Hrs)' before enabling the calibration check. The first zero and span check will start immediately and takes around 50 seconds to complete.

A progress bar is displayed during the automatic zero and span routine.



NOTE: During the automatic zero and span check the 'DATA NOT VALID' will be activated with the messages 'ZERO CHECK ACTIVE' followed by 'SPAN CHECK ACTIVE'.

At the end of the routine a countdown timer appears showing the time until the next automatic zero and span check.



NOTE: *If power is lost to the instrument with 'Enable Auto' selected, on restoration of power the automatic zero and span check will re-start 1 hour after the 'STABILISATION MODE' fault clears.*

Zero Drift (mg/m³)

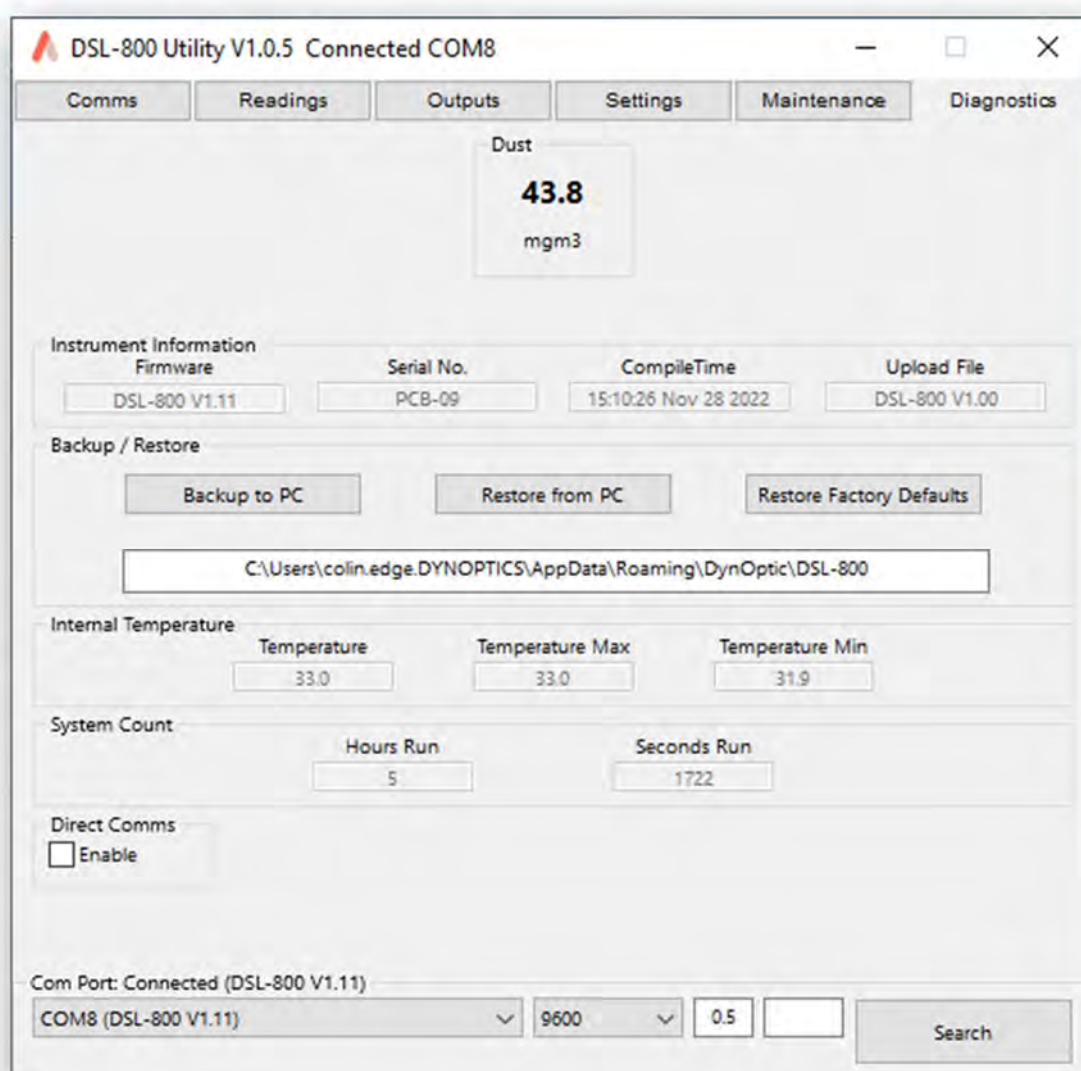
This displays the zero drift, in units of mg/m³, as measured at the most recent automatic zero check. The measured zero drift will typically be <1mg/m³.

Span Drift (%)

This displays the span drift as a percentage of the span value, as measured at the most recent automatic span check. The measured span drift will typically be <±2%. If the measured span drift is >±90% the warning message 'HI SPAN DRIFT' will be triggered.

The Diagnostics tab

The Diagnostics tab has a number of information only boxes and some features for backup and restore. See the following screenshot.



Instrument Information

This section provides basic information about the CPU board within the instrument and its firmware version.

Firmware

This provides information about the version of the firmware in the CPU board.

Serial No.

This is the serial number of the CPU board.

Compile Time

This is the date/time on which the firmware was compiled.

Upload File

This provides information about the version of the upload file used to define basic settings in the CPU board.

Backup / Restore

This section allows the user to save a backup file or to restore the instrument settings using an old backup file.

Backup to PC

Clicking this button will cause the current instrument settings to be saved in a backup text file, the location of this file is shown in the 'folder location' box. The filename of the backup file will incorporate the electronic serial number of the interface board, the date, and the time of the backup.

It is highly recommended that you use this button to take a backup of your instrument settings after successfully completing the set up and commissioning, before and after any maintenance and if there is an instrument fault. The backup file can be used to restore the instrument settings, should the live settings ever be lost or become corrupted, they can also be used to help diagnose faults or installation issues.

NOTE: A backup file should be downloaded and archived immediately after the successful set up and commissioning of this device, or whenever the settings and/or parameters are permanently altered.

Restore from PC

Clicking this button will allow you to choose a backup file from which to restore your instrument settings.

NOTE: Restoring from a backup file will permanently overwrite ALL settings and parameters in the instrument.

NOTE: A reserve backup file should be downloaded and archived immediately prior to restoring a previous backup file.

Restore Factory Defaults

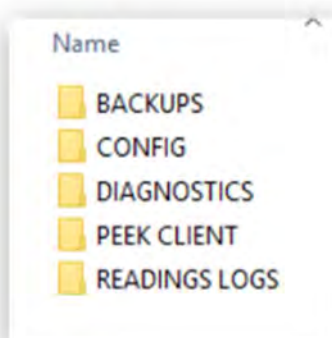
Clicking this button will restore all settings and parameters to default values as determined at our factory.

NOTE: Restoring factory defaults will permanently overwrite ALL settings and parameters in the instrument.

NOTE: A reserve backup file should be downloaded and archived immediately prior to restoring factory defaults.

Folder Location

This box shows the location on the PC of the folder that contains all of the data stored by the Utility. Left clicking on this filename takes you to the DSL-800 folder location on the PC, Right clicking opens the folder to show the following subfolders.



BACKUPS: Is where the instrument backups are stored (see 'Backup to PC' on Diagnostics tab)

READINGS LOGS: Is where the reading logs are stored (see 'Log Reading' on Readings

tab)

Additional folders, such as CONFIG, DIAGNOSTICS and PEEK CLIENT may also be present, but they will be empty.

System Count

System count shows the cumulative total time (in hours and seconds) that the instrument has been powered.

Direct Comms

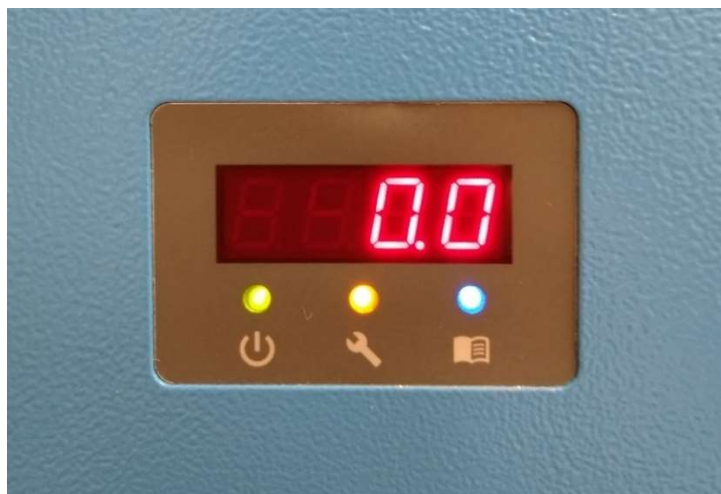
Selecting the 'Enable' box opens up an input box, as shown below, which provides direct communication access to the instrument. This can be used to read and edit additional system parameters. This feature should only be used with help from Dynoptic.



Instrument Display

The DSL-800 contains a display on the back of the instrument. This consists of a four-digit LED display and three coloured LED status indicator lights.

The display shows the live reading as measured by the instrument. This is the same reading that will be provided by any of the instrument outputs (analogue or Modbus) and displayed on the Utility or a connected DSCU.



The display has 4 digits and so shows a maximum reading of 9999 mg/m³.

The status LEDs are as follows:

GREEN=Power. LED is on when power is applied to the instrument.

YELLOW=Fault. LED is on when an instrument fault has been detected ('DATA NOT VALID').

BLUE=User configurable. Default setting is a Maintenance Mode or Calibration Mode status LED.

NOTE: Entering the Maintenance or Calibration Mode also activates 'DATA NOT VALID' and so both the Yellow and Blue indicator LEDs will be turned on.

It is possible to change the function that triggers the BLUE indicator LED on the 'Outputs Tab' of the Utility.

Set Up and Commissioning

WARNING: When powered this instrument emits a visible laser beam (Class 3R). Avoid direct eye exposure. Always disconnect the power (or turn on Maintenance Mode) before removing the instrument for repair or maintenance.

WARNING: It is essential that all components of the instrument have been properly installed, both physically and electrically, before proceeding with the set up and commissioning.

CAUTION: Care must be taken when working on a stack. In particular hot surfaces and hot stack gases. Take all necessary precautions.

NOTE: Applying power to the instrument activates the 'DATA NOT VALID' and 'STABILISATION MODE' fault messages. This can last for 20 minutes.

The DSL-800 is a single-sided instrument and so does not require any precise angular alignment to the stack. The minimum set up and commissioning is simply to configure the outputs as required by the data acquisition system.

If accurate readings of particle density are required it will be necessary to calibrate the instrument against a standard reference measurement (SRM), such as isokinetic sampling (e.g., ISO 9096).

The list below is an overview of the steps required to set up and commission a DSL-800. Detailed descriptions of each stage are provided later.

1. **Configure the Instrument** – When the instrument has been installed and running for at least 20 minutes, the settings and outputs can be configured as required for integration into the wider plant systems. This requires either setting the Modbus comms or scaling the analogue and alarm outputs.
2. **Test the Interface** – After configuration the integration of the instrument into the plant data acquisition system can be checked using the Reading Override Mode.

3. **Calibrate the Instrument** – Three different types of ‘calibration’ can be performed to commission the DSL-800. A simple zero offset compensation, a zero and single-point span calibration or a full, multi-point linearisation correlation using an independent reference measurement (such as isokinetic sampling). The completion of the calibration is to manually set the slope (a_1) and intercept (a_0) parameters of the ‘Linear’ output function.
4. **Automatic Zero and Span Check** – The automatic zero and span check function should only be enabled after the instrument has been configured and calibrated. This is generally set to operate every 24 hours.
5. **Save a Backup File** – When the set up and commissioning is complete a backup file should be saved to provide a record of the instrument settings. This backup file can be used to restore the instrument settings, should the live settings ever be lost or become corrupted.

Configure the Instrument

The DSL-800 provides the user with a range of settings and output options, which can be configured to match the requirements of the data acquisition system. These settings can all be seen and adjusted using the walk-up USB comms and Utility software.

Detailed descriptions of all of the adjustable parameters are provided in the section “Using the Utility Software”. The following is a brief overview of the more common configured parameters.

Damping (Readings Tab)

The damping time is the rolling average applied to the reading. The T90% response time is approximately three times the damping. The default damping is 3 sec and so the default response time is ~10 sec.



This value can be changed by manually entering a new number into the 'Damping (s)' box on the Readings tab of the Utility.

Negative Clipping (Readings Tab)

If negative clipping is turned on, then the instrument will automatically clip any negative reading to 0.0 mg/m³. The default setting is negative clipping turned off, so the instrument provides a true reading (positive and negative) with a live zero.



The status can be changed by checking the 'Neg Clip' box on the Readings tab of the Utility.

Analogue Output (Outputs Tab)

If the data acquisition system requires the reading to be supplied as an analogue output the settings can be configured from the Outputs tab of the Utility. The instrument has only one analogue output and its default settings are shown in the following screenshot.

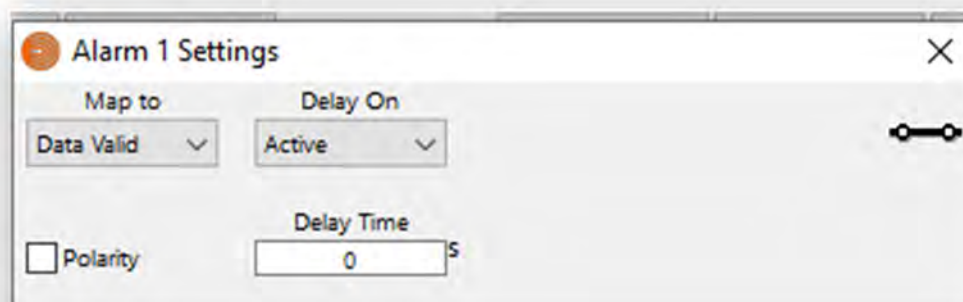


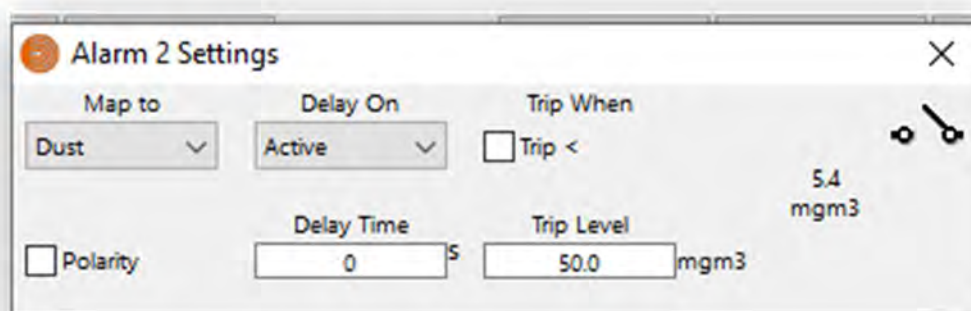
The parameters in this 'settings' window can be changed to match the requirements of the data acquisition system. The default example shown provides a 4 to 20 mA reading scaled to 0 to 50 mg/m³. The 'Clipping' has been selected and so the analogue output will not go below 4 mA.

NOTE: If the analogue output is not being used it should be mapped to 'Disabled'.

Alarm Relays (Outputs Tab)

The DSL-800 has two alarm relay outputs that can be configured from the Outputs tab of the Utility. The default settings are shown in the following screenshots.





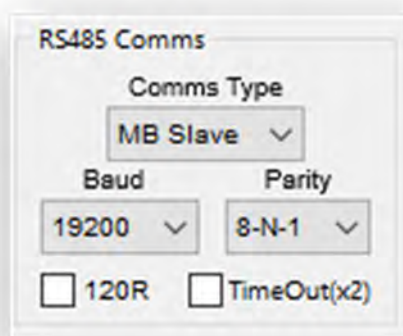
The parameters in these two ‘settings’ windows can be changed to match the requirements of the data acquisition system. The default examples show Alarm 1 as a ‘Data Not Valid’ (fault) relay, set in failsafe mode. Alarm 2 is set as a threshold relay that will close if the reading goes above 50 mg/m³.

NOTE: Any alarm relay not being used should be mapped to ‘Disabled’.

Modbus RS485 (Comms Tab)

If the data acquisition system requires the use of Modbus over RS485 as the output from the DSL-800 then the settings can be configured from the Comms tab of the Utility.

The default RS485 setting is ‘DSL Slave’, to set it to Modbus use the drop-down menu ‘Comms Type’ and select ‘MB Slave’. The standard Modbus settings are shown in the screenshot below. These can be configured as required.



For further information on using Modbus to communicate with this instrument and details of the Modbus tables please contact Dynoptic.

NOTE: Selecting Modbus (MB Slave) will prevent any connected DSCU from communicating with the instrument.

NOTE: Further information on using the Modbus comms is provided in “Appendix B: Modbus Configuration” .

Units mg/m^3 or mg/Nm^3 (Settings Tab)

By default, the DSL-800 provides the reading in mg/m^3 . Some applications require this reading to be scaled from the actual stack gas temperature to a standard temperature (either 0 degC or 20 degC).

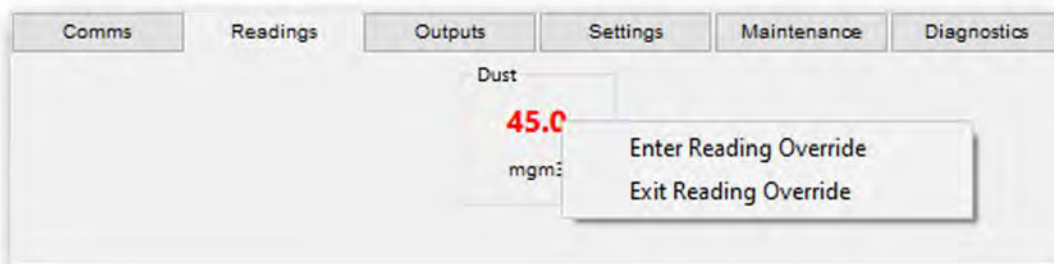
Selecting ‘ mg/Nm^3 ’ on the Settings tab of the Utility enables the user to select the required standard temperature (0 degC or 20 degC) and also enter the actual stack gas temperature. The reading is then automatically scaled from mg/m^3 to mg/Nm^3 .



Test the Interface

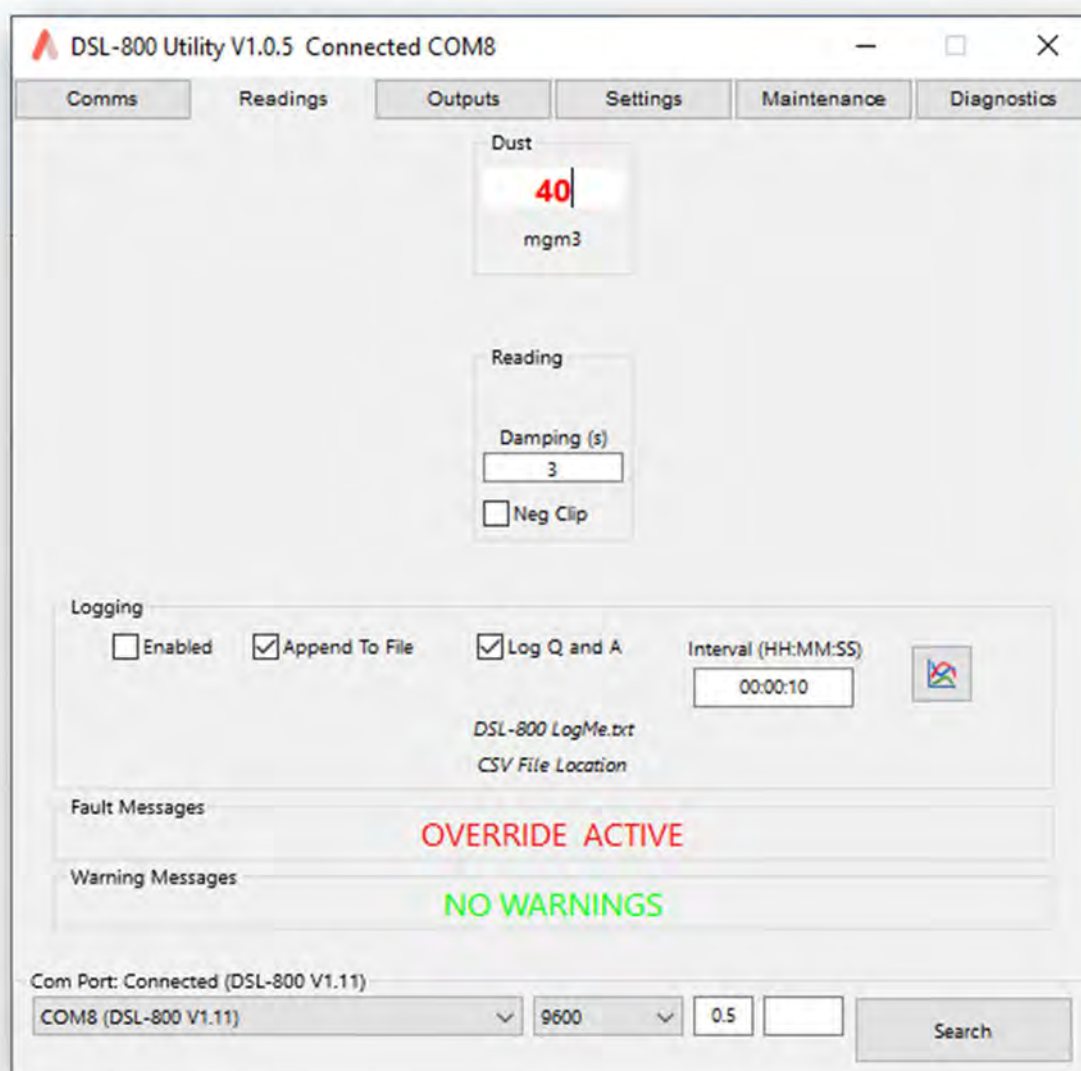
The Reading Override Mode can be used to send different signal levels from the DSL-800 to the data acquisition system to confirm correct operation.

To enter Reading Override Mode, use the Readings tab of the Utility and right clicking on the measurement reading box. A popup menu will appear, from this select ‘Enter Reading Override’.



The measurement reading text box will be highlighted in white (see below). Typing a value into this measurement reading box and then pressing enter forces the reading to the value entered. This feature can be used to confirm that the same reading value appears at the data acquisition system, setting a high value can confirm correct operation of any threshold alarm relays.

The following screenshot shows a manually entered reading of 40mg/m³.



To exit the Reading Override Mode right click on the highlighted measurement reading and select 'Exit Reading Override' from the popup menu. Alternatively, the instrument will automatically return to normal readings after 5 minutes of inactivity.

Calibration of the Instrument

The relationship between the measured backscatter signal and particulate concentration is approximately linear but the slope (α_1) will vary for each installation, depending on the physical properties of the dust particles in the gas stream (e.g., shape, size and colour).

The default settings for the DSL-800 have been set for typical stack dust. However, if

accurate particulate concentration readings are required the slope (a_1) value must be determined on site, through comparison of the instrument readings with an independent reference measurement.

The nationally recommended standard reference measurement (SRM) method should be used. This is likely to be a gravimetric analysis based on isokinetic sampling of the gas stream (e.g., ISO 9096).

The following summarises three alternative calibration approaches, of increasing complexity.

Calibration 1: Zero Offset

The simplest method is not to perform a span check, instead the default slope value ($a_1=1.00$), or an estimate of the expected slope setting, is used. This method can be used in applications where only indicative measurements of particulate concentration are required.

On the Settings tab select 'Linear' make sure the intercept parameter, a_0 , is set to 0.0, and that the slope parameter, a_1 , is set to the required value (e.g., 1.00).

Linearisation

☒ Linear
☐ Logarithmic
☐ Exponential
☐ Polynomial
☐ Power

$Y = a_0 + (a_1 \cdot X)$

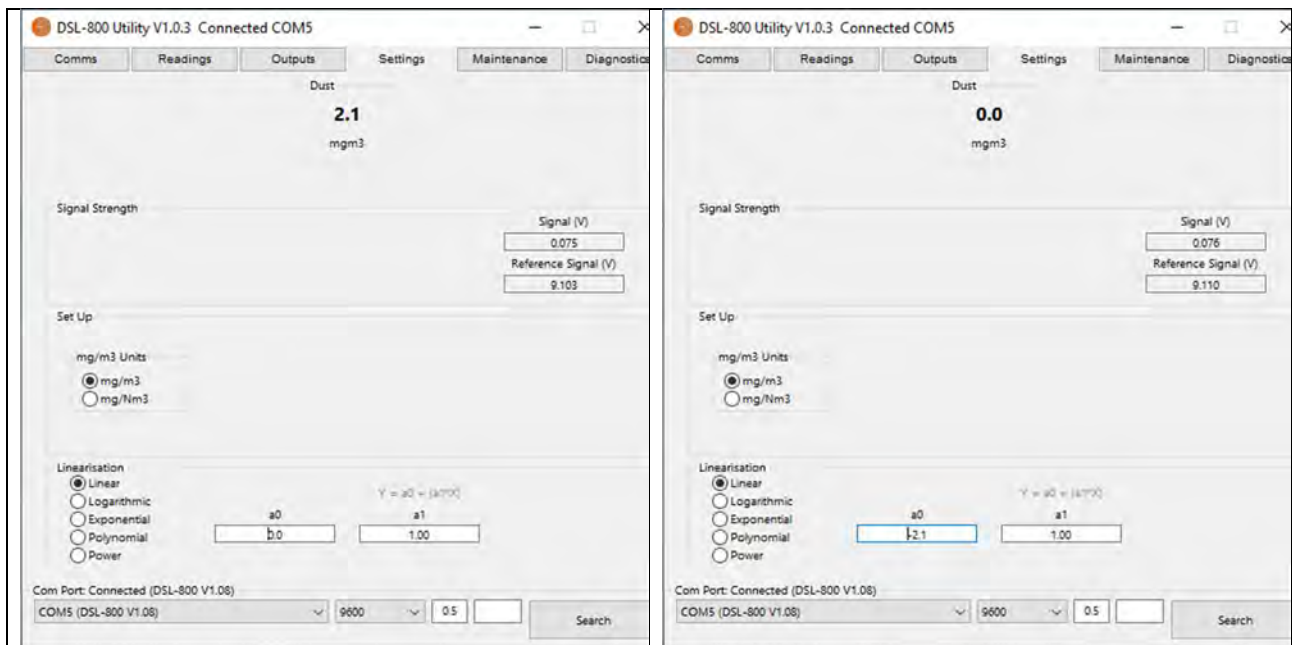
a_0	a_1
0.0	1.00

Make sure the stack is clear of dust (e.g., the process turned off), record the zero reading over 10 to 20 minutes and calculate the average zero offset reading (Z_M). It is likely that this reading will be slightly positive, due to internal electrical noise and spurious optical scattering inside the stack.

The measured zero offset reading (Z_M) should be less than $10\text{mg}/\text{m}^3$, a high zero reading could be due to several reasons.

- TRX Head not correctly earthed.
- TRX Head is too close to the far stack wall for the instrument path length variant.
- Blockage or build-up of dirt inside the flange tube.
- TRX head mounting nuts are not uniformly tightened.
 - This could result in laser beam not being parallel to the flange tube and so could be clipping the inside of the flange tube.

To remove the measured zero offset reading (Z_M) set the intercept parameter, a_0 , to the negative of the average zero offset reading. This is illustrated in the following two screen shots where the average zero offset reading is $Z_M=2.1 \text{ mg/m}^3$ and so the intercept parameter has been set to $a_0=-2.1$.



Calibration 2: Zero- and Single-Point Span

A more typical calibration approach is to measure both a zero and span point and use these to derive the two linear coefficients, a_0 and a_1 .

The average zero reading is measured as described above to give Z_M .

A standard reference method should be used to perform the span measurement. The reference measurement must be taken at the same time, and from a place in the stack very close to the DSL-800, so that the two measurements are coincident.

The coincident measurements should be performed when the particulate

concentration in the stack is at its highest possible level, since this improves the accuracy of the calibration. Calibrations performed with reference measurements close to 0 mg/m³ are not valid.

In order to perform a particulate concentration calibration, the following conditions for the DSL-800 must be met:

1. The instrument must have been properly and permanently installed.
2. The instrument must have been correctly configured (see above).
3. The instrument has been powered up and left to settle for a minimum of 1 hour.
4. The plant must be running and the particulate concentration in the stack must be at a high level.

The sampling probe for the standard reference measurement must be inserted into the duct close to, but not actually in, the optical path of the DSL-800. In this way it will be sampling a very similar gas stream.

During the time period over which the reference sample is taken (typically 30 mins) the particulate concentration displayed by the DSL-800 instrument should be logged and the average reading calculated (PC_M).

If the particulate concentration measured by the standard reference method is PC_{REF} then the new intercept (α0₂) and slope (α1₂) values can be derived using the following expressions

$$\alpha_{1_2} = \alpha_{1_1} \times PC_{REF} / (PC_M - Z_M)$$

$$\alpha_{0_2} = (\alpha_{0_1} - Z_m) \times (\alpha_{1_2} / \alpha_{1_1})$$

Where:

Z_M = the average zero level measured by the DSL-800

PC_M = the average span measured by the DSL-800

PC_{REF} = the average span level measured by the SRM.

α0₁ = is the starting intercept value (default setting =0.0).

α0₂ = is the new calibrated intercept value.

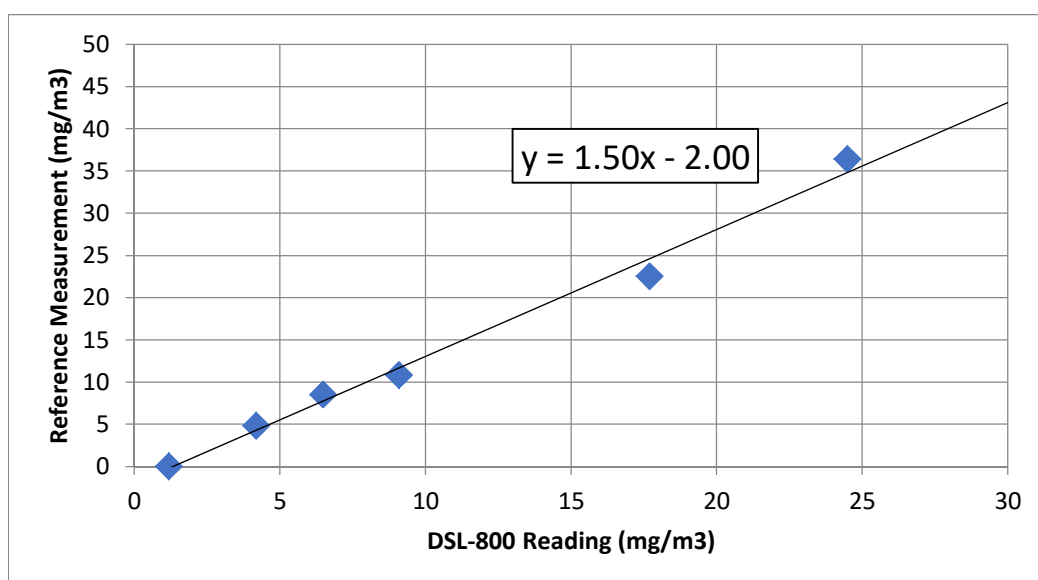
α1₁ = is the starting slope value (default setting =1.0).

α1₂ = is the new calibrated slope value.

Calibration 3: Multi-Point Full Range

A more precise linear calibration can be achieved by performing the previously described coincident measurement at a range of different particulate densities in the stack, from very low to close to maximum. The usual requirement is at least 5 different particulate density levels, approximately equally distributed across the required measurement range. This type of calibration testing generally has to be performed over several days or weeks to get the required range of particulate densities.

From this set of coincident readings, the best fit linear coefficients a_0 (intercept) and a_1 (slope) can be derived using linear regression. An example of this is illustrated below where $a_0 = -2.0$ and $a_1 = 1.50$



If a multi-point calibration test has been performed spanning a large range of dust concentrations, it may be found that the best fit correlation plot is non-linear. This generally only occurs at very high dust levels (e.g., $>200 \text{ mg/m}^3$). In such cases one of the alternative linearisation functions available on the Settings tab could be used.

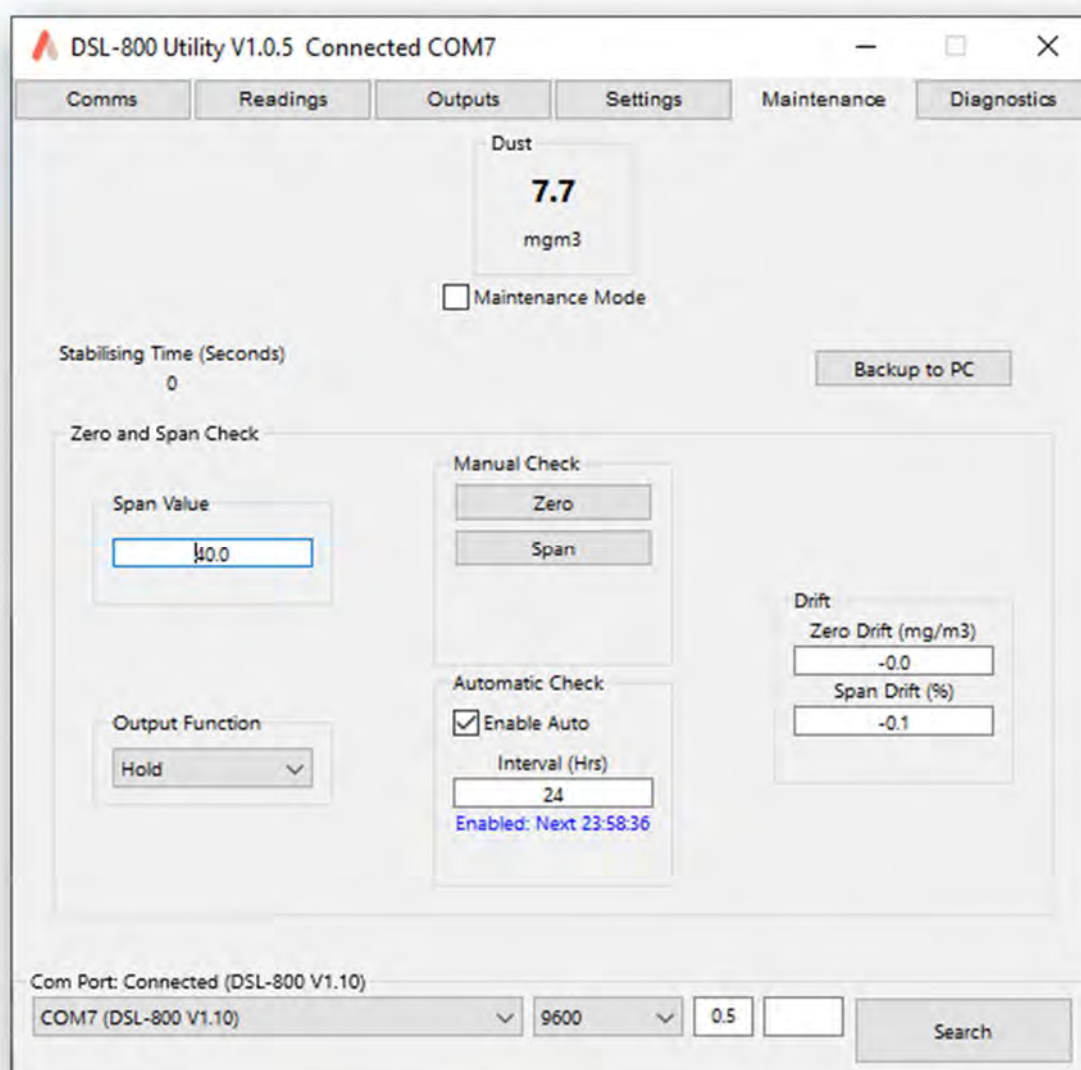
Automatic Zero and Span Check

The DSL-800 has the ability to perform an internal, electronic, zero and span check. The zero check involves turning off the laser and the span check injects a fixed, known amount of light into the detection circuit. These tests check the instrument for any internal long-term drifting, but do not measure any effects due to dirt build up on the optics. The external instrument performance can be checked by a manual

external audit check using the optional calibration head and audit filters (see Appendix A).

NOTE: The automatic zero and span check should only be enabled after the instrument has been configured and calibrated.

The automatic zero and span check is configured using the Maintenance Tab of the Utility. An example is illustrated in the following screenshot.



To set up the automatic zero and span checking it is best to set the parameters in the following order.

First set the 'Span Value' by entering the value into the 'Span Value' box. In the

example above it has set to be 40.0mg/m³.

The maximum and minimum allowed setting for this span value depend on the linearisation settings. If the span value entered is outside of the allowed range the value will change to the closest allowed value.

NOTE: *Do not change the Span Value if the instrument is performing a manual span check or an automatic zero and span check.*

Second set the 'Output Function'. In the example above it has been set to 'Hold'. This settings means that during the automatic zero and span check the instrument output will be held at the last reading before the routine started.

The alternative setting is 'Track', if this is selected the outputs and displays from the instrument will track the readings that occur during the automatic zero and span check.

Third set the 'Interval (Hrs)'. In the example above it has been set to a typical value of '24', which means the automatic zero and span check will be repeated every 24 hours.

NOTE: *The accuracy of the internal clock means that the actual time interval can drift by up to ±30 minutes / year.*

Finally, when you are ready to perform the first automatic zero and span check select the 'Enable Auto' check box. The automatic zero and span check will start immediately, during this time the 'Data Not Valid' flag will be on and a progress bar will be visible. The check lasts about 50 seconds, when complete the 'Zero Drift' and 'Span Drift' values are updated and a timer appears below the 'Interval (Hrs)' box, counting down to the next automatic zero and span check.

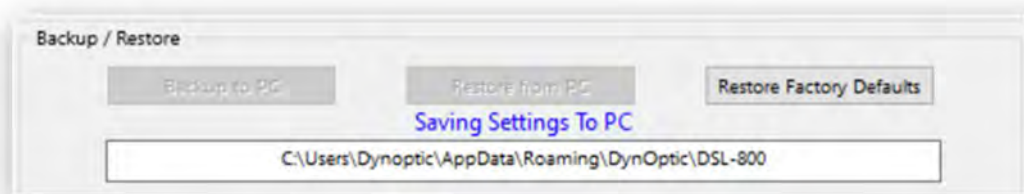
If the instrument is power cycled the automatic zero and span check will re-start 1 hour after the end of the Stabilisation Mode.

The automatic zero and span check can be turned off at any time by de-selecting the 'Enable Auto' check box.

Save a Backup File

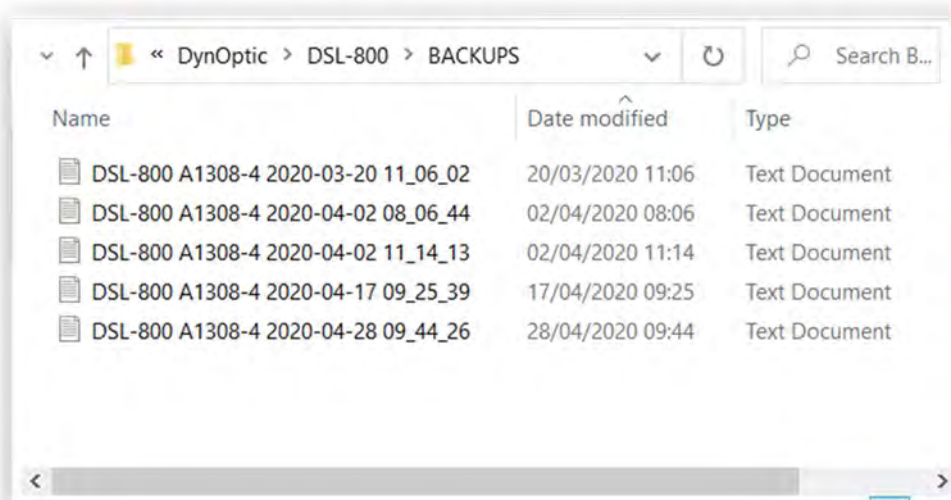
Whenever instrument settings or parameters are changed it is recommended that a backup file is saved. This text file contains all of the current instrument settings. These backup files can be used to restore the instrument settings, should the live settings ever be lost or become corrupted, they can also be used to help diagnose faults or installation issues.

The backup file is produced by clicking the 'Backup to PC' button on the Diagnostics tab of the Utility. It may take several seconds to save the file, the message 'Saving Settings To PC' appears on the Utility tab during the saving process.



The backup text file is saved in a folder called 'BACKUPS'. The location of this folder on the PC is shown in the folder location box. Clicking on this text will open the location.

Each backup text file is saved with the instrument CPU serial number, date and time of the backup. An example of files in the backup folder is shown below.



The backup is a text file containing a list of questions (parameters) and their values.

The following shows the first few lines of a typical DSL-800 backup file.

```
2020-03-20 11_06_02 DynOptic-DSL-800 BackUp file
#VER?,          DSL-800 V1.07
#COMPILETIME?,  11:16:29 Feb 13 2020
#SERNO?,        A1308-4
#UPVER?,        DSL-800 V1.00
MHSTRN?,        DSL-800
MTYPE?,         0
#HWARE?0,       12010120
#HWARE?1,       00000040
EHWARE?,        1
#ID?A,          1
#MBADDR?A,      1
ID?,            1
#DUST?,         61.6,4.99,0000,0101
#VAL?,          61.6
#VAL?1,         61.6
#RDG?4 0,       61.599
#STAT?A,        0000,0000
#STAT?F,        0000,0000
#STAT?W,        0000,0000
#STAT?V,        0000,0000
#STAT?I,        0000,0000
#MA?,           4.99,4.99
#RLY?,          0101
#RLYFW?,        00
#HRSRUN?,       25
#MAINTMODE?,    0
```

General Maintenance

The DSL-800 is a continuous emissions monitor and, as such, is designed to operate unattended for many weeks, or even months, between maintenance. The period between maintenance is often defined by local requirements.

The only regular maintenance required is inspection and cleaning of the optical surfaces. The level of optical contamination will depend on the design and reliability of the purge air, as well as the level of dust in the stack and in the atmosphere around the plant. This means that the optimum maintenance interval is application specific.

An example maintenance procedure is summarised in the table below. The stages in this table are only a guide, any actual maintenance procedure will depend on the installation and any local requirements.

Maintenance Step	Summary
(1) Visual Inspection	- Visually check the TRX head, cables, purge body and mounting flange for any sign of damage or wear. - Inspect the inlet filter on the purge blower and replace if required.
(2) Fault / Warning Check	- Use the UTILITY software to look for any Faults & Warnings and to check the basic performance parameters (e.g., reference signal level, span drift). - Save a Backup file. Compare with a previous Backup file.
(3) Clean Optics	- Use the UTILITY software to set the instrument into Maintenance Mode (turns off the laser). - Unclip the TRX head from the purge-body. - Inspect the optical window and lens, if required remove any dust and wipe away grease marks. - Check the inside of the purge-body and flange tube for excessive dirt build up, clean if required. - Replace the TRX head, turn off Maintenance Mode.
(4) Audit Filter Check (If required)	Perform an audit filter check to see if there has been any significant performance change (see Appendix A).
(5) Test the Interface (if required)	- Use the UTILITY software to set the instrument into Reading Override Mode. - Manually set the instrument reading and confirm correct reading at the data acquisition.

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	○ This test only needs to be performed occasionally (e.g., every 12 months) and only when using the analogue outputs. • Exit Reading Override Mode.
(6) Save final Backup file	• When all tests and changes have been made save a final Backup file.

Table 1: Example Maintenance Procedure

Repairs and Replacements

The DSL-800 TRX head is a sealed unit and should not be opened when power is applied or when it is mounted on a stack. The DSL-800 head should only be opened by trained Dynoptic personnel.

The DSL-800 has been designed to be quickly removed and replaced. All electrical interfaces to the instrument are plugs (apart from the earth connection) and the head simply clips onto the purge-body. If an instrument is faulty or requires repair the recommendation is to replace it with a spare one (of the same variant). The faulty instrument can then be returned to Dynoptic for investigation and repair.

After replacing a DSL-800 TRX head it is good practice to repeat the set up and commissioning procedure.

The DSL-800 contains a laser source which will degrade with time and will need replacing. The laser replacement should only be performed by trained Dynoptic personnel.

The laser power is continuously monitored (Reference Signal Level) and if it drops below 1.5V the instrument activates the 'DATA NOT VALID' and 'LO REF SIGNAL' fault messages. It is recommended that the 'Reference Signal Level' is monitored at each instrument maintenance, to provide an early indication of when an instrument is approaching a 'LO REF SIGNAL' fault.

Fault and Warning Messages

Fault and Warning messages can appear either on the Readings tab of the Utility software, or on the display of the optional DSCU.

When messages appear in the Utility software, they can remain permanently displayed on the Readings tab until the fault has been cleared and the operator has clicked on the text to acknowledge the fault. This means that even if a fault occurs only temporarily and has been resolved, the fault message may remain visible until the operator has clicked on the text to acknowledge it. Fault message text can only be cleared if the fault has been resolved.

Where messages appear on the optional DSCU they will flash intermittently on the display. This means that the operator can still navigate and use the DSCU but will be constantly reminded of the fault by the flashing message. Flashing messages will automatically disappear when the fault is resolved.

Fault messages indicate there is a fault with the instrument and that the readings are not valid. Warning messages indicate that there may be a problem, and the warning should be investigated.

The DSL-800 TRX Head is the intelligent centre of the instrument, and all measurements, calculations, and data storage are done in the TRX Head. The optional DSCU is a terminal which presents data from the instrument and provides an interface through which the operator can alter settings and parameters in the instrument.

The Utility software runs on a PC or laptop which can be connected via USB to either the TRX Head, or to the optional DSCU. If a PC is connected to the DSCU instead of the RX head, the DSCU simply becomes a conduit through which data is passed between the PC and the TRX head. In this scenario, if the DSCU has a fault, or loses communications with the instrument, or loses power, the Utility software will be unable to retrieve the required data from the instrument, and the Utility software will disconnect.

The following section describes all of the fault and warning messages that can

appear in DSL-800 instrument. Potential causes of each fault are outlined and actions are highlighted that can be taken to try and resolve the issue. If the actions described do not solve the problem, then further advice should be sought from Dynoptic.

Fault Messages

LO REF SIGNAL

This message indicates that the measured 'Reference Signal Level' has dropped below 1.5V. This signal is a direct measure laser brightness.

This message can be caused by a dim or failed Laser source (all lasers diminish with time and ultimately fail, though expected lifetimes are measured in years), a failure of the laser drive circuit, or a failure of the reference signal measurement circuit.

If the message appears during normal operation the following actions should be considered:

- Replace the TRX head.
- Send the failed head back to Dynoptic for investigation and repair.

In the event that the TRX head is replaced, it is advisable to repeat the set up and commissioning procedures.

STABILISING (self-clearing error message)

This fault message occurs when the instrument has undergone a major change and requires time to stabilise. The changes include power on, performing an internal zero or span check and turning off Maintenance Mode Calibration Mode or Reading Override Mode. The stabilisation fault message will remain for a period of time ranging from 1 to 20 minutes, to give the instrument readings time to settle, and then the fault message will self-clear.

When the instrument is in Stabilisation Mode it is not possible to perform some Utility functions such as internal zero or span checking.

If an instrument is producing the 'STABILISATION' fault it will self-clear at the end of

the stabilisation period. A stabilisation period countdown (in seconds) is provided in the Maintenance tab of the Utility.

MAINTENANCE MODE (self-clearing error message)

Before performing any maintenance on the instrument, it must first be set into Maintenance Mode. This will turn off the laser and activate the 'MAINTENANCE MODE' fault.

When Maintenance Mode is turned off the fault will self-clear. However, the STABILISATION fault will be activated and will stay in place for 1 minute.

If the instrument is power cycled this will also turn off the Maintenance Mode.

If the Utility is disconnected from the instrument without exiting Maintenance Mode then the instrument will automatically exit Maintenance Mode after 1 hour.

CALIBRATION MODE (self-clearing error message)

Before performing a calibration or audit check on the instrument, it must first be set into Calibration Mode. This will activate the 'CALIBRATION MODE' fault.

When Calibration Mode is turned off the fault will self-clear. However, the STABILISATION fault will be activated and will stay in place for 1 minute.

If the instrument is power cycled this will also turn off the Calibration Mode.

If the Utility is disconnected from the instrument without exiting Calibration Mode then the instrument will automatically exit Calibration Mode after 1 hour.

OVERRIDE ACTIVE (self-clearing error message)

During set up, commissioning or maintenance of the instrument it may be necessary to set the DSL-800 into Reading Override Mode. This feature is used to confirm the correct operation of the interface to the data logging system. When in this mode the fault relay is activated and 'OVERRIDE ACTIVE' fault message is displayed. The fault self-clears when the unit returns to normal reading mode.

If this message appears during normal operation, then the instrument has been accidentally switched into Reading Override Mode. It should turn itself off after

5 minutes.

ZERO CHECK ACTIVE (self-clearing error message)

This message indicates that the instrument is performing an internal zero check. If this is part of the periodic automatic zero and span checking, then the fault message will clear at the end of the routine.

The same fault message appears when the Utility has been used to switch on the manual zero check. The message will remain until the manual zero check has been turned off or the instrument has been power cycled.

SPAN CHECK ACTIVE (self-clearing error message)

This message indicates that the instrument is performing an internal span check. If this is part of the periodic automatic zero and span checking, then the fault message will clear at the end of the routine.

The same fault message appears when the Utility has been used to switch on the manual span check. The message will remain until the manual span check has been turned off or the instrument has been power cycled.

Warning Messages

HI SPAN DRIFT (self-clearing error message)

This message indicates that the last, internal automatic zero and span check has measured an internal span drift of more than $\pm 90\%$. This level of span drift can only occur if there is a fault with the span optical source. A faulty span optical source will not affect the dust measuring performance of the instrument and so this condition activates a Warning message and not a Fault (or Data Not Valid) message.

This issue can only be solved by returning the instrument to Dynoptic for investigation and repair.

TEMP OUT OF RANGE (self-clearing error message)

The DSL-800 monitors the CPU board temperature. The Diagnostics tab of the Utility displays the measured board temperature, as well as the minimum and maximum

temperatures. The 'TEMP OUT OF RANGE' warning will be activated if the temperature reading goes outside of the operating range of -15 to +70 degC.

This warning message can be an indication that the TRX head has been subjected to excessive heat (or cold) outside of the recommended operating range.

The fault could also be caused by a failure of the temperature sensor or the surrounding circuitry. Contact Dynoptic for further advice.

MEMORY FAULT (self-clearing error message)

The DSL-800 employs CRC checking on the EEPROM memory. Each time the instrument is powered up the CRC checking compares the state of the EEPROM that it last knew, against the state of the EEPROM memory on power up.

If the instrument finds a discrepancy between the two EEPROM memory states the 'MEMORY FAULT' warning is activated.

On registering the 'MEMORY FAULT' the instrument performs a self-checking routine which analyses each byte of memory and ensures that the values stored there are within an acceptable range. Values outside the acceptable range are considered to be corrupted values and the memory byte is replaced with a default value from FLASH memory. Essentially the instrument heals itself.

Memory corruption is rare, but can occur in certain circumstances, for example if the instrument suffers a power loss whilst writing to its EEPROM memory.

Discrepancies can also be caused after performing a software upgrade to the instrument.

On encountering a 'MEMORY FAULT' warning operators should always power cycle the instrument, i.e., switch it off, then on again. This action should clear the warning because the self-healing process will have installed a new value in the corrupted memory byte, and the compared states of the power off memory and the power on memory should now match.

If a power cycle does not clear the warning it is likely that there is a more serious problem and you should contact Dynoptic for further assistance.

NOTE: *When the instrument self-heals by replacing corrupted data with default data from FLASH memory, some operational settings and values may have been overwritten. It is therefore essential that all parameters, settings, and values are checked for validity by the operator immediately after resolving a 'MEMORY FAULT'.*

COMMS CRC FAULT (self-clearing error message)

This warning message can only occur when the instrument is connected to a an optional DSCU. The communication between the DSL-800 and a DSCU uses a check signal (CRC) to test the quality of the line. This warning message appears if the communication check fails.

This message may indicate one of the following: -

- The communications line is too noisy, this may be due to the close proximity of high power or AC voltage cables.
- There is a problem with the set-up of the DSL-800 to DSCU communication cable (e.g., poor grounding, bad connection).

ALARM ACTIVE (self-clearing error message)

If one of the instrument alarm relays has been set to be a threshold relay, then this warning message will appear whenever the reading triggers the threshold relay.

Appendix A: Performing an Audit Check

This appendix describes how to perform a manual audit filter check on an installed DSL-800 instrument.

Performing an audit check requires the optional calibration check kit (DSL-8-AFK) which contains a calibration head and three audit filters. Examples of these items are shown below.



Audit Filter

The audit filter introduces a low-level scatter very close to the instrument. This low level of scatter light is detected by the instrument and displayed as a dust reading (mg/m^3). The value of the dust reading will depend on the instrument and the calibration values set during commissioning. Therefore, it is not possible to provide a fixed calibration value for the audit filters.

After a DSL-800 has been installed and 'calibrated', the first audit filter

measurement will provide the reference reading for this instrument. Subsequent readings with the same audit filter can then be used to track long term drifts and variations in the performance of the instrument.

The audit filter is provided as a set of three (low, mid and high). The 'mid range' filter provides a reading approximately x3 higher than the 'low range' filter. The 'high range' filter provides a reading approximately x3 higher than the 'mid range' filter. This set of filters should provide at least one filter that will produce a reading within most application ranges.

WARNING: *The performance of the filter is very sensitive to contamination of the optical surface. When not in use the filter should always be stored in the packaging in which it was supplied.*

WARNING: *The exposed optical surface should never be touched. Any loose dust or dirt should be removed with a puff of clean dry air or brushed off with a camera lens brush.*

NOTE: *If the optical surface shows signs of scratches or contamination that cannot be removed the filter should no longer be used.*

NOTE: *The audit filter will produce a different reading value for each instrument. The filter is used to track long term performance changes, not to calibrate an instrument.*

Test Method

Before performing a performance check the instrument must have been correctly installed, set up, commissioned and running for at least 30 minutes.

The basic performance check procedure is summarised in the following pictures and steps.

	Start with the DSL-800 mounted on a stack (1) Use the Utility to turn on Maintenance Mode (to turn off the laser)
	(2) Unclip the instrument from the purge body, Attach the Calibration head to the purge body, Clip the instrument to the Calibration head. (3) Use the Utility to turn on Calibration Mode. Wait at least 2 minutes.
	(4) Insert the Audit filter into the calibration head (Optical surface towards the instrument) (5) Wait at least 10 min then record the reading
	(6) Remove the Audit filter (7) Use the Utility to turn off Calibration Mode and then turn on Maintenance Mode (to turn off the laser).
	(8) Unclip the instrument from the calibration head, remove the calibration head from the purge body, clip the instrument back onto the purge body. (9) Use the Utility to turn off Maintenance Mode

1. Use the UTILITY to set the instrument into 'Maintenance Mode'
 - a. This turns off the laser making it safe to remove the instrument from the purge body.
2. Unclip the instrument from the purge body, attach the calibration head to the purge body and clip the instrument to the calibration head.
3. Use the UTILITY to turn on 'Calibration Mode'. Wait at least 2 minutes.
 - a. Do not perform an audit filter test if there is a possibility of the automatic zero and span turning on during the test. Check the countdown timer on the Maintenance tab of the UTILITY.
 - b. It can be helpful to log the readings during the test. Use the Readings tab of the UTILITY to enable the 'logging' running at 10 sec interval.
4. Insert the audit filter into the calibration head (optical surface towards the instrument).
5. Wait at least 10minutes then record the reading (from the UTILITY or the instrument display).
 - a. If this is the first audit filter measurement then this reading is used as the reference.
 - b. All future readings made using the same DSL-800 instrument and same audit filter should be compared to the reference reading.
6. Remove the audit filter.
7. Use the UTILITY to set the instrument off 'Calibration Mode' and then turn on 'Maintenance Mode'.
 - a. If 'logging' is being used then turn 'logging' off before setting the instrument into 'Maintenance Mode'.
8. Unclip the instrument from the calibration head, remove the calibration head, then clip the instrument back onto the purge body.
9. Use the UTILITY to turn off 'Maintenance Mode'

Appendix B: Modbus Configuration

Protocol Version:	Modbus RTU Over RS485
Baud Rate:	9600, 19200 , 28800, 38400, 57600
Data-Parity-Stop Bits:	8-None-1, 8-None-2, 8-Even-1 , 8-Odd-1:
ID:	1 -247 (Via Programmable ID) (Bold Indicates Default Values)

Modbus Information

Unit Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Unit Version	04	10000	String	13	Read Only
Serial No.	04	10013	String	6	Read Only
Code Version	04	10019	String	13	Read Only
Reading Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Particulate (mg/m3)	04	10500	Float	2	Read Only
Output/Alarm Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Output 1	04	10520	Float	2	Read Only
Relay/Alarm Status	04	10580	Word	1	Read Only Bitwise Flags (Alarm: 0=No Alarm, 1= Alarm) (Data Valid: 0=Not Valid, 1=Valid) Bit00-RLY1-ALARM/DV STATUS Bit01-RLY2-ALARM/DV STATUS Bits02-07-SPARE (0=Relay Open, 1=Relay Closed) Bit08-RLY1-RELAY STATUS Bit09-RLY2-RELAY STATUS Bit10-15-SPARE

Fault / Warning Information					
Description	Function Code	Address	Data Type	Length (Words)	Comment
Unit Status	04	10590	Word	1	Read Only Bitwise Flags Bit00-ZERO ADJUST ACTIVE Bit01-ZERO CHECK ACTIVE Bit02-SPAN CHECK ACTIVE Bit03-SPAN CAL ACTIVE Bit04-HIGH SPAN DRIFT Bits05-11-SPARE Bit12-MEMORY FAULT Bit13-COMMS CRC FAULT Bit14-SPARE Bit15-STABILISING
Unit Status	04	10591	Word	1	Read Only Bitwise Flags Bit00-ZAP ACTIVE Bit01-LOW REFERENCE Bit02-MAINTENANCE MODE Bit03-DATA INVALID Bit04-ALARM ACTIVE Bit05-TEMP OUT OF RANGE Bit06-SPARE Bit07-AGA ACTIVE Bit08-OVERRIDE ACTIVE Bit09-AGC ACTIVE Bit10-ARGA ACTIVE Bits11-15-SPARE

Examples

The following examples are for illustrative purposes and the Data received could be different dependent on the type of instrument.

String Read	
Unit Version Read	ModBus Addr:1, Address:10000, Length:13
Master Message (Command)	01 04 27 10 00 0D 3A BE
Instrument Message (Reply)	01 04 1A 56 31 2E 31 37 00 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 2A 00 2A 82 FB
Received String	V1.17*****
Word Read	
Unit Status Read	ModBus Addr:1, Address:10590, Length:1
Master Message (Command)	01 04 29 5E 00 01 58 44
Instrument Message (Reply)	01 04 02 C4 80 EA 50
Received Word	50304 (C480 Hexadecimal)
Float Read	
Reading 1 Read	ModBus Addr:1, Address:10500, Length:2
Master Message (Command)	01 04 29 04 00 02 38 56
Instrument Message (Reply)	01 04 04 41 58 C8 CE B8 3F
Received Word	13.55

Notes

Disclaimers

All technical details and specifications are subject to change without notice.

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CUSTOMER:

Ingredion (Thailand) Co., Ltd.

LOCATION:

Kalasin

PROJECT TITLE:

Project SPRINT 18K TPA Spray Dry

CUSTOMER P.O. NO.:

7001730092

PACKAGE:

Dust Analyzer

EPM PROJECT NO:

6626013

DOCUMENT TITLE:

Installation, Operation and Maintenance Manual

3.Purge Air Requirements

Air Supply for Optical Stack Monitor Purge Systems

Technical Note V1.2

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Introduction

The performance of stack monitors based on open-path optical measurement methods can be impaired by the build-up of dirt on the exposed optical surfaces from dust and soot in the flue gas. To minimise this effect it is common practice to provide an air-purge system, which typically consists of an air-purge body, positioned between the optical instrument and the stack, supplied with a source of clean and dry purge air. The flow of air through the air-purge body keeps the hot, dirty and corrosive flue gas away from the optical surfaces and also helps to cool the instrument head. An illustration of such a system, using an air blower to supply purge air to two instrument heads, is shown in Figure 1.

The design of the air-purge body varies considerably from different manufacturers and to ensure correct operation the purge air supply must be correctly matched to the air-purge body. This technical note summarises some of the general issues that need to be considered when selecting an air supply and provides detailed requirements for use with Dynoptic air-purge bodies. Our engineering team are also available to assist with the selection or definition of a suitable purge air supply.

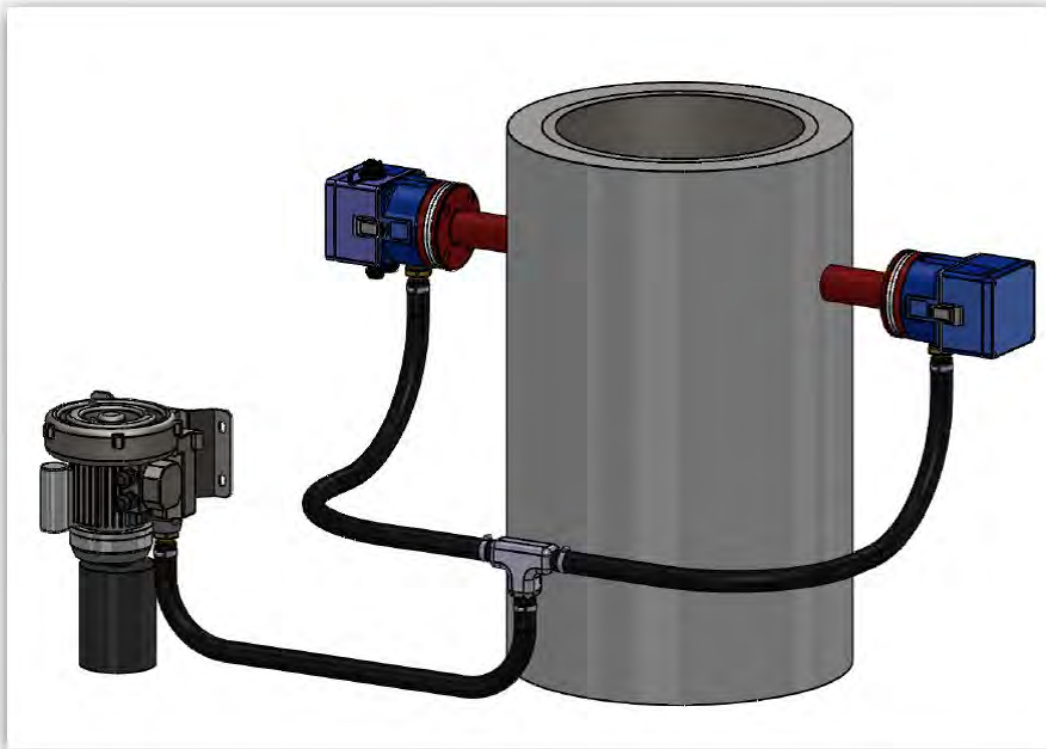


Figure 1: Dynoptic stack mounted monitor, with two air-purge bodies, fed by a single blower

Purge Air Requirements

Dynoptic air-purge bodies are designed to operate with an air flow into each body in the region of 50 to 200 litres/min. The lower flow rate is sufficient to overcome typical stack gas flow fluctuations and the upper flow rate is to prevent the flow of purge air from affecting the optical measurement.

Many optical stack monitors require two optical heads, one either side of the stack, as illustrated in Figure 1. In such a system it is common practice to use a single air supply to provide the purge air to both air-purge bodies. In this configuration the source of purge air must be able to supply 100 to 400 litres/min.

In some applications the stack, or measurement chamber, can be under a positive pressure. The effect of this is to force the hot, dirty, corrosive flue gas up the flange and onto the optical head. Under such conditions the purge air supply must be capable of providing the required flow rate and also have sufficient pressure to overcome the pressure of the stack. This requires the use of compressed air.

Dynoptic recommend two alternative sources of air purge, an air blower, or compressed air. The basic advantages and issues with each type are summarised below and further design issues are discussed in the following sections.

Air Blower	Compressed Air
Low pressure high volume source of air. Well matched to the air-purge body requirements.	High pressure lower volume source of air. A large compressor reservoir capacity is needed to provide the required volume air flow.
Low cost standalone system.	Pressure regulation and in-line particle and oil mist filters must be used to provide the required clean and dry purge air.
Uses ambient air, requires a good filter to keep it clean. Inlet air must be dry (low humidity).	Flow restriction required at the air-purge body.
Blower must be fixed relatively close to the purge bodies (<50m).	Capable of providing a high pressure flow for use in positive stack pressure applications.

Table 1: Summary of alternative sources of purge air

Air Blower

An air blower consists of a rotating fan arranged to suck air in to an inlet port and push air out of the outlet port. The performance of a blower is generally specified by two parameters, the maximum differential pressure it can provide at the outlet port with no air flow, and the maximum air flow with no pressure loss. Ideally the blower specification will also provide a pressure flow performance graph, an example of which is illustrated in Figure 2.

A typical requirement for a Dynoptic purge air source is to be able to provide an air flow of at least 100 litre/min. For the blower performance illustrated below this flow rate can be achieved provided the total pressure loss in the system is less than ~90 mbar.

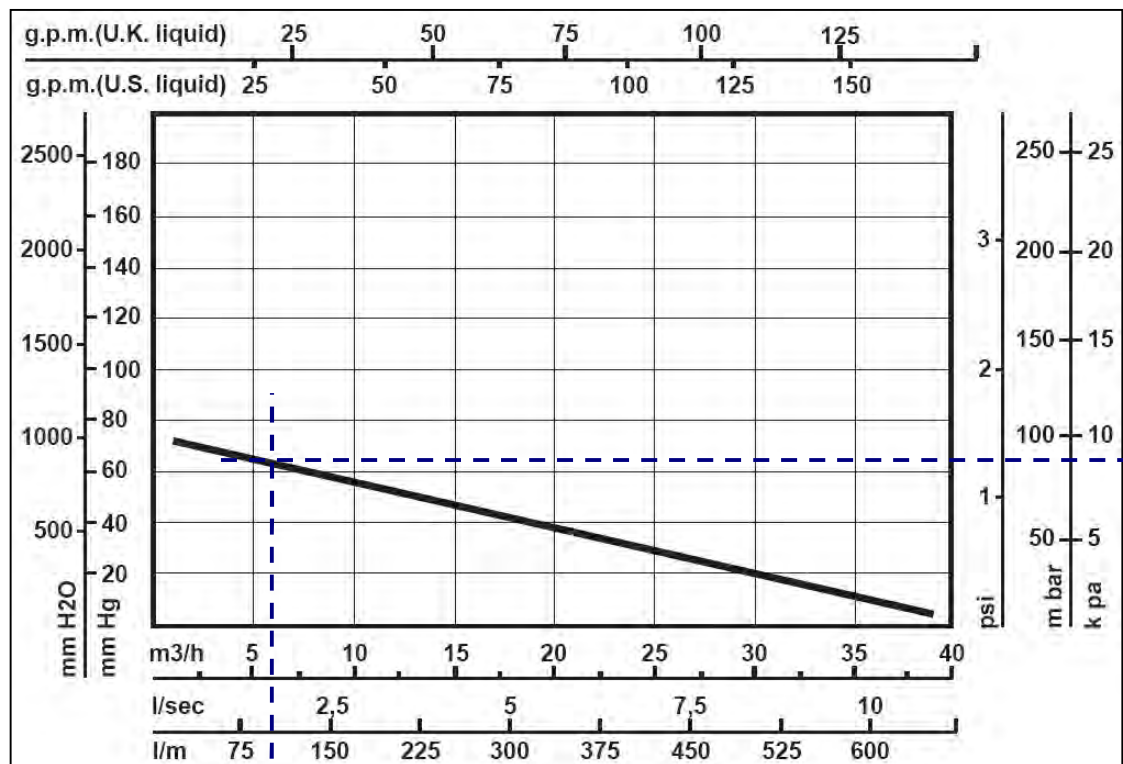


Figure 2: Example of a blower performance curve

General Blower Requirements

The main contribution to the pressure loss in an air-purge system is the internal design of the air-purge body. However, some pressure is also lost in the tubing from the blower to the air-purge bodies. To minimise the system pressure loss the tubing should have a large internal diameter (i.e. 25mm) and the total length must be as short as practical.

Table 2 summarises the minimum blower performance recommended for use with two Dynoptic air-purge bodies. This minimum requirement assumes that the tubing has an internal diameter of 25mm and a maximum length of 50m.

	Value
Maximum Flow Rate	>0.007m³/s >24m³/hr >400 litres/min >14 cfm
Maximum Differential Pressure	>3.0x10³ Pa >30 mbar >0.4 PSI

Table 2: Blower performance required to supply two Dynoptic air purge bodies

Blower Inlet Filter

The purge air produced by the blower must be clean, dry and free from dust and aerosols, in order to prevent it contributing to the measured stack dust or contaminating the optics. It is recommended that the air going into the blower comes from a relatively clean and dry environment. In addition a good air filter is required. The recommended performance of a filter for the blower inlet is summarised in Table 3.

Parameter	Value
Particle Filter Size	<5µm
Maximum working flow rate	>0.005 m³/s >18 m³/hr >300 l/min >10 cfm
Pressure Drop (at working flow rate)	<200 Pa <2 mbar <0.03 PSI

Table 3: Recommended performance of blower inlet filter

Dynoptic Blower Kit (DSL-BK40B)

Dynoptic offers a standard blower kit for use with our air-purge bodies. This unit is designed for applications with a total tubing length up to 50m. A summary of the blower kit is provided below.

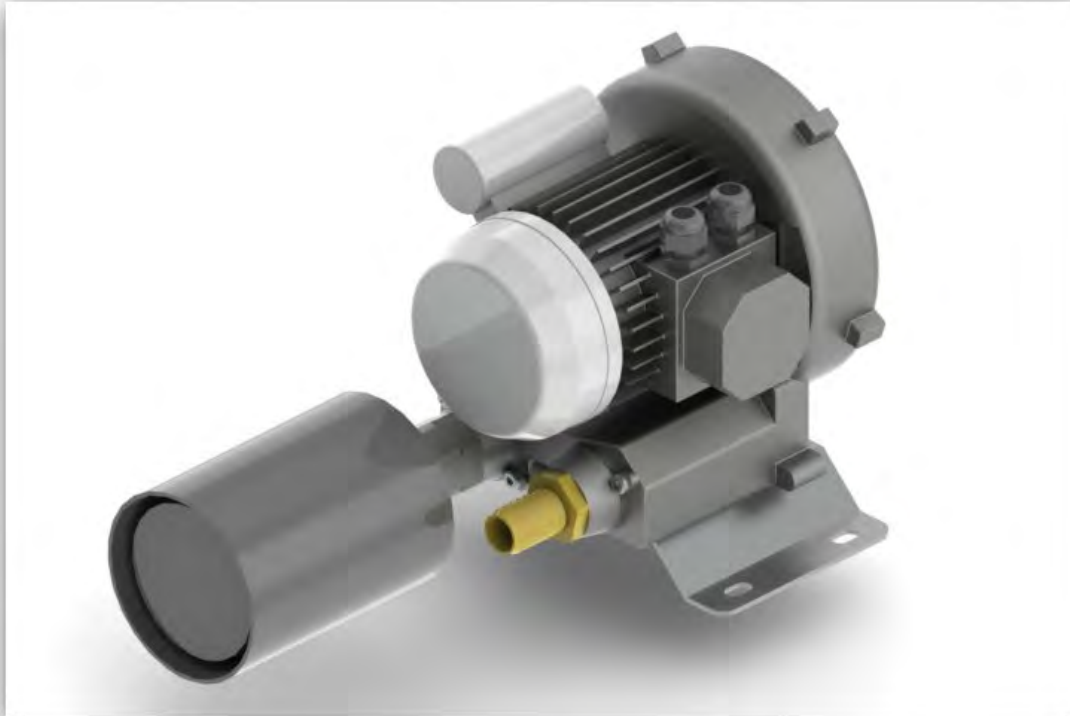
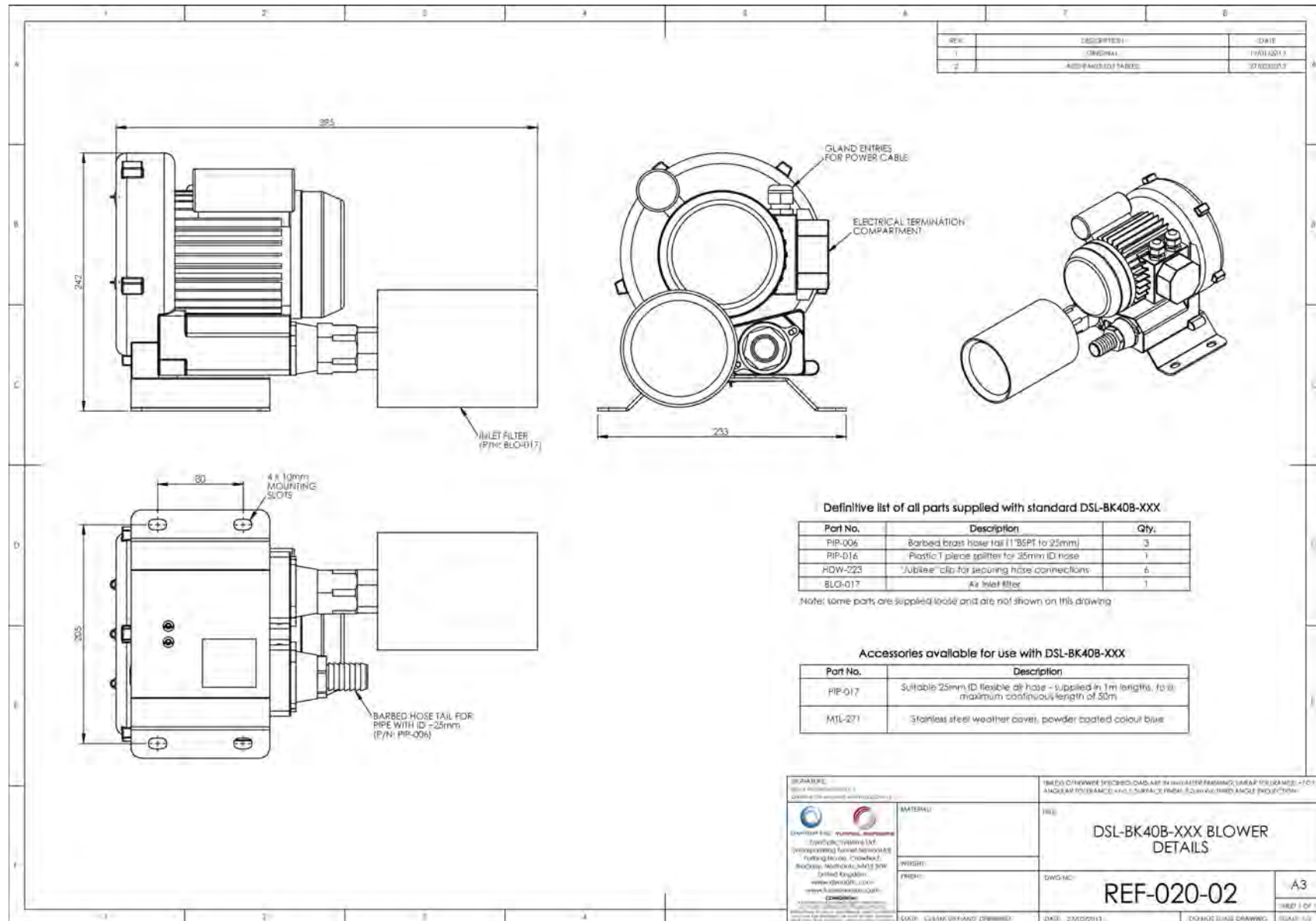


Figure 3: DSL-BK40B Blower

Part No.	Description	Quantity
Parts supplied with blower DSL-BK40B		
PIP-006	Barbed brass hoesetail (1" BSPT to 25mm)	3
PIP-016	T-piece splitter for 25mm ID hose	1
HDW-223	"Jubilee" clip for securing hose connections	6
BLO-017	Air inlet filter	1
Accessories available for blower DSL-BK40B		
PIP-017	25mm ID flexible air hose. Supplied in lengths from 0.5 up to 50m.	
MTL-271	Stainless steel weather cover, powder coated, colour blue.	

Table 4: Parts Supplied & Accessories Available for the BK40B

Air Supply for Optical Stack Monitor Purge Systems



Blower Installation Issues

When defining the location of a blower based air-purge system some other factors need to be considered.

Filter Position

The inlet air filter supplied with the blower has a weather proof cover. When the blower is mounted this filter must be positioned pointing downwards to minimise the effect of large particles falling into the filter. If the filter is to be exposed to rain position the filter away from flat surfaces to avoid rain splashing up into the filter.

T-Piece

If the air flow from the blower needs to be split, to feed two air-purge bodies on either side of the stack, then this can be achieved using the supplied T-piece. To minimise air flow losses it is recommended that the T-piece is mounted as close to the blower as possible. Ideally the length of any tubing from the blower outlet port to the T-piece should be less than 2m.

Tube Length Balance

If the blower is supplying air to two air-purge bodies then the tubing from the T-piece to each one should be the same length. In this way the air flow will be balanced in both air-purge bodies. If one tube length is significantly shorter than the other then the pressure losses in that tubing will be less and its flow rate will be higher.

It is not always practical to maintain equal lengths of tubing. The recommendation is that for short lengths the difference between the tube lengths should be no more than 5m and for long lengths the relative lengths of the tubing in the two sections should be less than 2:1.

Height Above Blower

If the air-purge body is significantly higher than the blower then some of the available differential pressure will be used to overcome the height difference. To prevent this reducing the efficiency of the system it is recommended that any height difference should be less than 20m.

Compressed Air

If a suitable source of compressed air is available at the stack then this can be used as the purge air, provided careful consideration is taken to cleaning and controlling the air flow.

Compressed air should never be supplied directly to the inlet of an air-purge body. The high pressure will produce very high airflows and the contamination in the compressed air is very likely to affect the optics. To use compressed air it is necessary to correctly filter the air to remove particulates and any oil mist before it enters the air purge body. At the air-purge body a flow restrictor must be used to achieve the required flow rate from the high pressure air.

A practical system arrangement is illustrated in the figure below for a single sided installation; all of the parts required to implement this system are available from Dynoptic.

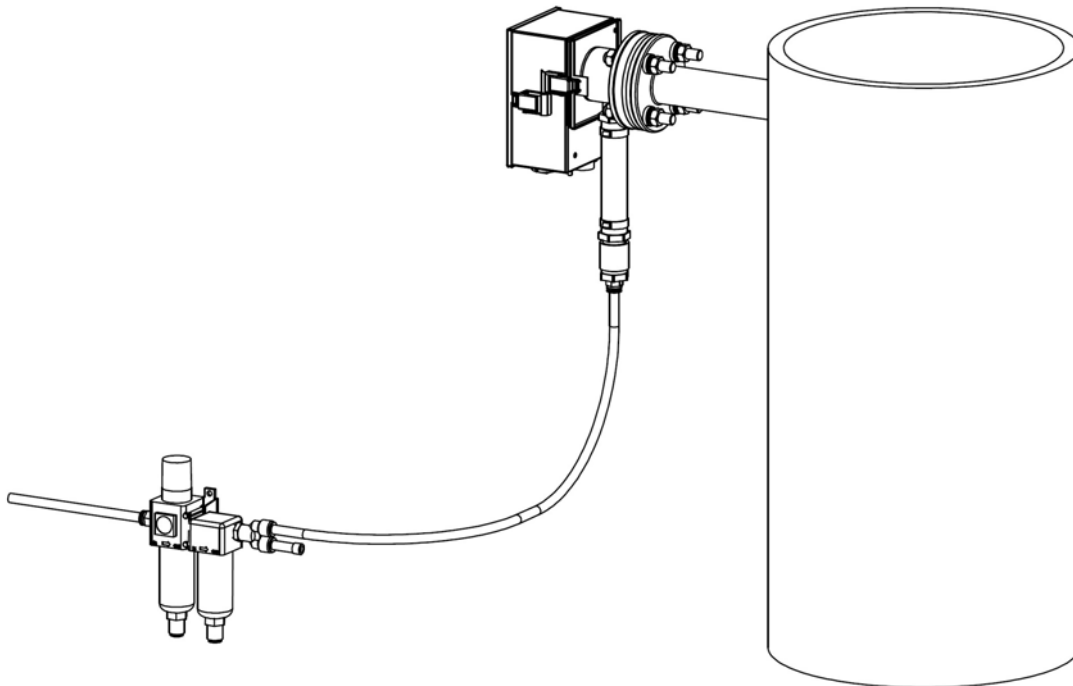


Figure 4: Illustration of a single sided Dynoptic stack mounted monitor, with purge air supplied by compressed air

Flow Restrictor Assembly

The pressure available from a compressed air system is generally very high (>5bar). If this was applied directly to the inlet port of an air-purge body the air flow would be too high. To control the air flow into an air-purge body a flow restrictor assembly is available from Dynoptic (part no. DSL-CAFR).

Air Supply for Optical Stack Monitor Purge Systems

The flow restrictor assembly attaches to inlet of the two styles of Dynoptic air-purge body as illustrated in the following drawing.

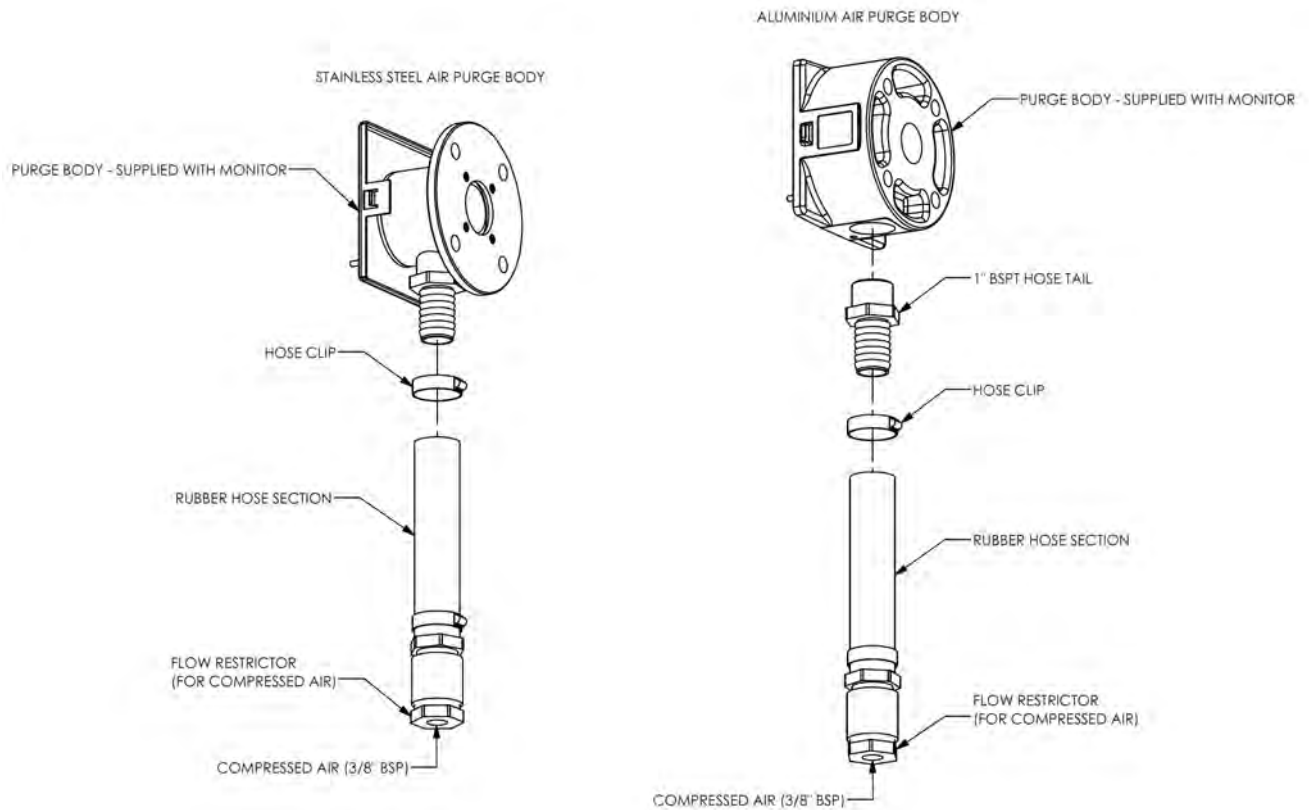


Figure 5: Installation of Flow Restrictor Assembly onto a Stainless Steel or Aluminium Air-Purge Body

With the flow restrictor in place the desired airflow is achieved by controlling the pressure of the compressed air at the inlet. The typical variation of air flow through the air-purge body as a function of the pressure of the supplied compressed air is shown below. The recommended operating pressure is 3 to 5 bar.

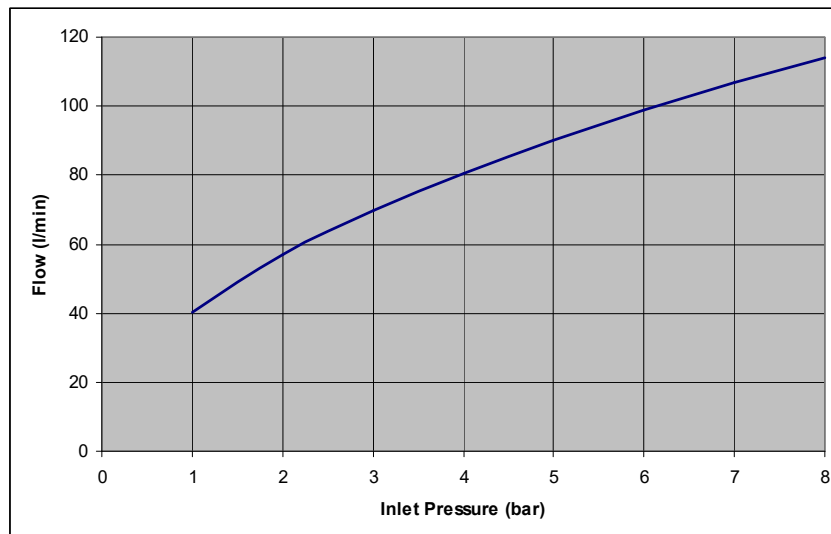


Figure 6: Airflow in a Dynoptic air-purge body with flow restrictor, as a function of input air pressure

If the system is operating on a positive pressure stack then a higher operating pressure will be required to overcome the gas pressure within the stack.

Pressure Regulator

To control the pressure, and hence the air flow, at the inlet to the air-purge body, it is recommended that a pressure regulator is used at the input to the filtering stage. The basic requirements for such a pressure regulator are summarised below.

Parameter	Value
Minimum Adjustment Range	3 to 5 bar (44 to 73 psi)
Flow Rate	>0.003 m ³ /s >12 m ³ /hr >200 l/min >7 cfm

Table 5: Recommended performance of pressure regulator

Filtering

Industrial compressed air is not generally clean enough to use directly as purge air, it can contain water and oil as well as particulates, all of which can affect the optical measurement. It is therefore necessary to provide a well filtered compressed air supply for air-purge applications. The ideal filtration stage should consist of a 5µm particulate filter, to remove the larger particles, followed by a fine oil mist filter. The purity level and flow rate required at the output of the filtration stage is summarised below.

Parameter	Value
Particle Filter Size	0.01µm
Filtered Oil Level	0.1 mg/m ³
Flow Rate	>0.003 m ³ /s >12 m ³ /hr >200 l/min >7 cfm

Table 6: Recommended performance of compressed air in-line filter

Dynoptic Compressed Air Kit (DSL-CAK)

Dynoptic provide a kit of parts that simplifies the design and implementation of a compressed air based air-purge system. The basic kit contains a regulated air filter assembly (ASY-180) as illustrated below, as well as the flow restrictor assemblies (DSL-CAFR). The regulator dial measures the line pressure in units of MPa (1bar = 0.1MPa) and the line connection ports are 12mm push fit.



Figure 7: Regulated filter assembly (ASY-180)

Air Supply for Optical Stack Monitor Purge Systems

Part No.	Description	Quantity
Parts supplied with compressed air kit DSL-CAK-1 or DSL-CAK-2		
PIP-019	Regulated air filter assembly	1
PIP-020	Micro mist separator assembly	1
PIP-021	Mounting bracket assembly	1
PIP-022	Y-piece tube fitting (3/8" BSP male to 12mm push-fit)	1
PIP-026	Straight tube fitting (3/8" BSP male to 12mm push-fit)	3
PIP-030	Sealing bung for 12mm push fitting	1
DSL-CAFR	Flow restrictor assembly	1 or 2
Accessories available for the compressed air kit		
ASY-182	Replacement filter element pack. 5 off air filter elements and 5 of micro mist separator elements.	
PIP-025	20m reel of nylon pneumatic tubing. 12mm OD.	
PIP-033	Cutting tool for tubing.	

Table 7: Parts supplied with compressed air kits DSL-CAK-1 or DSL-CAK-2

Air Supply for Optical Stack Monitor Purge Systems

